Particles and Nuclei 2

- 4/21/2026
- Announcements:

- HW 6/Takehome Final First parts open- due Mon. May 4th.

- New Reading -- Handout from Wong, will be more covered in class, Thomson handout mostly just for interest.

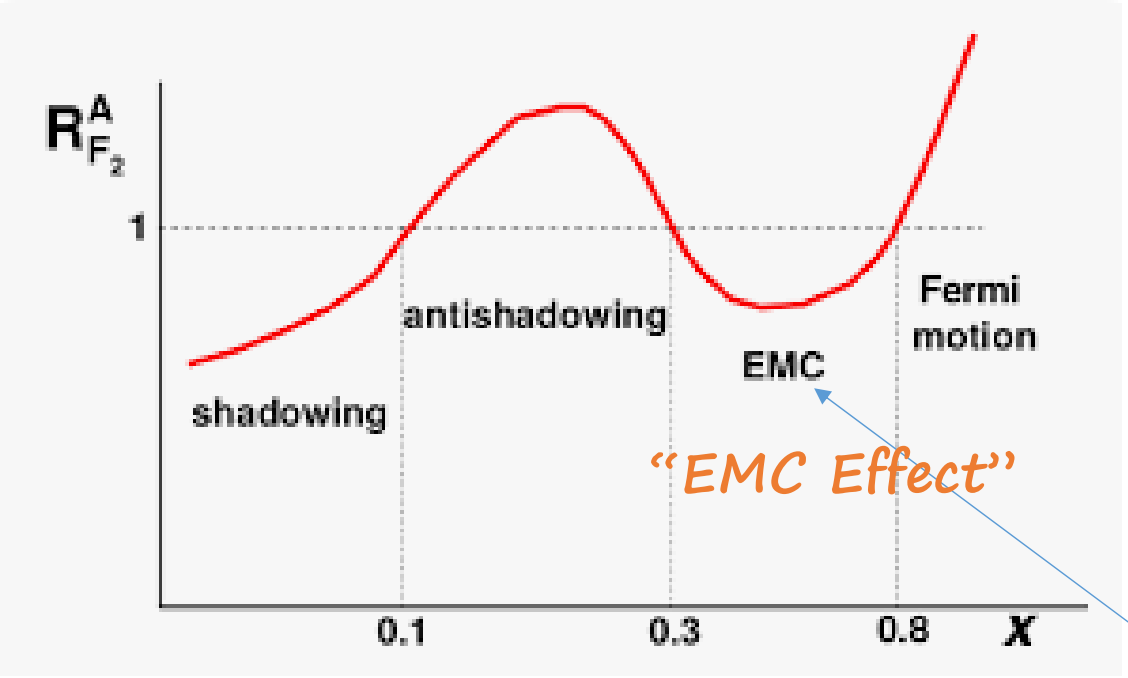
Next

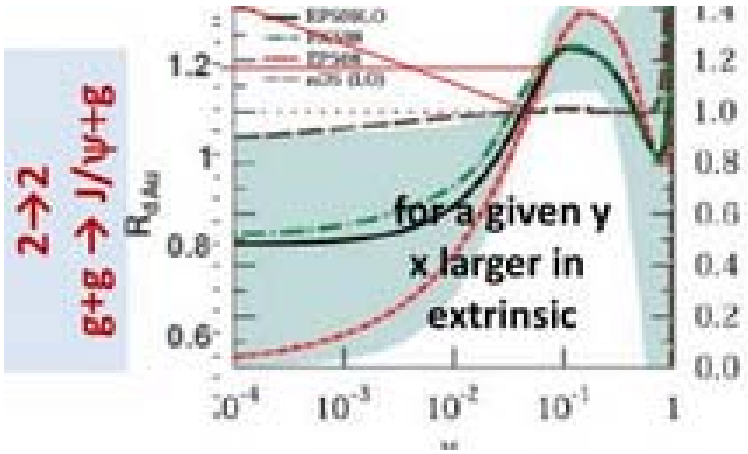
- Finish off QCD Factorization/ pdfs/ structure functions
  - Fragmentation Functions
  - Different cases of factorization
  - Hadronization Models Codes
  - Alt Factorizations Color Dipole model

- Weak Interaction
  - Already covered: Flavor structure and CKM Mixing (long ago)
  - Already covered: Basics of Feynman charge current scattering situations (F3) (last time)
  - TODAY: CC in DECAYS → Muon and Beta Decay
  - Z boson "Neutral Current" Interactions
  - More on Chiral fermion states & Electroweak Unification Weak Isospin
  - Back to CKM Neutrino Oscillations
  - Local Gauge Invariance, Spontaneous Symm Breaking and Higgs Mechanism

Nuclear PDF's

- Generally we expect PDF's to be a little different when p,n embedded in nuclei
- nPDF's. Similar global fits
- You would expect it would also be something for EIC !!!
  - HERA DIDN''T HAVE THAT!!!!!





There is evidence from measurements @ JLAB it's from "Short Range Correlations" SRC relevant for current Neutron Star calcs

Factorization- Universality

$$\sigma \sim \iiint F(x_1) \quad F(x_2) \quad \hat{\sigma} \quad D(z_1) \quad D(z_2).$$

- Factorization: the above integral– proven mathematically. Different versions for different processes above is for hadron-hadron goes to 2-jet.
  - pdfs (the F() 's) : for both incoming 2 partons
  - **New: Fragmentation Functions:** the D()'s: The REVERSE kind of function called Fragmentation Functions that describe probability distribution of different hadrons carrying fraction z of struck parton (jet) momentum
- Idea for PDF's,FF's is should be "universal"
  - ~Independent of Q^2 (turns out not to be true, but QCD can predict why exactly)
  - **Independent of process** (this too has run into problems as well, .e.g e+p vs. mu+p, vs. nu+p, but has been fixable)

- In some historical situations, without higher orders, pQCD's predictive power is limited but lately these uncertainties are better understood → Higg's discovery required precision knowledge of pdfs

Best way to Discover Quarks

- Lepton+Hadron Scattering
  - Much like discovering charge distribution w/ e+p/A
  - Crank up the energy
- Primarily e+p → there have been mu+p
- Also ν + p (more sensitive to sea)
- Generally this is the system quark PDF's originally defined within

$$l(k)\,h(p) \rightarrow l'(k') + X$$



t - channel



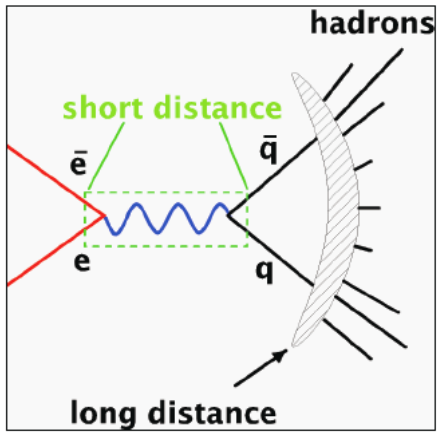
e.s jet

$$\sigma \propto \int F_2(x) \hat{\sigma}\, D_q(x)$$

Best Way to Study Fragmentation Functions

- WRITE SCHEMATIC OF FACTORIZATION IN THIS CASE

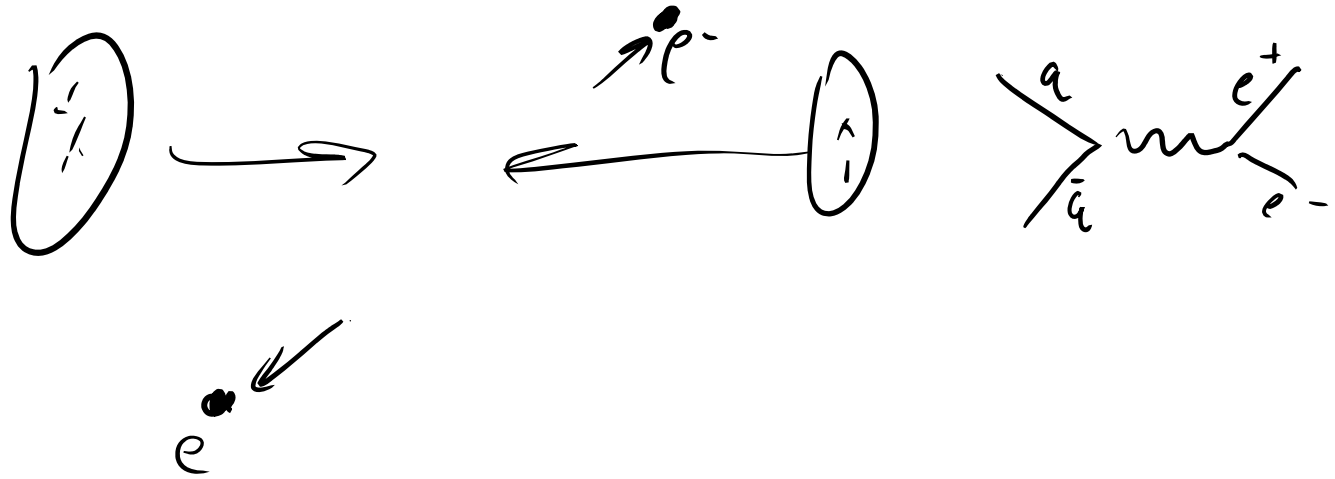
- e+ e- Scattering: Jets and Fragmentation Only



$\sigma \propto ?$

Good way to Focus on PDF's

- Drell-Yan Process (Inverse : p+p→ e+e-)



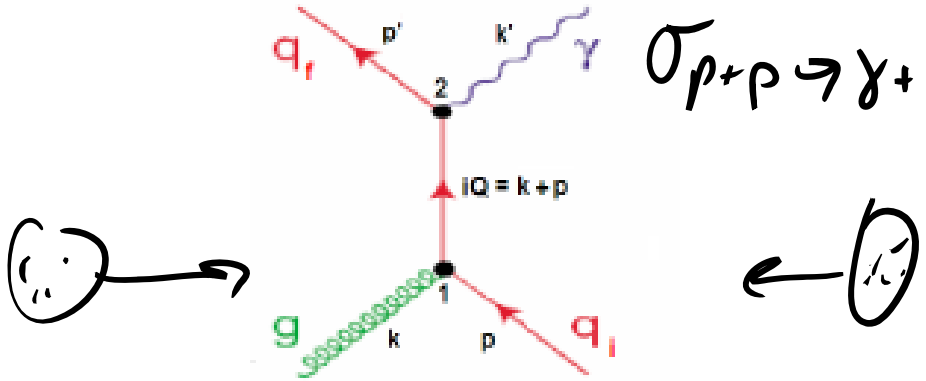
Creation + Decay of Photon

$\sigma \propto ?$

give factorization schematic

Many Simpler Versions

- Justin's Thesis: Direct Photons Compton Scattering $qg \rightarrow q\gamma$:



$\sigma_{p+p \to \gamma + X}$ parton dist fcn

$\sigma \propto ?$

(Schematic form)

The Master formula

- What does the master factor correspond to

$$\sigma \sim \iiint F(x_1) \quad F(x_2) \quad \hat{\sigma} \quad D(z_1) \quad D(z_2).$$

h+h

Switch to Deflorian

- FF slides from Deflorian talk

Review on Fragmentation Functions

Daniel de Florian

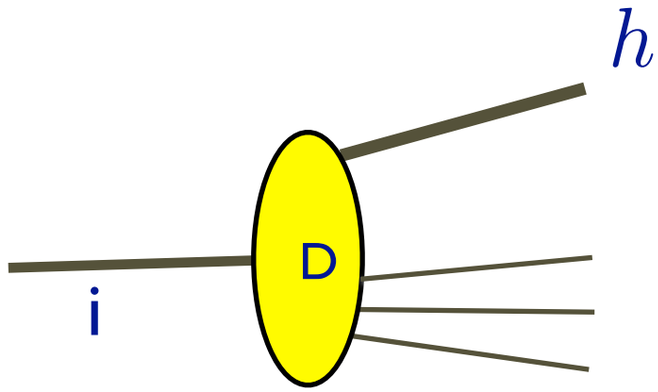
Dpto. de Física - FCEyN
Universidad de Buenos Aires

IWHSS 2011
Paris, April 4



Fragmentation functions

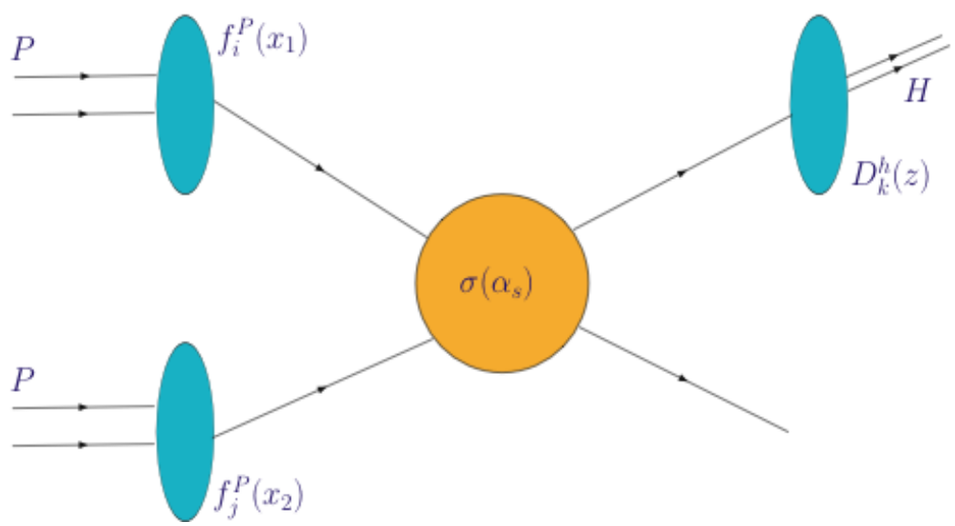
- Describe the collinear transition of a parton *i* into a massless hadron *h* carrying momentum fraction *z*



Represent the probability that a parton hadronizes

- Time-like version of PDFs : information on structure of the nucleon

- Relevant any time a hadron is produced in high energy collisions



Why care about hadrons?

- Hadrons : dominant final state in pp and ep collisions
- Heavy Ion Collisions : QGP by suppression in hadron-production

- The flavor-separation of polarized parton distributions depends crucially on SIDIS results (Hermes-SMC-Compass)
- Extraction of polarized gluon density depends much on PHENIX and STAR pion data

Different FFs → very different results for spin dependent distributions

DdeF, Navarro, Sassot (2005)

wrong D_f → wrong Δf

- Look into "unconventional targets" : Lambda polarized fragmentation functions
- Key elements for Target Fragmentation: evolution of fracture functions
- Fundamental role in understanding single spin asymmetries, transversity...
- TH challenge: NLO pQCD fails to describe fixed target experiments!

Some properties of $D_i^h(z,\mu^2)$

- ***Non-perturbative*** objects but universal : *can be obtained from global analysis!*
- ***Depend*** on energy fraction $z = \dfrac{E_h}{E_i}$
- ***Depend*** on factorization scale μ^2 : described by AP equations

$$\frac{d}{d\ln\mu^2} D_q^h(z,\mu^2) = \left[P_{qq}\otimes D_q^h + P_{gq} D_g^h\right](z,\mu^2)$$

NLO
partial NNLO: Moch, Vogt

Kernels very singular at small z :negative FFs ! $\quad P_{gg} \to \dfrac{2C_A}{z} - \dfrac{\alpha_s}{2\pi}\dfrac{4C_A^2}{z}\ln^2 z$

- Anyway, mass and higher twist effects not trivial : no systematic treatment

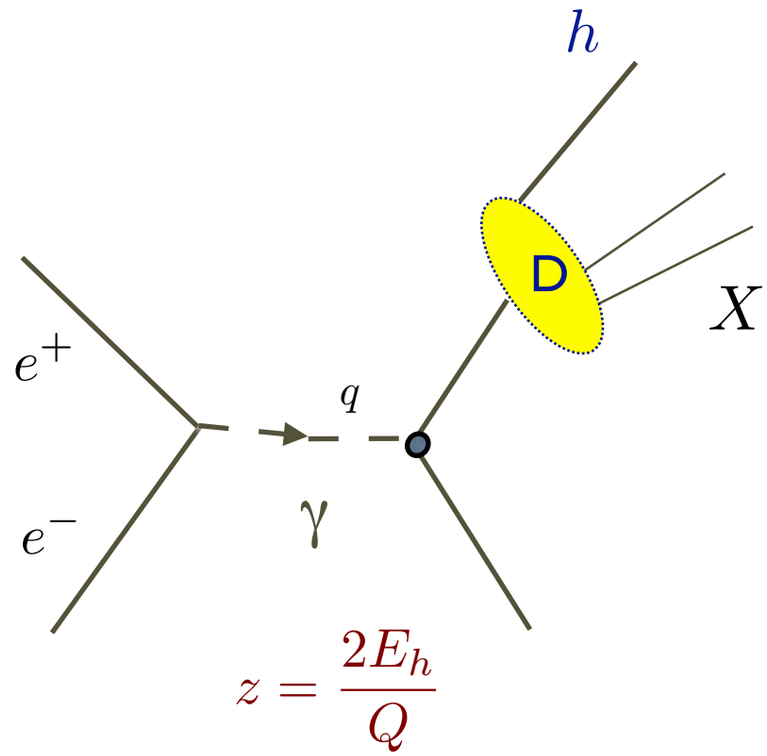
Limits use to z > 0.05 / 0.1

- ***Energy- Momentum conservation*** sum rule $\displaystyle\sum_h \int_0^1 z\, D_i^h(z,\mu^2)\, dz = 1$

without much practical use

e+e- (SIA) single-inclusive annihilation

$$e^+e^- \to (\gamma,\, Z) \to H$$



- Cross section depends on two structure functions

$$\frac{1}{\sigma_{tot}}\frac{d\sigma^H}{dz} = \frac{\sigma_0}{\sum_q \hat{e}_q^2}\left[2\,F_1^H(z,Q^2) + F_L^H(z,Q^2)\right]$$

$$\sigma_{tot} = \sum_q \hat{e}_q^2\;\sigma_0\left[1+\frac{\alpha_s(Q^2)}{\pi}\right]$$

- At NLO they can be written as

$$2F_1^H(z,Q^2) \;=\; \sum_q \hat{e}_q^2 \left\{ \left[D_q^H(z,Q^2) + D_{\bar{q}}^H(z,Q^2)\right] + \frac{\alpha_s(Q^2)}{2\pi}\left[C_q^1 \otimes (D_q^H + D_{\bar{q}}^H) \;+ C_g^1 \otimes D_g^H\right](z,Q^2)\right\}$$

Known up to NNLO

Coeff. functions
Calculable in pQCD

- Notice that SIA can only give information on the sum $D_q^H(z,Q^2) + D_{\bar{q}}^H(z,Q^2)$

e+e- (SIA) single-inclusive annihilation

Advantages

- Very precise data from LEP/SLD
- Heavy quark tagged data
- Only fragmentation functions enter (clean process)

Disadvantages

- SIA data dominated by precise LEP/SLD measurements at M_Z
- weak scale dependence (bad resolution for g fragmentation)
- mostly determine "singlet" distribution (high precision)

$$\Sigma = D_u + D_{\bar{u}} + D_d + D_{\bar{d}} + D_s + D_{\bar{s}} + D_c + D_{\bar{c}} + D_b + D_{\bar{b}}$$

- OPAL offers "flavor separated data": no trivial interpretation at NLO
- not precise at large z (relevant for pp collisions)
- Can not separate $D_q^h(z,Q^2)$ from $D_{\bar{q}}^h(z,Q^2)$

SIDIS

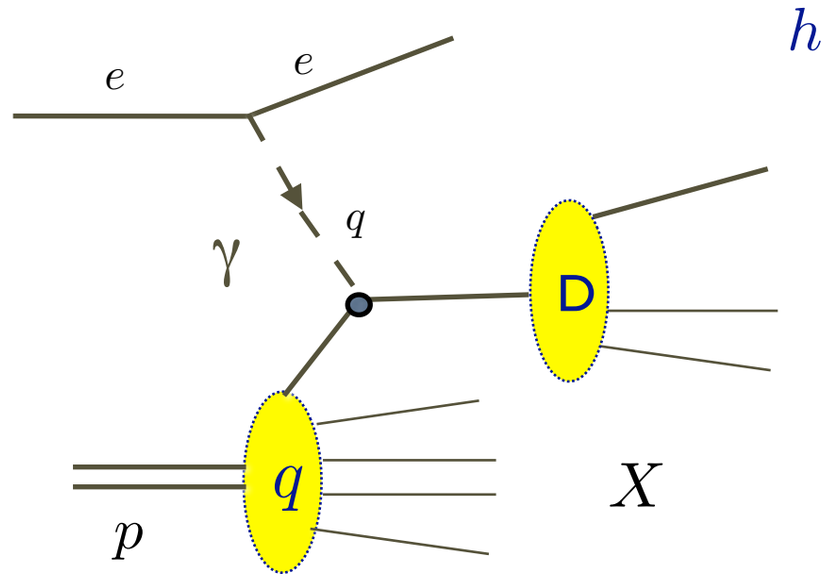
current fragmentation

- Distributions in *x* and *z*

$$\frac{d\sigma^H}{dx\,dQ^2\,dz_H} = \frac{2\pi\alpha^2}{xsQ^2}\left[\frac{1+(1-y)^2}{y}2F_1^H(x,z_H,Q^2) + \frac{2(1-y)}{y}F_L^H(x,z_H,Q^2)\right]$$

$$x = \frac{Q^2}{2P^N\cdot q} \qquad y = \frac{Q^2}{sx}$$

"scaling variable" $\quad z_H \equiv \dfrac{P^H\cdot P^N}{P^N\cdot q}$



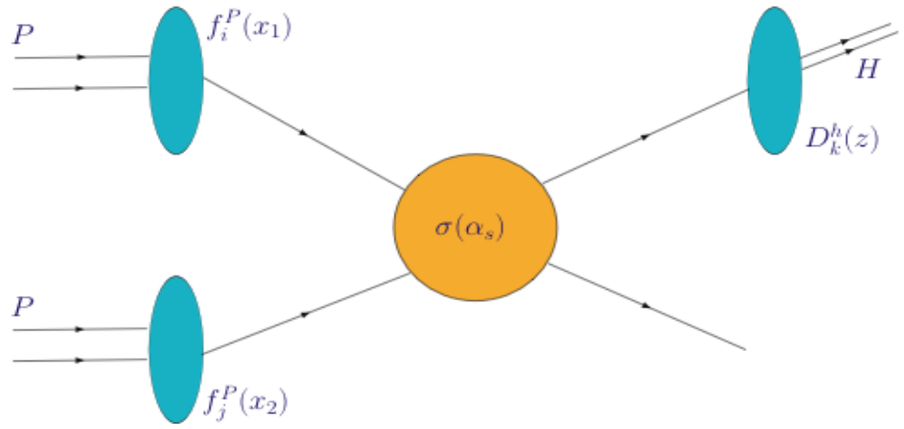
at LO $\quad 2F_1^H(x,z_H,Q^2) = \sum_{q,\overline{q}} e_q^2 \quad q(x,Q^2)D_q^H(z_H,Q^2)$ expression known up to NLO

"effective charge" : different for quarks and antiquarks!
Most valuable observable for flavor/charge separation!
Different targets (and x!) can be used

Hadron-Hadron collisions

Transverse momentum distribution

hard scale



$$q\bar{q} \longrightarrow g$$
$$qq \longrightarrow q$$
$$qg \longrightarrow g$$
$$qg \longrightarrow q$$
$$gg \longrightarrow q$$
$$gg \longrightarrow g$$

$$d\sigma(pp \to h) = \sum_{i,j,k} \int_0^1 dx_1\, f_i^P(x_1, \mu_{FI}^2) \int_0^1 dx_2\, f_j^P(x_2, \mu_{FI}^2) \int_0^1 dz\, D_k^h(z, \mu_{FF}^2)\, d\hat{\sigma}(ij \to k)$$

Advantages : allows (very little) flavor/charge separation (gluon dominated)
several subprocesses
much larger z
large (direct!) contribution from D_g
relevant for RHIC kinematics (polarized/heavy ion)
almost no HQ fragmentation

Disadvantages: problems for fixed target experiments, use only colliders
otherwise Threshold resummation needed (not in first approach)
larger TH uncertainty (scale dependence) DdeF, W.Vogelsang
larger dependence on pdfs
almost no HQ fragmentation

Frag Functions Isospin e.g. Constraints

- Like pdf's the FF's also constrained by the quark model expectations
- Although exceptions must be allowed for in many cases for both FF and pdf's in order to fit the experimental data
  - ANOTHER SIGN THAT QUARK MODEL IS TOO SIMPLE

$$D_{\bar{q}}^{h^+}(z,Q^2) = (1-z)\, D_q^{h^+}(z,Q^2)$$

$$D_{d+\bar{d}}^{\pi^+} = N\, D_{u+\bar{u}}^{\pi^+}$$

$$D_s^{\pi^+} = D_{\bar{s}}^{\pi^+} = N' D_{\bar{u}}^{\pi^+}$$

$$D_{\bar{u}}^{\pi^+} = D_{\bar{d}}^{\pi^+}$$

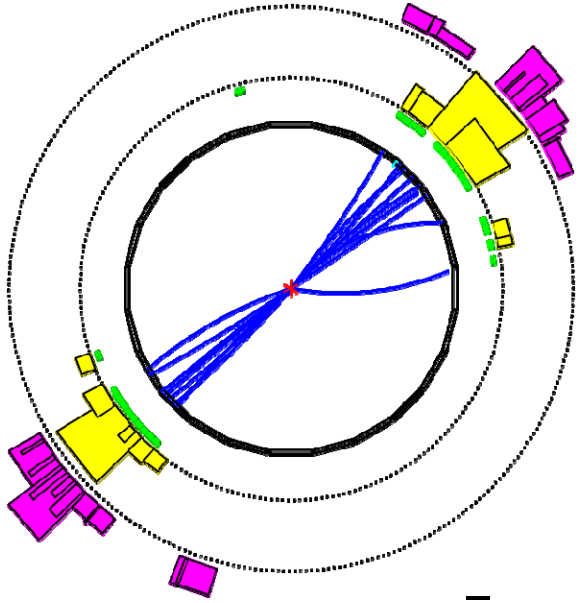
$$D_{\bar{u}}^{K^+} = D_s^{K^+} = D_{\bar{d}}^{K^+} = D_{\bar{d}}^{K^+}.$$

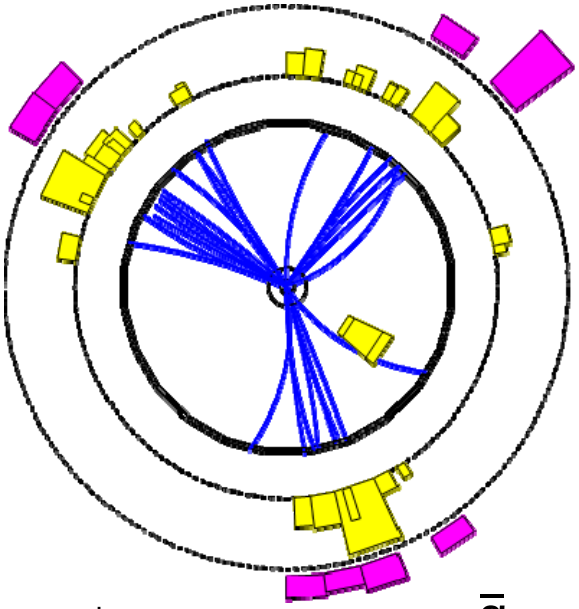
Jet production in e+e- Collisions

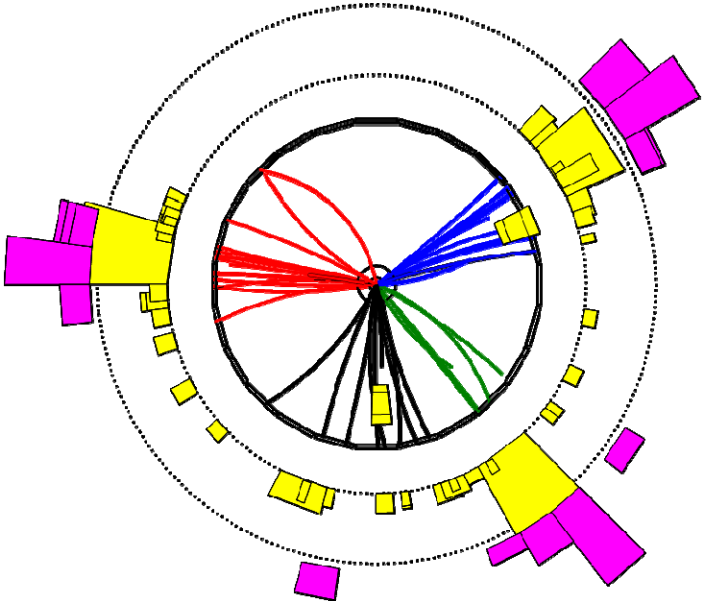
★e^+e^- colliders are also a good place to study gluons

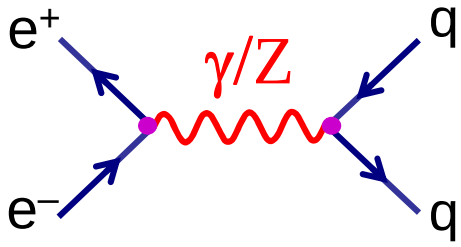
$e^+e^- \to q\bar{q} \to 2\text{jets}$ $e^+e^- \to q\bar{q}g \to 3\text{jets}$ $e^+e^- \to q\bar{q}gg \to 4\text{jets}$

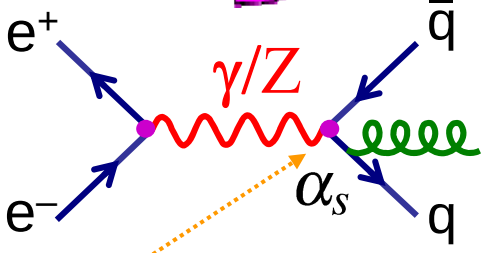
OPAL at LEP (1989-2000)

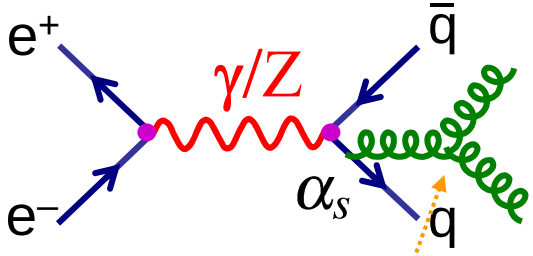












Experimentally:

- Three jet rate ➡ measurement of α_s
- Angular distributions ➡ gluons are spin-1
- Four-jet rate and distributions ➡ QCD has an underlying SU(3) symmetry

pp→2j Thomson book

$\hat{s} = x_1 x_2 s$

$E_{PCM} \rightarrow p_{PCM}$

- Thomson book good section on pp→2jets
- Rapidity and pT in action
- Conserve mom → 2 jets **ALWAYS** back to back in pT
  - for FAR Dominant LO
  - But not back to back in lab frame → previously discussed
- Net longitudental pz (Boost) of system:

$$p_{PCM\,z} = (x_1 - x_2)\sqrt{s}/2$$



Fig. 10.30 An example of a pp → two-jets + X event observed in the ATLAS detector at the LHC: (left) the transverse view (perpendicular to the beam direction) and (right) the *yz*-view with the *z*-axis along the beam direction. Reproduced courtesy of the ATLAS collaboration.

$\Delta\phi = \pi$

$\Delta y = y_1 - y_2 \neq 0$

Thomson Relation to Quark PDF's

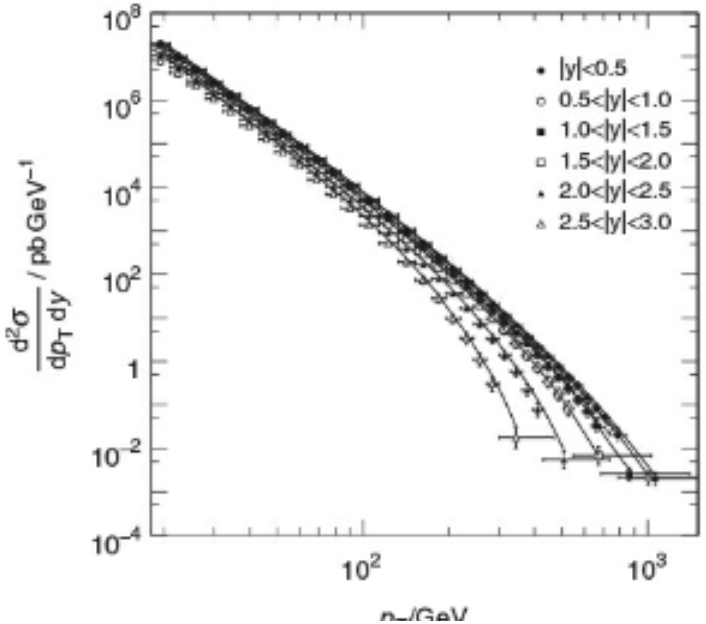
- Now we need to include gluon pdf's primarily at low x

$$\frac{\mathrm{d}\sigma}{\mathrm{d}Q^2} = \frac{4\pi\alpha_S^2}{9Q^4}\left[1+\left(1-\frac{Q^2}{sx_1x_2}\right)^2\right] g(x_1,x_2)\,\mathrm{d}x_1\,\mathrm{d}x_2.$$

- But if we can distinguish q vs gluon jets (now becoming more and more accurate) selecting only quark jets

$$g(x_1,x_2) = [u(x_1)\bar{u}(x_2) + u(x_1)\bar{d}(x_2) + \bar{d}(x_1)u(x_2) + \bar{d}(x_1)d(x_2)]\,.$$

- Theory w/ predefined pdfs matches data well



Some of extra reading

- show jet finding

Event Generators

$$\sigma \sim \iiint F(x_1)\ \ F(x_2)\ \ \hat{\sigma}\ \ D(z_1)\ \ D(z_2).$$

- Event Generators (code/programs) can do this for you
- Two Most important (LO only)
  - PYTHIA
  - HERWIG
- Use FF some, but also other hadronization models:
  - PYTHIA: String model in reading
- SHERPA: NLO
- Note pdfs/ FF's change slightly if including NLO or higher

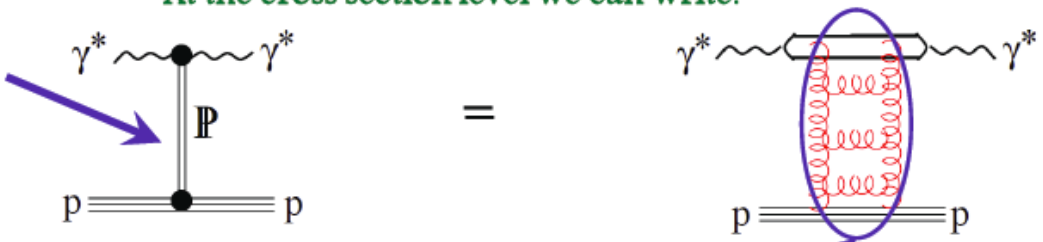
Alt factorizations

- Color dipole model "reverses" non-perturb part of factorization

Alternative model: color dipole approach to DIS

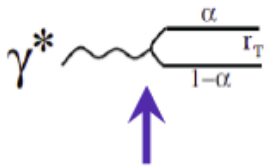
At the cross section level we can write:

QCD Pomeron (considered in Regge theory framework)



Fast virtual photon has a certain probability to fluctuate into the quark-antiquark pair which then scatters off the target

Elastic dipole–target scattering amplitude



The DIS cross section is then given by a convolution

$$\sigma_{tot}^{\gamma^* N}(x,Q^2) = \int_0^1 d\alpha \int d^2\rho |\Psi_{q\bar{q}}(\rho,\alpha)|^2 \sigma(\rho,x)$$

given by the light-cone wave function:

$$|\Psi_{q\bar{q}}(\rho,\alpha)|_T^2 \approx \frac{6\alpha_{em}}{(2\pi)^2}\sum_1^{N_f} Z_i^2[1-2\alpha(1-\alpha)]\,\epsilon^2 K_1^2(\epsilon\rho)$$

$$\epsilon^2 = \alpha(1-\alpha)Q^2 + m_q^2$$

GBW saturation model:

$$\sigma_{\bar{q}q}(r_T,x) = \sigma_0\left[1 - e^{-\frac{1}{4}r_T^2 Q_s^2(x)}\right]$$

universal dipole cross section

fitted to data!

saturation scale: $Q_s(x) = 1\,\text{GeV} \times (x_0/x)^{\lambda/2}$ and $\sigma_0 = 23.03\,\text{mb};\ \lambda = 0.288;\ x_0 = 3.04 \cdot 10^{-4}$

- There are other kinds of factorization for accessing non-perturb QCD:
  - Finite Temperature expansion→ very low E (~100 MeV)

Rest of class Tuesday Wk14

- Then we started discussing a little about the Weak interactions again, but more just quickly introducing some slides about neutrino oscillations, I include those in the slides from Thursday just after this.

- We didn't cover ALL of these in detail just flashed many but they are included for your reference.

Particles and Nuclei 2

- 4/23/2026 Last day
- Announcements:

- HW 6/Takehome Final due Mon. May 4th.
  - WILL BE FINALIZED BY TOMORROW

- Graded HW 4 (5?) returned tomorrow

- No New Reading – Handout from Wong, will be more covered in class, Thomson handout mostly just for interest, but some of the relations at the end part needed for final assignment.

Today

- First finish Weak Interaction and generalization to Higg's Mechanism which affects all of nuclear/particle physics

- Then at very end quickly highlight some of the various directions of Non-perturbative QCD → where Medium and high energy QCD "nuclear" physics still contains many unknowns driving experimental studies

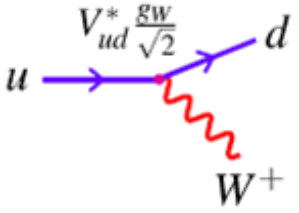
Guide : Weak/Related Physics

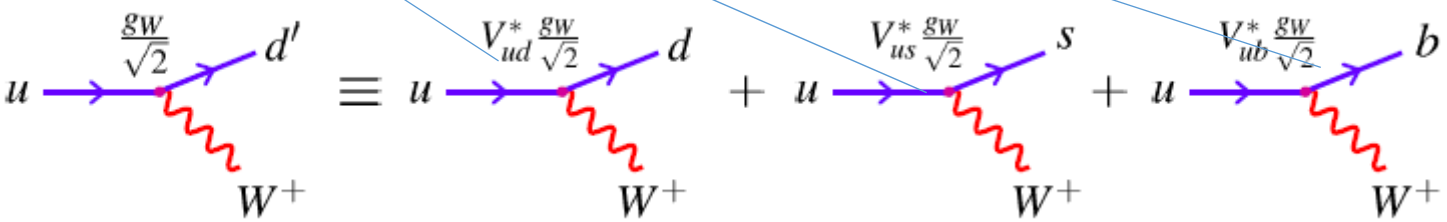
- Weak Interaction
  - Already covered: Flavor structure and CKM Mixing (long ago)
  - Already covered: Basics of Feynman charge current scattering situations (F3) (last time)
  - **(again) Back to CKM Neutrino Oscillations**
    - **A lot of slides included more for reference,**
  - Weak CC (Charge Current – W) in DECAYS → Muon and Beta Decay
  - Z boson "Neutral Current" Interactions
  - More on Chiral fermion states & Electroweak Unification Weak Isospin
- Local Gauge Invariance, Spontaneous Symm Breaking and Higgs Mechanism

Reminder Quark Ch Weak Vtx/CKM Mixing

- Remember vertex changes flavor, by "destroying/recreating" it (W carries no flavor → only charge) and any q w/charge can be used
- Charged weak vertex possiblilities:

$$\begin{pmatrix} u \\ d \end{pmatrix}, \begin{pmatrix} u \\ s \end{pmatrix}, \begin{pmatrix} u \\ b \end{pmatrix}, \begin{pmatrix} c \\ d \end{pmatrix}, \begin{pmatrix} c \\ s \end{pmatrix}, \begin{pmatrix} c \\ b \end{pmatrix}, \begin{pmatrix} t \\ d \end{pmatrix}, \begin{pmatrix} t \\ s \end{pmatrix}, \begin{pmatrix} t \\ b \end{pmatrix}$$





- Thomson: starts listing all 9 in Ch 1 intro. Later explains connection to Vud
- Note : confusing terminology :
  - coupling only "3 couplings no e.g. u to s'
  - d' to u, … etc. OK → no u', s', u'to d' (no prime prime vtx)
  - **W "creates" d' from u**
- CKM Matrix : defines relative probs of these pairings

$$\begin{pmatrix} u \\ d' \end{pmatrix}, \begin{pmatrix} c \\ s' \end{pmatrix}, \begin{pmatrix} t \\ b' \end{pmatrix}$$

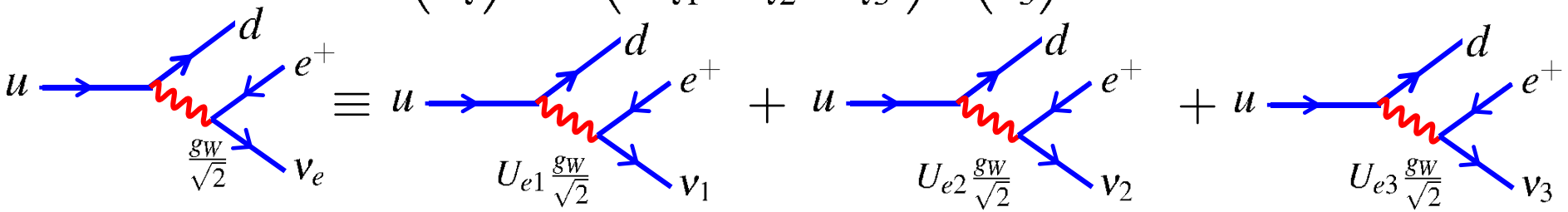
Neutrinos have the same thing

- "Nominal Couplings"

$$\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}, \begin{pmatrix} \nu_\mu \\ \mu^- \end{pmatrix}, \begin{pmatrix} \nu_\tau \\ \tau^- \end{pmatrix}$$

- Actual couplings → $\nu_1, \nu_2 \ldots$ (ν_e, ν_μ, already the prime [ν'] states)
- P-NMS: Pontecorvo–Maki–Nakagawa–Sakata - neutrinos

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$



- Leads of course to the oscillations:
- Time dependences determined by rest mass states v1,v2,v3
  - Usual QM phase oscillations for linear combinations:

$$|\nu_e\rangle = e^{-i\omega_1 t}|\nu_1\rangle + e^{-i\omega_2 t}|\nu_2\rangle + \ldots$$

CKM Values → Wolfenstein

- CKM:

$$\begin{bmatrix} |V_{ud}| & |V_{us}| & |V_{ub}| \\ |V_{cd}| & |V_{cs}| & |V_{cb}| \\ |V_{td}| & |V_{ts}| & |V_{tb}| \end{bmatrix} \begin{bmatrix} 0.97370 & 0.2245 & 0.00382 \\ 0.221 & 0.987 & 0.0410 \\ 0.0080 & 0.0388 & 1.013 \end{bmatrix}.$$

- No underlying reason from Standard Model for CKM values—parameterizations are just empirical:
- Two popular ways of parameterizing
- 1) "Wolfenstein → emphasizes diagnonalness

$$\begin{pmatrix} 1-\lambda^2/2 & \lambda & A\lambda^3(\rho-i\eta) \\ \lambda & 1-\lambda^2/2 & A\lambda^3 \\ A\lambda^3(1-\rho-i\eta) & -A\lambda^3 & 1 \end{pmatrix} + O(\lambda^4)$$

complex → needed for CP violation

- Leptons/ν's : P-NMS: not as diagonal
  - Pontecorvo–Maki–Nakagawa–Sakata - neutrinos

$$|U| = \begin{bmatrix} |U_{e1}| & |U_{e2}| & |U_{e3}| \\ |U_{\mu 1}| & |U_{\mu 2}| & |U_{\mu 3}| \\ |U_{\tau 1}| & |U_{\tau 2}| & |U_{\tau 3}| \end{bmatrix} = \begin{bmatrix} 0.801 \ldots 0.845 & 0.513 \ldots 0.579 & 0.143 \ldots 0.156 \\ 0.233 \ldots 0.507 & 0.461 \ldots 0.694 & 0.631 \ldots 0.778 \\ 0.261 \ldots 0.526 & 0.471 \ldots 0.701 & 0.611 \ldots 0.761 \end{bmatrix}$$

Standard CKM (!= CP Violation)

- CKM (NMS/ν) "mixing" is not really the "cause" of CP Violation or vice versa
  - Mixing Necessary for CP vio, but not "driver", CP vio doesn't affect mixing generally (exceptions)
- **STANDARD CKM Parameterization:**
  - Mixing is determined by modulus's of CKM matrix elements → determines "oscillations" e.g. like for ν_e
  - Amount of CP violation: a phase only in some CKM terms (mod 1), only detected through INTERFERENCE btw 2 processes

Weak Force Lagrangian term

$$(\overline{u}, \overline{c}, \overline{t})\gamma^{\mu}(1-\gamma^{5})V\begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

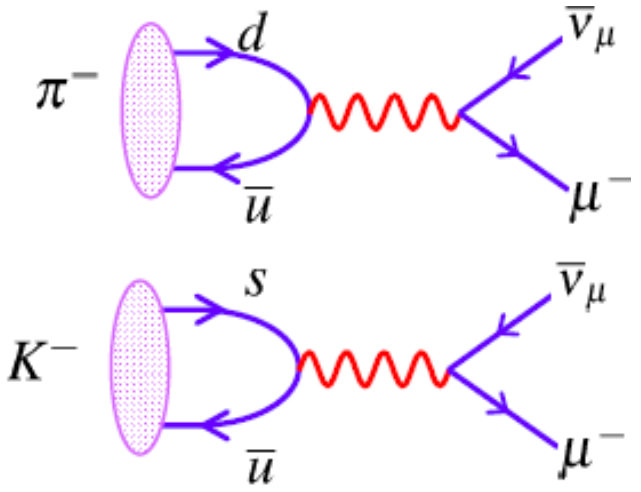
$$\begin{bmatrix} d' \\ s' \\ b' \end{bmatrix} = \begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} \begin{bmatrix} d \\ s \\ b \end{bmatrix}$$

c_{12} → $\cos(\theta_{12})$
etc 3 angles and a phase

$$\begin{bmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta_{13}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta_{13}} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta_{13}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta_{13}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta_{13}} & c_{23}c_{13} \end{bmatrix}.$$

History: Cabbibo Angle

- E.g. to explain why pi- decay rate much > than K-
  - (OPPOSITE EXPECTATION FROM PHASE SPACE/ENERGY)



$\cos\theta_C$

$\sin\theta_C$

$$\begin{pmatrix} d' \\ s' \end{pmatrix} = \begin{pmatrix} \cos\theta_c & \sin\theta_c \\ -\sin\theta_c & \cos\theta_c \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix}$$

Mini CKM:

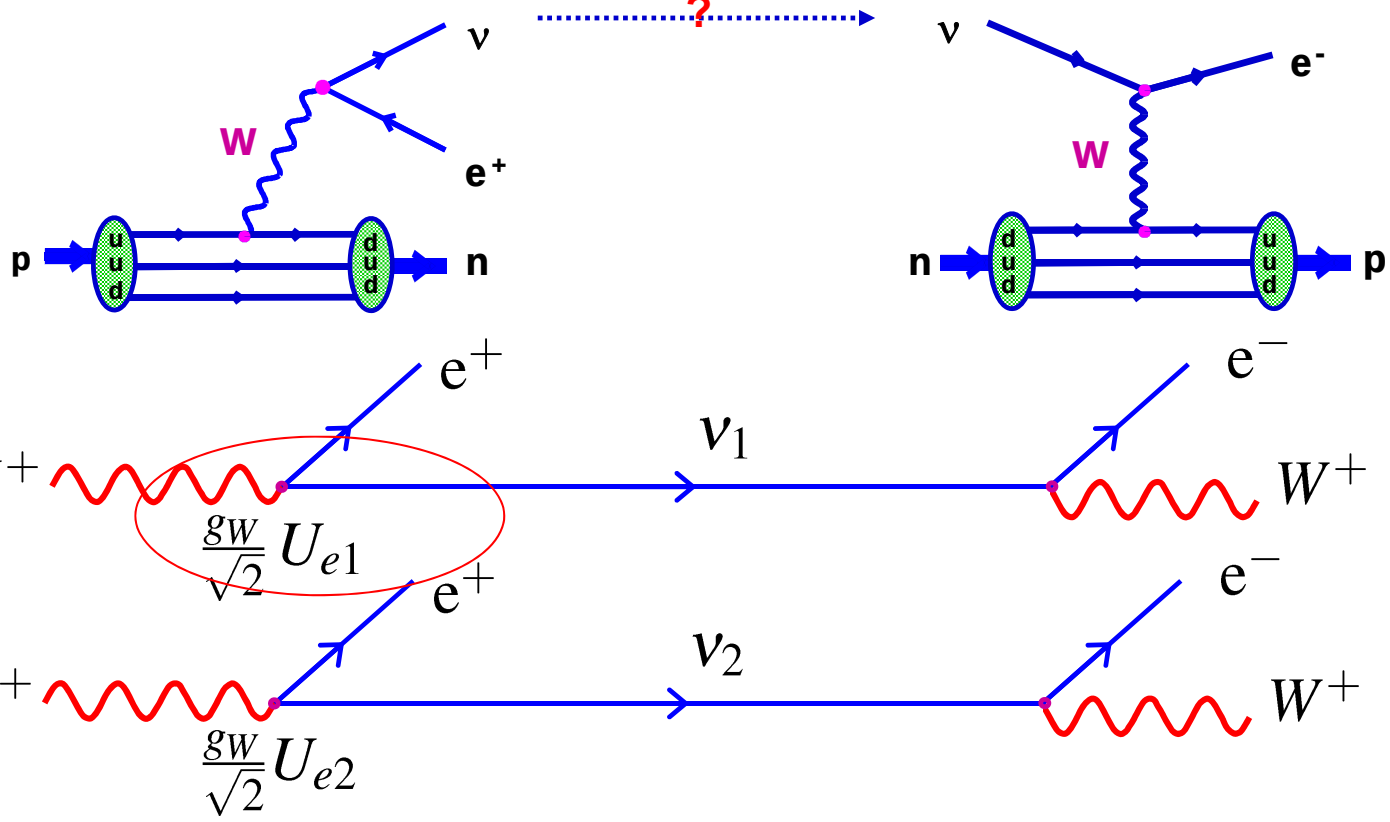
θc = 13 degree
(cC 0.97, sC 0.22)

4/23/26: TOO MANY "ANGLES" in Weak parameters try to forget→ think of as CKM elements (Vud,Vus,...)

Mass Eigenstates and Weak Eigenstates

★ The essential feature in understanding the physics of neutrino oscillations is to understand what is meant by weak eigenstates and mass eigenstates ν_1, ν_2

★ Suppose the process below proceeds via two fundamental particle states



★ Can't know which **mass eigenstate** (fundamental particle ν_1, ν_2) was involved

★ In Quantum mechanics treat as a coherent state $\psi = \nu_e = U_{e1}\nu_1 + U_{e2}\nu_2$

★ ν_e represents the wave-function of the coherent state produced along with an electron in the weak interaction, i.e. the **weak eigenstate**

Neutrino Oscillations for Two Flavours

★ Neutrinos are produced and interact as weak eigenstates, ν_e, ν_μ

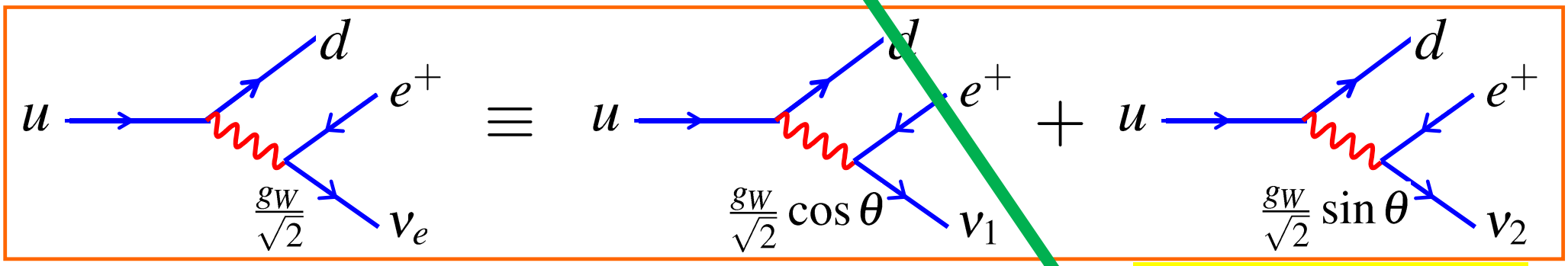
★ The weak eigenstates as coherent linear combinations of the fundamental "mass eigenstates" ν_1, ν_2

★ The mass eigenstates are the free particle solutions to the wave-equation and will be taken to propagate as plane waves

$$|\nu_1(t)\rangle = |\nu_1\rangle e^{i\vec{p}_1.\vec{x}-iE_1t} \qquad |\nu_2(t)\rangle = |\nu_2\rangle e^{i\vec{p}_2.\vec{x}-iE_2t}$$

★ The weak and mass eigenstates are related by the unitary 2x2 matrix

$$\begin{pmatrix}\nu_e\\ \nu_\mu\end{pmatrix} = \begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\nu_1\\ \nu_2\end{pmatrix} \qquad (1)$$



★ Equation (1) can be inverted to give

$$\begin{pmatrix}\nu_1\\ \nu_2\end{pmatrix} = \begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\nu_e\\ \nu_\mu\end{pmatrix}$$

Like Cabbibo θ but for ν's NO! why not use Uij elements?

Neutrino Oscillations for Two Flavours

★ Neutrinos are produced and interact as weak eigenstates, ν_e, ν_μ

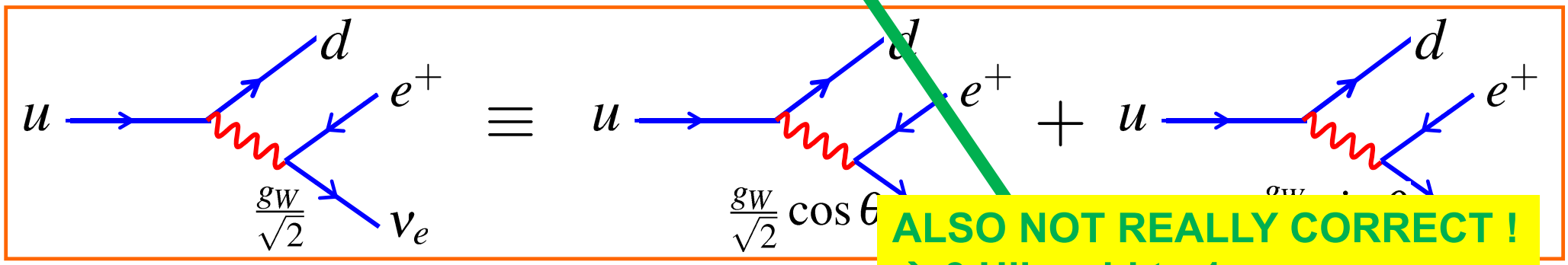
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★ Equation (1) can be inverted to give

$$\begin{pmatrix}\nu_1\\ \nu_2\end{pmatrix} = \begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\nu_e\\ \nu_\mu\end{pmatrix}$$

ALSO NOT REALLY CORRECT !
→ 3 Uijs add to 1
not cosθ^2 + sinθ^2
Limitation of "two flavor" simplification

- Suppose at time $t = 0$ a neutrino is produced in a pure ν_e state, e.g. in a decay $u \to d e^+ \nu_e$

$$|\psi(0)\rangle = |\nu_e\rangle = \cos\theta|\nu_1\rangle + \sin\theta|\nu_2\rangle$$

- Take the z-axis to be along the neutrino direction
- The wave-function evolves according to the time-evolution of the **mass eigenstates** (free particle solutions to the wave equation)

$$|\psi(t)\rangle = \cos\theta|\nu_1\rangle e^{-ip_1.x} + \sin\theta|\nu_2\rangle e^{-ip_2.x}$$

where $p_i.x = E_i t - \vec{p}_i.\vec{x} = E_i t - |\vec{p}_i| z$

- Suppose the neutrino interacts in a detector at a distance **L** and at a time **T**

$$\phi_i = p_i.x = E_i T - |\vec{p}_i| L$$

gives $$\boxed{|\psi(L,T)\rangle = \cos\theta|\nu_1\rangle e^{-i\phi_1} + \sin\theta|\nu_2\rangle e^{-i\phi_2}}$$

★ Expressing the mass eigenstates, $|\nu_1\rangle, |\nu_2\rangle$, in terms of weak eigenstates (eq 2)

$$|\psi(L,T)\rangle = \cos\theta(\cos\theta|\nu_e\rangle - \sin\theta|\nu_\mu\rangle)e^{-i\phi_1} + \sin\theta(\sin\theta|\nu_e\rangle + \cos\theta|\nu_\mu\rangle)e^{-i\phi_2}$$

$$\boxed{|\psi(L,T)\rangle = |\nu_e\rangle(\cos^2\theta e^{-i\phi_1} + \sin^2\theta e^{-i\phi_2}) + |\nu_\mu\rangle \sin\theta\cos\theta(-e^{-i\phi_1} + e^{-i\phi_2})}$$

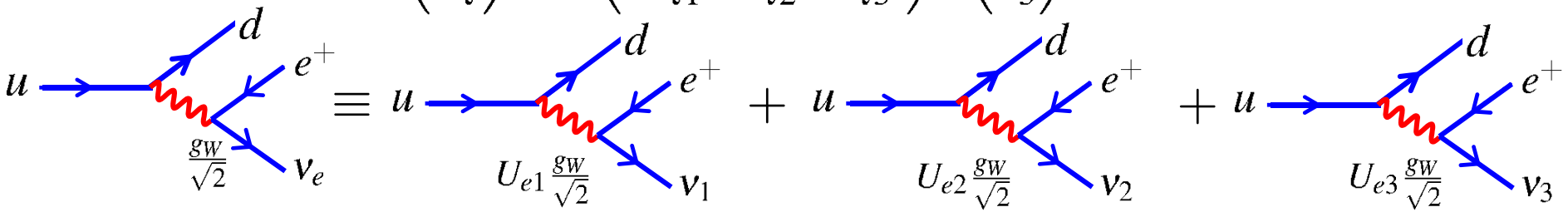
Neutrinos have the same thing

- "Nominal Couplings"

$$\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}, \begin{pmatrix} \nu_\mu \\ \mu^- \end{pmatrix}, \begin{pmatrix} \nu_\tau \\ \tau^- \end{pmatrix}$$

- Actual couplings → $\nu_1, \nu_2 \ldots$ (ν_e, ν_μ, already the prime [ν'] states)
- P-NMS: Pontecorvo–Maki–Nakagawa–Sakata - neutrinos

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$



- Leads of course to the oscillations:
- Time dependences determined by rest mass states v1,v2,v3
  - Usual QM phase oscillations for linear combinations:

$$|\nu_e\rangle = e^{-i\omega_1 t}|\nu_1\rangle + e^{-i\omega_2 t}|\nu_2\rangle + \ldots$$

★ If the masses of $|\nu_1\rangle, |\nu_2\rangle$ **are the same**, the mass eigenstates remain in phase, $\phi_1 = \phi_2$, and the state remains the linear combination corresponding to $|\nu_e\rangle$ and in a weak interaction will produce an electron

★ If the masses are different, the wave-function no longer remains a pure $|\nu_e\rangle$

$$\begin{aligned} P(\nu_e \to \nu_\mu) &= |\langle \nu_\mu | \psi(L,T)\rangle|^2 \\ &= \cos^2\theta \sin^2\theta(-e^{-i\phi_1} + e^{-i\phi_2})(-e^{+i\phi_1} + e^{+i\phi_2}) \\ &= \frac{1}{4}\sin^2 2\theta(2 - 2\cos(\phi_1 - \phi_2)) \\ &= \sin^2 2\theta \sin^2\left(\frac{\phi_1 - \phi_2}{2}\right) \end{aligned}$$

★ The treatment of the phase difference

$$\Delta\phi_{12} = \phi_1 - \phi_2 = (E_1 - E_2)T - (|p_1| - |p_2|)L$$

in most text books is dubious. Here we will be more careful…

★ One could assume $|p_1| = |p_2| = \mathrm{p}$ in which case

$$\Delta\phi_{12} = (E_1 - E_2)T = \left[(\mathrm{p}^2 + m_1^2)^{1/2} - (\mathrm{p}^2 + m_2^2)^{1/2}\right] L \qquad L \approx (c)T$$

$$\Delta\phi_{12} = \mathrm{p}\left[\left(1+\frac{m_1^2}{\mathrm{p}^2}\right)^{1/2} - \left(1+\frac{m_2^2}{\mathrm{p}^2}\right)^{1/2}\right] L \approx \frac{m_1^2 - m_2^2}{2\mathrm{p}} L$$

★ **However we have neglected that fact that for the same momentum, different mass eigenstates will propagate at different velocities and be observed at different times**

★ **The full derivation requires a wave-packet treatment and gives the same result**

★ **Nevertheless it is worth noting that the phase difference can be written**

$$\Delta\phi_{12} = (E_1 - E_2)T - \left(\frac{|p_1|^2 - |p_2|^2}{|p_1| + |p_2|}\right) L$$

$$\Delta\phi_{12} = (E_1 - E_2)\left[T - \left(\frac{E_1 + E_2}{|p_1| + |p_2|}\right) L\right] + \left(\frac{m_1^2 - m_2^2}{|p_1| + |p_2|}\right) L$$

★ **The first term on the RHS vanishes if we assume** $E_1 = E_2$ **or** $\beta_1 = \beta_2$

in all cases

$$\Delta\phi_{12} = \frac{m_1^2 - m_2^2}{2\mathrm{p}} L = \frac{\Delta m^2}{2E} \mathbf{L}$$

★ **Hence the two-flavour oscillation probability is:**

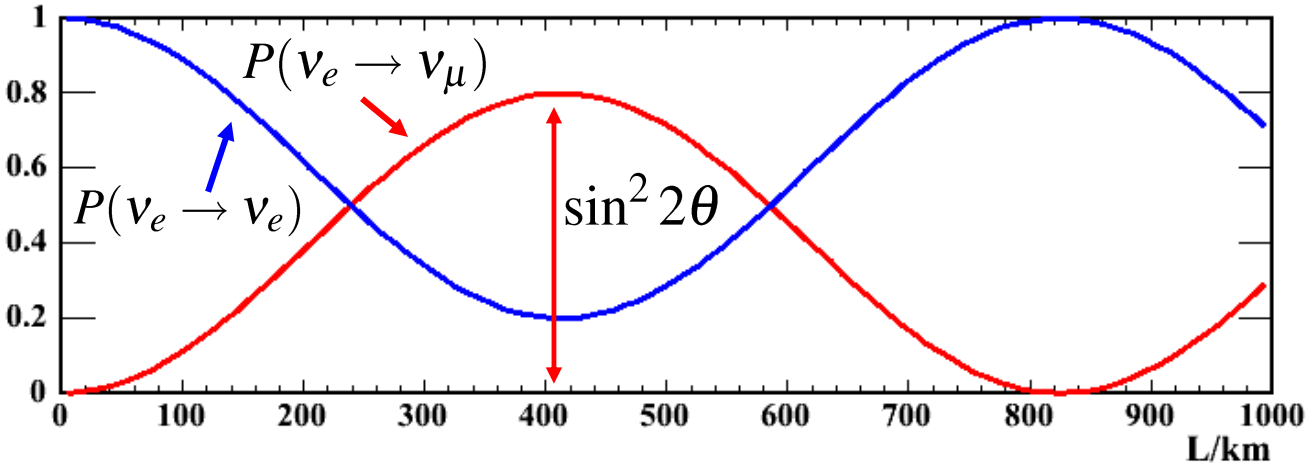
$$P(\nu_e \to \nu_\mu) = \sin^2 2\theta \sin^2\left(\frac{\Delta m^2_{21} L}{4E}\right)$$

with $\Delta m^2_{21} = m^2_2 - m^2_1$

★ **The corresponding two-flavour survival probability is:**

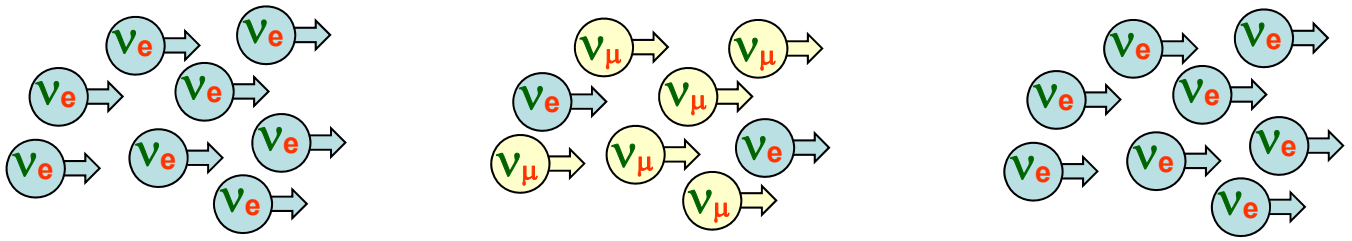
$$P(\nu_e \to \nu_e) = 1 - \sin^2 2\theta \sin^2\left(\frac{\Delta m^2_{21} L}{4E}\right)$$

•**e.g.** $\Delta m^2 = 0.003\,\text{eV}^2, \quad \sin^2 2\theta = 0.8, \quad E_\nu = 1\,\text{GeV}$



•**wavelength**

$$\lambda_{\text{osc}} = \frac{4\pi E}{\Delta m^2}$$

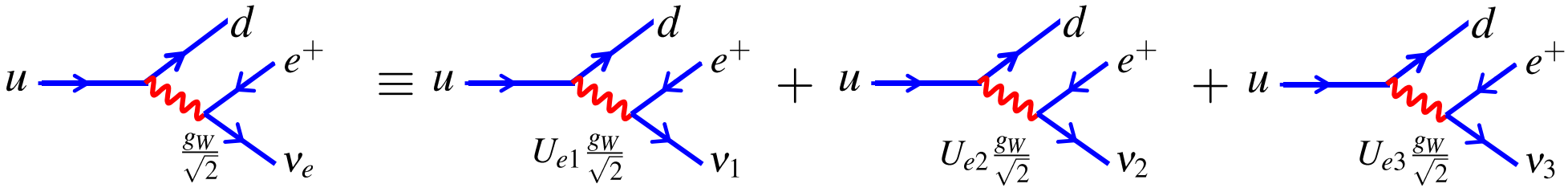


Neutrino Oscillations for Three Flavours

★ It is simple to extend this treatment to three generations of neutrinos.

★ In this case we have:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$



★ The 3x3 Unitary matrix U is known as the Pontecorvo-Maki-Nakagawa-Sakata matrix, usually abbreviated **PMNS**

★ Note : has to be unitary to conserve probability

•Using $U^\dagger U = I \quad \Rightarrow \quad U^{-1} = U^\dagger = (U^*)^T$

gives

$$\begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix} = \begin{pmatrix} U^*_{e1} & U^*_{\mu 1} & U^*_{\tau 1} \\ U^*_{e2} & U^*_{\mu 2} & U^*_{\tau 2} \\ U^*_{e3} & U^*_{\mu 3} & U^*_{\tau 3} \end{pmatrix} \begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix}$$

Unitarity Relations

★**The Unitarity of the PMNS matrix gives several useful relations:** $UU^\dagger = I \Rightarrow$

$$\begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} \begin{pmatrix} U^*_{e1} & U^*_{\mu 1} & U^*_{\tau 1} \\ U^*_{e2} & U^*_{\mu 2} & U^*_{\tau 2} \\ U^*_{e3} & U^*_{\mu 3} & U^*_{\tau 3} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

gives:

$$U_{e1}U^*_{e1} + U_{e2}U^*_{e2} + U_{e3}U^*_{e3} = 1 \quad \textbf{(U1)}$$

$$U_{\mu 1}U^*_{\mu 1} + U_{\mu 2}U^*_{\mu 2} + U_{\mu 3}U^*_{\mu 3} = 1 \quad \textbf{(U2)}$$

$$U_{\tau 1}U^*_{\tau 1} + U_{\tau 2}U^*_{\tau 2} + U_{\tau 3}U^*_{\tau 3} = 1 \quad \textbf{(U3)}$$

$$U_{e1}U^*_{\mu 1} + U_{e2}U^*_{\mu 2} + U_{e3}U^*_{\mu 3} = 0 \quad \textbf{(U4)}$$

$$U_{e1}U^*_{\tau 1} + U_{e2}U^*_{\tau 2} + U_{e3}U^*_{\tau 3} = 0 \quad \textbf{(U5)}$$

$$U_{\mu 1}U^*_{\tau 1} + U_{\mu 2}U^*_{\tau 2} + U_{\mu 3}U^*_{\tau 3} = 0 \quad \textbf{(U6)}$$

★**To calculate the oscillation probability proceed as before…**

•**Consider a state which is produced at** $t = 0$ **as a** $|\nu_e\rangle$ **(i.e. with an electron)**

$$|\psi(t=0)\rangle = |\nu_e\rangle = U_{e1}|\nu_1\rangle + U_{e2}|\nu_2\rangle + U_{e3}|\nu_3\rangle$$

- **The wave-function evolves as:**

$$|\psi(t)\rangle = U_{e1}|\nu_1\rangle e^{-ip_1.x} + U_{e2}|\nu_2\rangle e^{-ip_2.x} + U_{e3}|\nu_3\rangle e^{-ip_3.x}$$

where $p_i.x = E_i t - \vec{p}_i.\vec{x} = E_i t - |\vec{p}|z$

z axis in direction of propagation

- **After a travelling a distance** L

$$|\psi(L)\rangle = U_{e1}|\nu_1\rangle e^{-i\phi_1} + U_{e2}|\nu_2\rangle e^{-i\phi_2} + U_{e3}|\nu_3\rangle e^{-i\phi_3}$$

where $\phi_i = p_i.x = E_i t - |\vec{p}|L = (E_i - |\vec{p}_i|)L$

- **As before we can approximate**

$$\phi_i \approx \frac{m_i^2}{2E_i}L$$

- **Expressing the mass eigenstates in terms of the weak eigenstates**

$$\begin{aligned}|\psi(L)\rangle &= U_{e1}(U_{e1}^*|\nu_e\rangle + U_{\mu 1}^*|\nu_\mu\rangle + U_{\tau 1}^*|\nu_\tau\rangle)e^{-i\phi_1} \\ &+ U_{e2}(U_{e2}^*|\nu_e\rangle + U_{\mu 2}^*|\nu_\mu\rangle + U_{\tau 2}^*|\nu_\tau\rangle)e^{-i\phi_2} \\ &+ U_{e3}(U_{e3}^*|\nu_e\rangle + U_{\mu 3}^*|\nu_\mu\rangle + U_{\tau 3}^*|\nu_\tau\rangle)e^{-i\phi_3}\end{aligned}$$

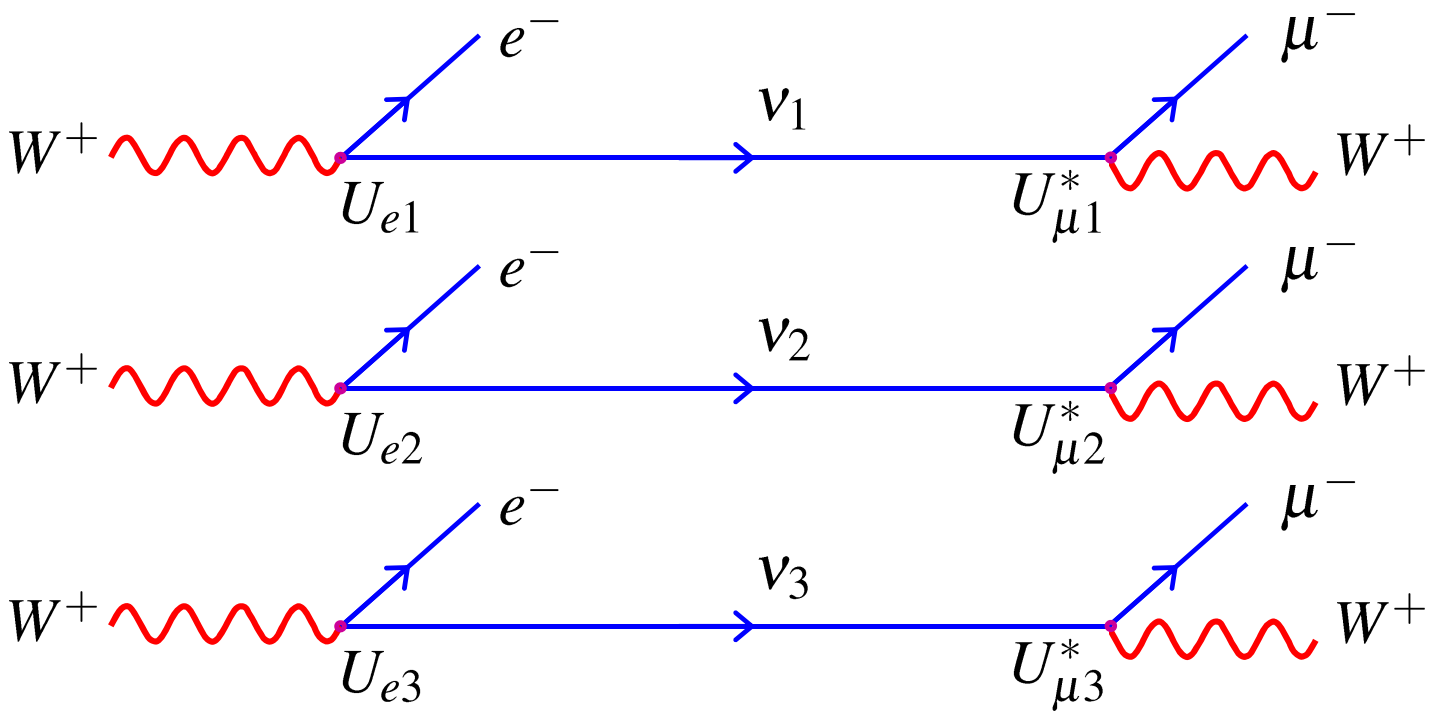
- **Which can be rearranged to give**

$$\begin{aligned}|\psi(L)\rangle &= (U_{e1}U_{e1}^*e^{-i\phi_1} + U_{e2}U_{e2}^*e^{-i\phi_2} + U_{e3}U_{e3}^*e^{-i\phi_3})|\nu_e\rangle \\ &+ (U_{e1}U_{\mu 1}^*e^{-i\phi_1} + U_{e2}U_{\mu 2}^*e^{-i\phi_2} + U_{e3}U_{\mu 3}^*e^{-i\phi_3})|\nu_\mu\rangle \\ &+ (U_{e1}U_{\tau 1}^*e^{-i\phi_1} + U_{e2}U_{\tau 2}^*e^{-i\phi_2} + U_{e3}U_{\tau 3}^*e^{-i\phi_3})|\nu_\tau\rangle\end{aligned} \quad (3)$$

- **From which**

$$P(\nu_e \to \nu_\mu) = |\langle \nu_\mu | \psi(L) \rangle|^2$$
$$= |U_{e1}U^*_{\mu 1} e^{-i\phi_1} + U_{e2}U^*_{\mu 2} e^{-i\phi_2} + U_{e3}U^*_{\mu 3} e^{-i\phi_3}|^2$$

- **The terms in this expression can be represented as:**



- **Because of the unitarity of the PMNS matrix we have (U4):**

$$U_{e1}U^*_{\mu 1} + U_{e2}U^*_{\mu 2} + U_{e3}U^*_{\mu 3} = 0$$

and, consequently, unless the phases of the different components are different, the sum of these three diagrams is zero, i.e., require different neutrino masses for osc.

•**Evaluate**

$$P(\nu_e \to \nu_\mu) \quad = \quad |U_{e1}U^*_{\mu 1}e^{-i\phi_1} + U_{e2}U^*_{\mu 2}e^{-i\phi_2} + U_{e3}U^*_{\mu 3}e^{-i\phi_3}|^2$$

using $|z_1+z_2+z_3|^2 \equiv |z_1|^2+|z_2|^2+|z_3|^2+2\Re(z_1z_2^*+z_1z_3^*+z_2z_3^*)$ **(4)**

which gives:

$$P(\nu_e \to \nu_\mu) \quad = \quad |U_{e1}U^*_{\mu 1}|^2+|U_{e2}U^*_{\mu 2}|^2+|U_{e3}U^*_{\mu 3}|^2+ \qquad \textbf{(5)}$$

$$2\Re(U_{e1}U^*_{\mu 1}U^*_{e2}U_{\mu 2}e^{-i(\phi_1-\phi_2)} + U_{e1}U^*_{\mu 1}U^*_{e3}U_{\mu 3}e^{-i(\phi_1-\phi_3)} + U_{e2}U^*_{\mu 2}U^*_{e3}U_{\mu 3}e^{-i(\phi_2-\phi_3)})$$

•**This can be simplified by applying identity (4) to |(U4)|²**

$$|U_{e1}U^*_{\mu 1}+U_{e2}U^*_{\mu 2}+U_{e3}U^*_{\mu 3}|^2 = 0$$

$$\Rightarrow \quad |U_{e1}U^*_{\mu 1}|^2+|U_{e2}U^*_{\mu 2}|^2+|U_{e3}U^*_{\mu 3}|^2 =$$

$$-2\Re(U_{e1}U^*_{\mu 1}U^*_{e2}U_{\mu 2}+U_{e1}U^*_{\mu 1}U^*_{e3}U_{\mu 3}+U_{e2}U^*_{\mu 2}U^*_{e3}U_{\mu 3})$$

•**Substituting into equation (5) gives**

$$\begin{aligned} P(\nu_e \to \nu_\mu) \quad &= \quad 2\Re\{U_{e1}U^*_{\mu 1}U^*_{e2}U_{\mu 2}[e^{-i(\phi_1-\phi_2)}-1]\} \\ &+ \quad 2\Re\{U_{e1}U^*_{\mu 1}U^*_{e3}U_{\mu 3}[e^{-i(\phi_1-\phi_3)}-1]\} \\ &+ \quad 2\Re\{U_{e2}U^*_{\mu 2}U^*_{e3}U_{\mu 3}[e^{-i(\phi_2-\phi_3)}-1]\} \end{aligned} \qquad \textbf{(6)}$$

★ This expression for the electron survival probability is obtained from the coefficient for $|\nu_e\rangle$ in eqn. (3):

$$P(\nu_e \to \nu_e) = |\langle \nu_e | \psi(L) \rangle|^2$$
$$= |U_{e1}U_{e1}^* e^{-i\phi_1} + U_{e2}U_{e2}^* e^{-i\phi_2} + U_{e3}U_{e3}^* e^{-i\phi_3}|^2$$

which using the unitarity relation (U1)

$$|U_{e1}U_{e1}^* + U_{e2}U_{e2}^* + U_{e3}U_{e3}^*|^2 = 1$$

can be written

$$\begin{aligned} P(\nu_e \to \nu_e) = 1 &+ 2|U_{e1}|^2|U_{e2}|^2\Re\{[e^{-i(\phi_1-\phi_2)} - 1]\} \\ &+ 2|U_{e1}|^2|U_{e3}|^2\Re\{[e^{-i(\phi_1-\phi_3)} - 1]\} \\ &+ 2|U_{e2}|^2|U_{e3}|^2\Re\{[e^{-i(\phi_2-\phi_3)} - 1]\} \end{aligned} \quad (7)$$

★ This expression can simplified using

$$\Re\{e^{-i(\phi_1-\phi_2)} - 1\} = \cos(\phi_2 - \phi_1) - 1$$
$$= -2\sin^2\left(\frac{\phi_2 - \phi_1}{2}\right)$$
$$= -2\sin^2\left(\frac{(m_2^2 - m_1^2)L}{4E}\right)$$

with $\phi_i \approx \frac{m_i^2}{2E}L$

Phase of mass eigenstate i at $z = L$

•**Define:** $$\Delta_{21} = \frac{(m_2^2 - m_1^2)L}{4E} = \frac{\Delta m_{21}^2 L}{4E}$$ **with** $$\Delta m_{21}^2 = m_2^2 - m_1^2$$

NOTE: $\Delta_{21} = (\phi_2 - \phi_1)/2$ **is a phase difference (i.e. dimensionless)**

•**Which gives the electron neutrino survival probability**

$$P(\nu_e \to \nu_e) = 1 - 4|U_{e1}|^2|U_{e2}|^2 \sin^2\Delta_{21} - 4|U_{e1}|^2|U_{e3}|^2 \sin^2\Delta_{31} - 4|U_{e2}|^2|U_{e3}|^2 \sin^2\Delta_{32}$$

•**Similar expressions can be obtained for the muon and tau neutrino survival probabilities for muon and tau neutrinos.**

★ **Note that since we only have three neutrino generations there are only two independent mass-squared differences, i.e.**

$$m_3^2 - m_1^2 = (m_3^2 - m_2^2) + (m_2^2 - m_1^2)$$

and in the above equation only two of the Δ_{ij} **are independent**

★**All expressions are in Natural Units, conversion to more useful units here gives:**

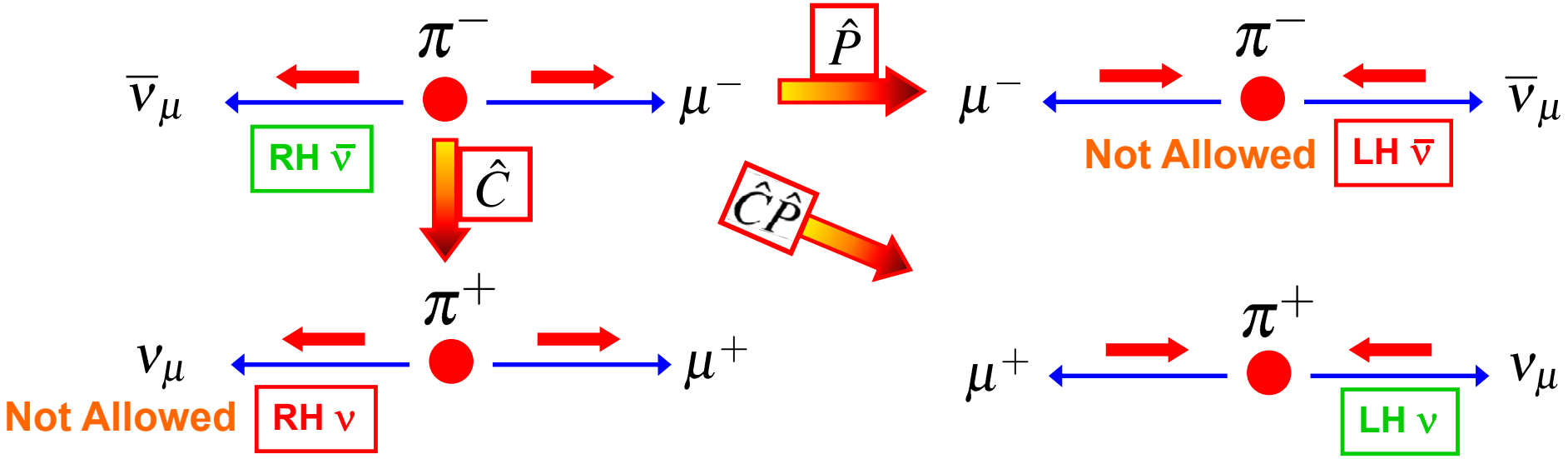
$$\Delta_{21} = 1.27\frac{\Delta m_{21}^2(\mathrm{eV}^2)L(\mathrm{km})}{E(\mathrm{GeV})}$$ **and** $$\lambda_{\mathrm{osc}}(\mathrm{km}) = 2.47\frac{E(\mathrm{GeV})}{\Delta m^2(\mathrm{eV}^2)}$$

CP and CPT in the LEPTON Weak Interaction

★ In addition to parity there are two other important discrete symmetries:

Parity	$\hat{P}:$	$\vec{r} \rightarrow -\vec{r}$
Time Reversal	$\hat{T}:$	$t \rightarrow -t$
Charge Conjugation	$\hat{C}:$	**Particle ⟷ Anti-particle**

★ The weak interaction violates parity conservation, but what about **C** ? Consider pion decay remembering that the neutrino is ultra-relativistic and only **left-handed neutrinos and right-handed anti-neutrinos participate in WI**



★ Hence weak interaction also **violates charge conjugation** symmetry but appears to be invariant under combined effect of **C** and **P**

CP and T Violation in Neutrino Oscillations

- Previously derived the oscillation probability for $\nu_e \rightarrow \nu_\mu$

$$\begin{aligned} P(\nu_e \rightarrow \nu_\mu) &= 2\Re\{U_{e1}U_{\mu 1}^* U_{e2}^* U_{\mu 2}[e^{-\imath(\phi_1-\phi_2)} - 1]\} \\ &+ 2\Re\{U_{e1}U_{\mu 1}^* U_{e3}^* U_{\mu 3}[e^{-i(\phi_1-\phi_3)} - 1]\} \\ &+ 2\Re\{U_{e2}U_{\mu 2}^* U_{e3}^* U_{\mu 3}[e^{-i(\phi_2-\phi_3)} - 1]\} \end{aligned}$$

- The oscillation probability for $\nu_\mu \rightarrow \nu_e$ can be obtained in the same manner or by simply exchanging the labels $(e) \leftrightarrow (\mu)$

$$\begin{aligned} P(\nu_\mu \rightarrow \nu_e) &= 2\Re\{U_{\mu 1}U_{e1}^* U_{\mu 2}^* U_{e2}[e^{-i(\phi_1-\phi_2)} - 1]\} \\ &+ 2\Re\{U_{\mu 1}U_{e1}^* U_{\mu 3}^* U_{e3}[e^{-i(\phi_1-\phi_3)} - 1]\} \\ &+ 2\Re\{U_{\mu 2}U_{e2}^* U_{\mu 3}^* U_{e3}[e^{-i(\phi_2-\phi_3)} - 1]\} \end{aligned} \quad (8)$$

★ Unless the elements of the PMNS matrix are real (see note below)

$$P(\nu_e \rightarrow \nu_\mu) \neq P(\nu_\mu \rightarrow \nu_e) \quad (9)$$

- If any of the elements of the PMNS matrix are complex, neutrino oscillations are not invariant under time reversal $t \rightarrow -t$

NOTE: can multiply entire PMNS matrix by a complex phase without changing the oscillation prob. T is violated if one of the elements has a different complex phase than the others

- Consider the effects of T, CP and CPT on neutrino oscillations

T	$\nu_e \to \nu_\mu$	$\xrightarrow{\hat{T}}$	$\nu_\mu \to \nu_e$
CP	$\nu_e \to \nu_\mu$	$\xrightarrow{\hat{C}\hat{P}}$	$\overline{\nu}_e \to \overline{\nu}_\mu$
CPT	$\nu_e \to \nu_\mu$	$\xrightarrow{\hat{C}\hat{P}\hat{T}}$	$\overline{\nu}_\mu \to \overline{\nu}_e$

Note C alone is not sufficient as it transforms LH neutrinos into LH anti-neutrinos (not involved in Weak Interaction)

- If the weak interactions is invariant under CPT

$$P(\nu_e \to \nu_\mu) = P(\overline{\nu}_\mu \to \overline{\nu}_e)$$

and similarly

$$P(\nu_\mu \to \nu_e) = P(\overline{\nu}_e \to \overline{\nu}_\mu) \qquad (10)$$

- If the PMNS matrix is not purely real, then (9)

$$P(\nu_e \to \nu_\mu) \neq P(\nu_\mu \to \nu_e)$$

and from (10)

$$\boxed{P(\nu_e \to \nu_\mu) \neq P(\overline{\nu}_e \to \overline{\nu}_\mu)}$$

★Hence unless the PMNS matrix is real, CP is violated in neutrino oscillations!

Future experiments, e.g. "a neutrino factory", are being considered as a way to investigate CP violation in neutrino oscillations. However, CP violating effects are well below the current experimental sensitivity. In the following discussion we will take the PMNS matrix to be real. (question 22)

Neutrino Mass Hierarchy

i.e. we don't know if diff is negative or positive!

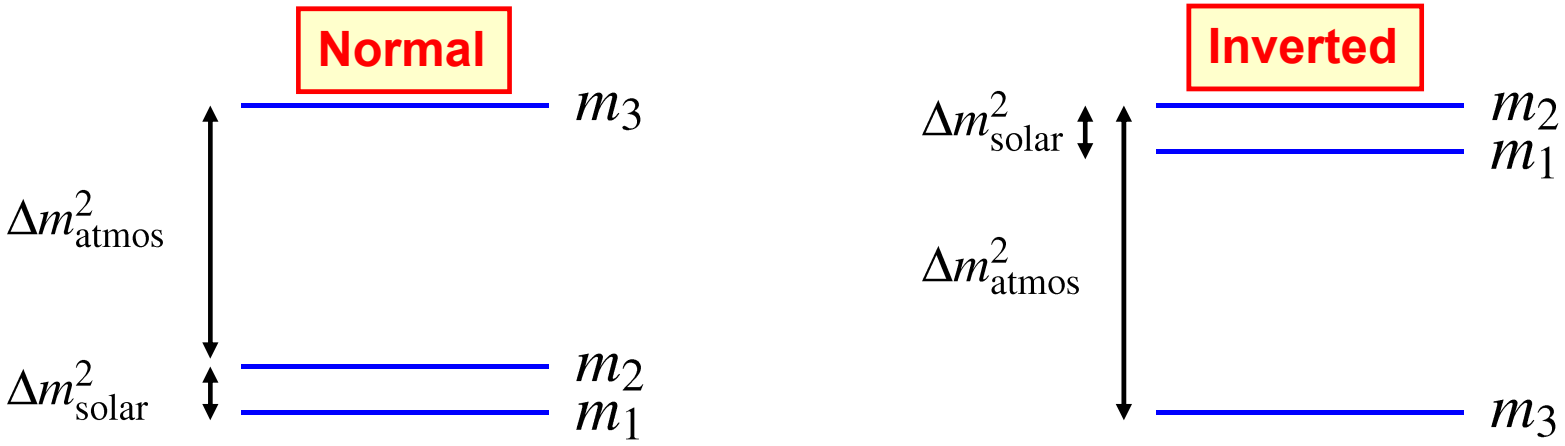
★ To date, results on neutrino oscillations only determine

$$|\Delta m^2_{ji}| = |m^2_j - m^2_i|$$

★ Two distinct and very different mass scales:

- Atmospheric neutrino oscillations : $|\Delta m^2|_{\rm atmos} \sim 2.5 \times 10^{-3}\,{\rm eV}^2$
- Solar neutrino oscillations: $|\Delta m^2|_{\rm solar} \sim 8 \times 10^{-5}\,{\rm eV}^2$

•Two possible assignments of mass hierarchy:



•In both cases: $\Delta m^2_{21} \sim 8 \times 10^{-5}\,{\rm eV}^2$ (solar)

$$|\Delta m^2_{31}| \approx |\Delta m^2_{32}| \sim 2.5 \times 10^{-3}\,{\rm eV}^2 \quad \text{(atmospheric)}$$

•Hence we can approximate $\Delta m^2_{31} \approx \Delta m^2_{32}$

Three Flavour Oscillations Neglecting CP Violation

- Neglecting CP violation considerably simplifies the algebra of three flavour neutrino oscillations. Taking the PMNS matrix to be real, equation (6) becomes:

$$P(\nu_e \to \nu_\mu) = -4U_{e1}U_{\mu 1}U_{e2}U_{\mu 2}\sin^2\Delta_{21} - 4U_{e1}U_{\mu 1}U_{e3}U_{\mu 3}\sin^2\Delta_{31} - 4U_{e2}U_{\mu 2}U_{e3}U_{\mu 3}\sin^2\Delta_{32}$$

with $$\Delta_{ji} = \frac{(m_j^2 - m_i^2)L}{4E} = \frac{\Delta m_{ji}^2 L}{4E}$$

- Using: $\Delta_{31} \approx \Delta_{32}$ (see p. 365)

$$P(\nu_e \to \nu_\mu) \approx -4U_{e1}U_{\mu 1}U_{e2}U_{\mu 2}\sin^2\Delta_{21} - 4(U_{e1}U_{\mu 1} + U_{e2}U_{\mu 2})U_{e3}U_{\mu 3}\sin^2\Delta_{32}$$

- Which can be simplified using (U4) $U_{e1}U_{\mu 1}^* + U_{e2}U_{\mu 2}^* + U_{e3}U_{\mu 3}^* = 0$

$$\Rightarrow \boxed{P(\nu_e \to \nu_\mu) \approx -4U_{e1}U_{\mu 1}U_{e2}U_{\mu 2}\sin^2\Delta_{21} + 4U_{e3}^2U_{\mu 3}^2\sin^2\Delta_{32}}$$

- Can apply $\Delta_{31} \approx \Delta_{32}$ to the expression for electron neutrino survival probability

$$P(\nu_e \to \nu_e) = 1 - 4U_{e1}^2U_{e2}^2\sin^2\Delta_{21} - 4U_{e1}^2U_{e3}^2\sin^2\Delta_{31} - 4U_{e2}^2U_{e3}^2\sin^2\Delta_{32}$$
$$\approx 1 - 4U_{e1}^2U_{e2}^2\sin^2\Delta_{21} - 4(U_{e1}^2 + U_{e2}^2)U_{e3}^2\sin^2\Delta_{32}$$

- Which can be simplified using (U1) $U_{e1}^2 + U_{e2}^2 + U_{e3}^2 = 1$

$$\Rightarrow \boxed{P(\nu_e \to \nu_e) \approx 1 - 4U_{e1}^2U_{e2}^2\sin^2\Delta_{21} - 4(1 - U_{e3}^2)U_{e3}^2\sin^2\Delta_{32}}$$

★ **Neglecting CP violation (i.e. taking the PMNS matrix to be real) and making the approximation that $|\Delta m^2_{31}| \approx |\Delta m^2_{32}|$ obtain the following expressions for neutrino oscillation probabilities:**

$$P(\nu_e \to \nu_e) \approx 1 - 4U_{e1}^2 U_{e2}^2 \sin^2\Delta_{21} - 4(1-U_{e3}^2)U_{e3}^2 \sin^2\Delta_{32} \quad (11)$$

$$P(\nu_\mu \to \nu_\mu) \approx 1 - 4U_{\mu 1}^2 U_{\mu 2}^2 \sin^2\Delta_{21} - 4(1-U_{\mu 3}^2)U_{\mu 3}^2 \sin^2\Delta_{32} \quad (12)$$

$$P(\nu_\tau \to \nu_\tau) \approx 1 - 4U_{\tau 1}^2 U_{\tau 2}^2 \sin^2\Delta_{21} - 4(1-U_{\tau 3}^2)U_{\tau 3}^2 \sin^2\Delta_{32} \quad (13)$$

$$P(\nu_e \to \nu_\mu) = P(\nu_\mu \to \nu_e) \approx -4U_{e1}U_{\mu 1}U_{e2}U_{\mu 2} \sin^2\Delta_{21} + 4U_{e3}^2 U_{\mu 3}^2 \sin^2\Delta_{32} \quad (14)$$

$$P(\nu_e \to \nu_\tau) = P(\nu_\tau \to \nu_e) \approx -4U_{e1}U_{\tau 1}U_{e2}U_{\tau 2} \sin^2\Delta_{21} + 4U_{e3}^2 U_{\tau 3}^2 \sin^2\Delta_{32} \quad (15)$$

$$P(\nu_\mu \to \nu_\tau) = P(\nu_\tau \to \nu_\mu) \approx -4U_{\mu 1}U_{\tau 1}U_{\mu 2}U_{\tau 2} \sin^2\Delta_{21} + 4U_{\mu 3}^2 U_{\tau 3}^2 \sin^2\Delta_{32} \quad (16)$$

★ **The wavelengths associated with $\sin^2\Delta_{21}$ and $\sin^2\Delta_{32}$ are:**

"SOLAR" $\lambda_{21} = \dfrac{4\pi E}{\Delta m^2_{21}}$ **"Long"-Wavelength**

and

$\lambda_{32} = \dfrac{4\pi E}{\Delta m^2_{32}}$ **"ATMOSPHERIC"** **"Short"-Wavelength**

PMNS Matrix

★ The PMNS matrix is usually expressed in terms of 3 rotation angles $\theta_{12}, \theta_{23}, \theta_{13}$ and a complex phase δ, using the notation $s_{ij} = \sin\theta_{ij}, \quad c_{ij} = \cos\theta_{ij}$

$$\begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix}} \times \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{-i\delta} & 0 & c_{13} \end{pmatrix} \times \underbrace{\begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}}$$

Dominates: "Atmospheric" "Solar"

• Writing this out in full:

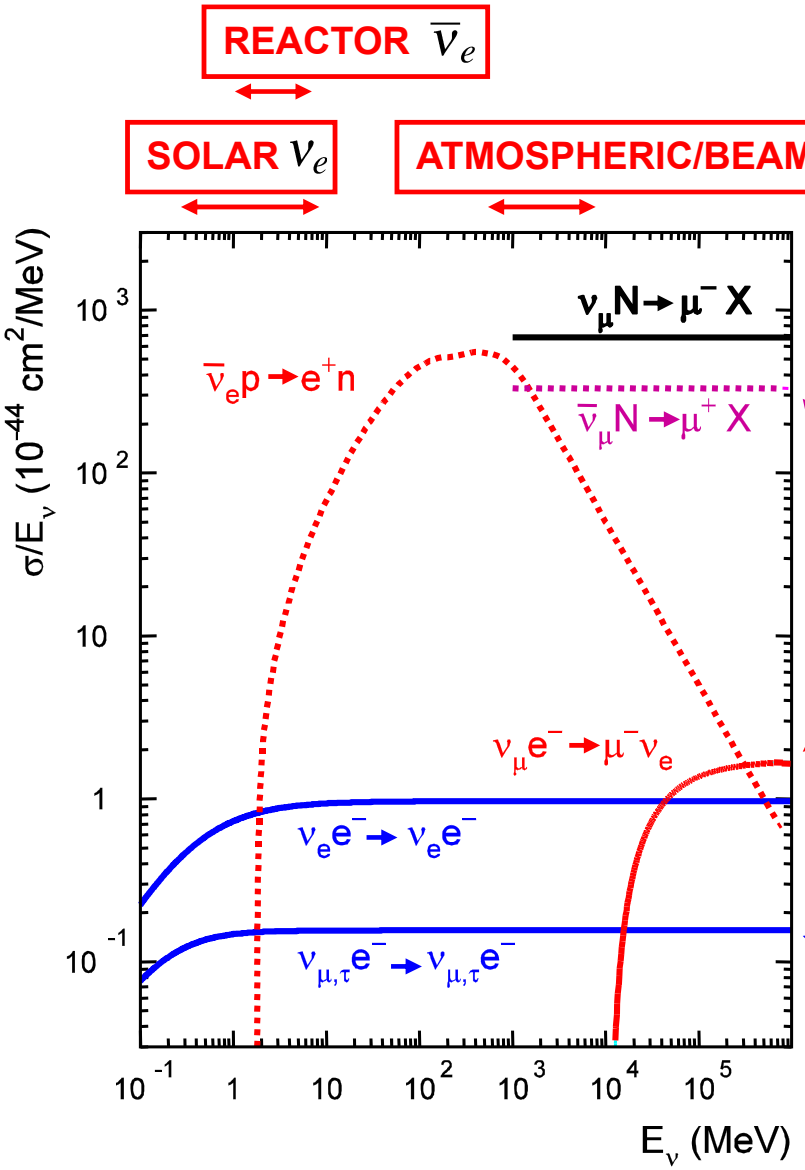
$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}$$

★ There are six **SM parameters** that can be measured in ν oscillation experiments

$\lvert\Delta m_{21}\rvert^2 = \lvert m_2^2 - m_1^2\rvert$	θ_{12}	Solar and reactor neutrino experiments
$\lvert\Delta m_{32}\rvert^2 = \lvert m_3^2 - m_2^2\rvert$	θ_{23}	Atmospheric and beam neutrino experiments
	θ_{13}	Reactor neutrino experiments + future beam
	δ	Future beam experiments

Neutrino Detection

★ The detector technology/interaction process depends on type of neutrino and energy



$$\sigma_{\nu_\mu N} = 0.22 \frac{2 m_p G_F^2 E_\nu}{\pi}$$

$$\sigma_{\overline{\nu}_\mu N} = 0.33 \frac{2 m_p G_F^2 E_\nu}{3\pi}$$

Deep Inelastic Scattering (p. 332)

$$\sigma_{\nu_\mu e^-} = \frac{2 m_e G_F^2 E_\nu}{\pi}$$

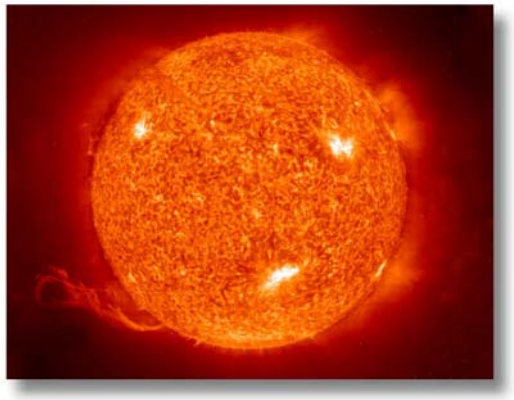
CC only Threshold 11 GeV

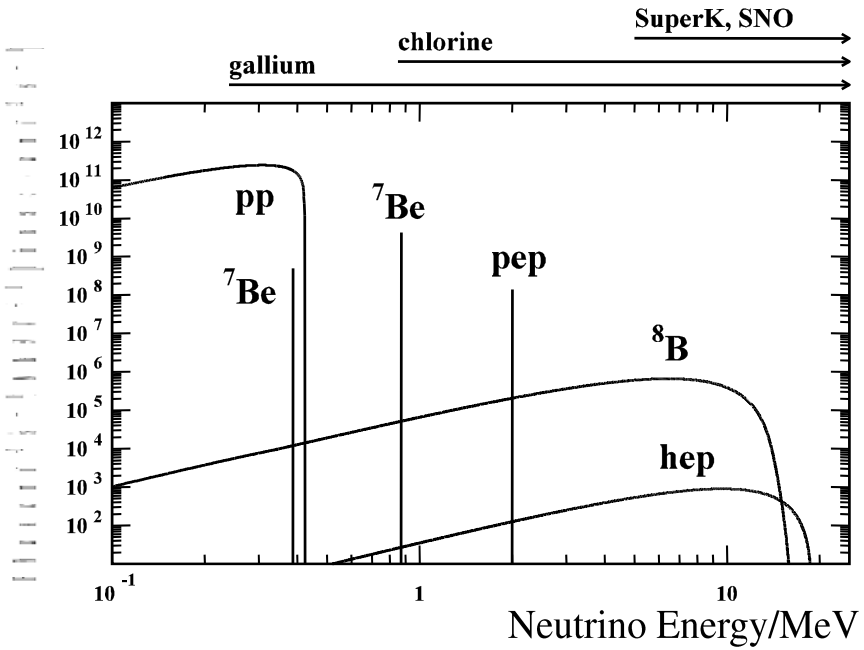
$$\sigma_{\nu_e e^-} \approx 0.6 \frac{2 m_e G_F^2 E_\nu}{\pi}$$

NC + CC

NC only (see handout 13)

2) Solar Neutrinos





- The Sun is powered by the weak interaction – producing a very large flux of electron neutrinos

$$2 \times 10^{38}\, \nu_e\, \mathrm{s}^{-1}$$

- Several different nuclear reactions in the sun ⇨ complex neutrino energy spectrum

$$p + p \rightarrow d + e^+ + \nu_e \qquad E_\nu < 0.5\,\mathrm{MeV}$$

$$^8B \rightarrow ^8Be^* + e^+ + \nu_e \qquad E_\nu \sim 5\,\mathrm{MeV}$$

$$p + e^- + p \rightarrow d + \nu_e$$

$$^7Be + e^- \rightarrow ^7Li + \nu_e$$

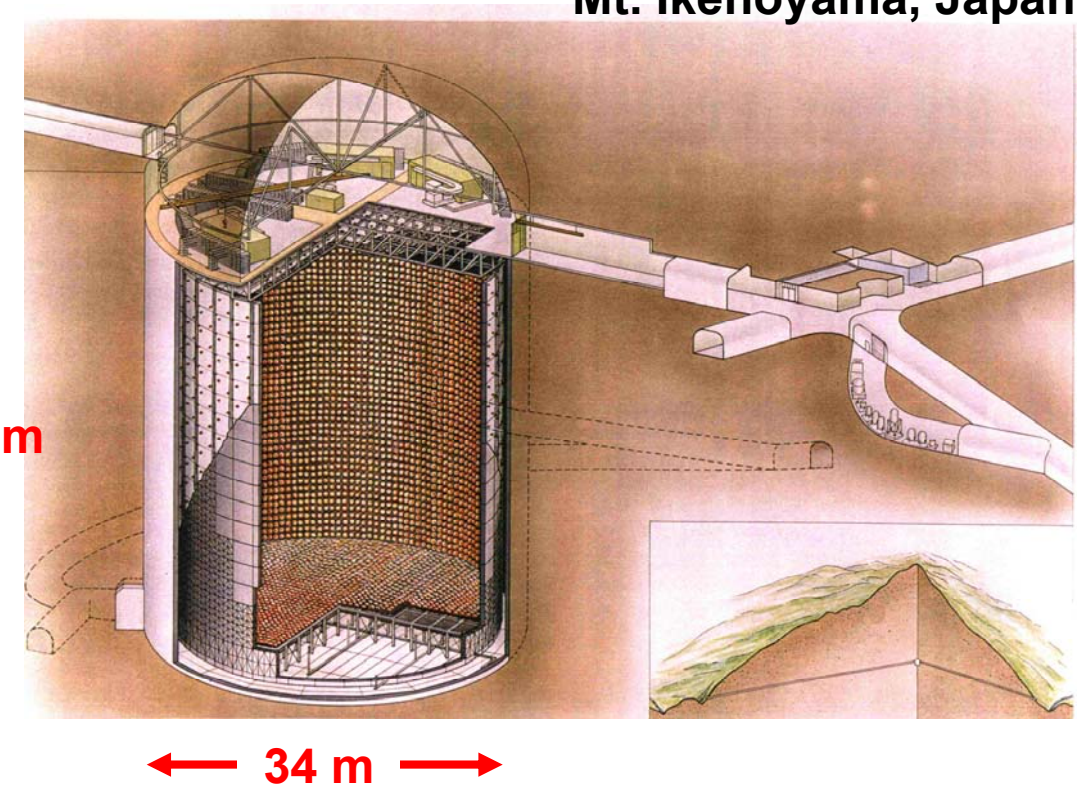
$$^3He + p \rightarrow ^4He + e^+ + \nu_e$$

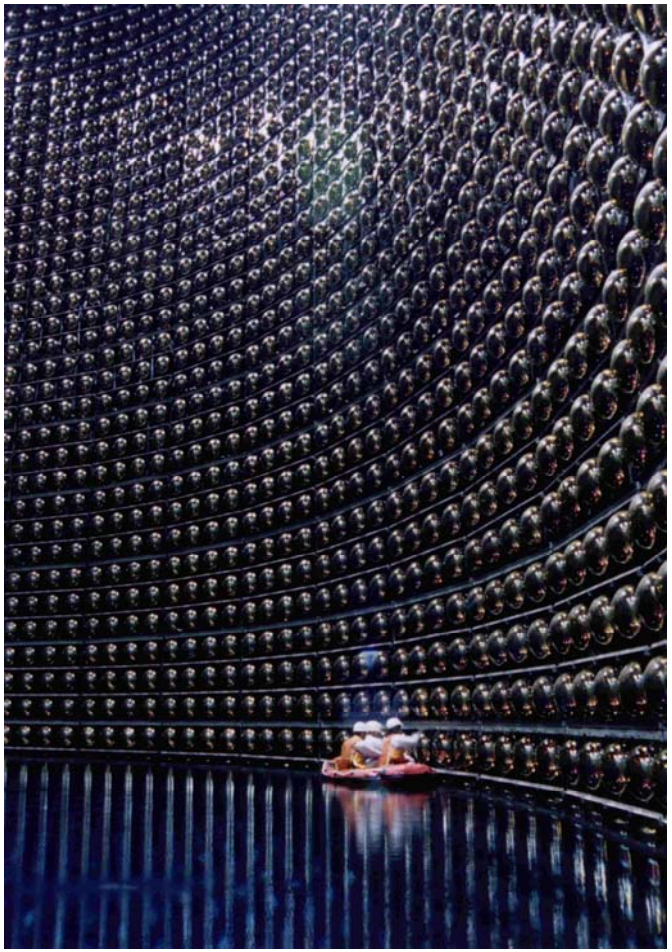
- All experiments saw a deficit of electron neutrinos compared to experimental prediction – the SOLAR NEUTRINO PROBLEM
- e.g. Super Kamiokande

Solar Neutrinos I: Super Kamiokande

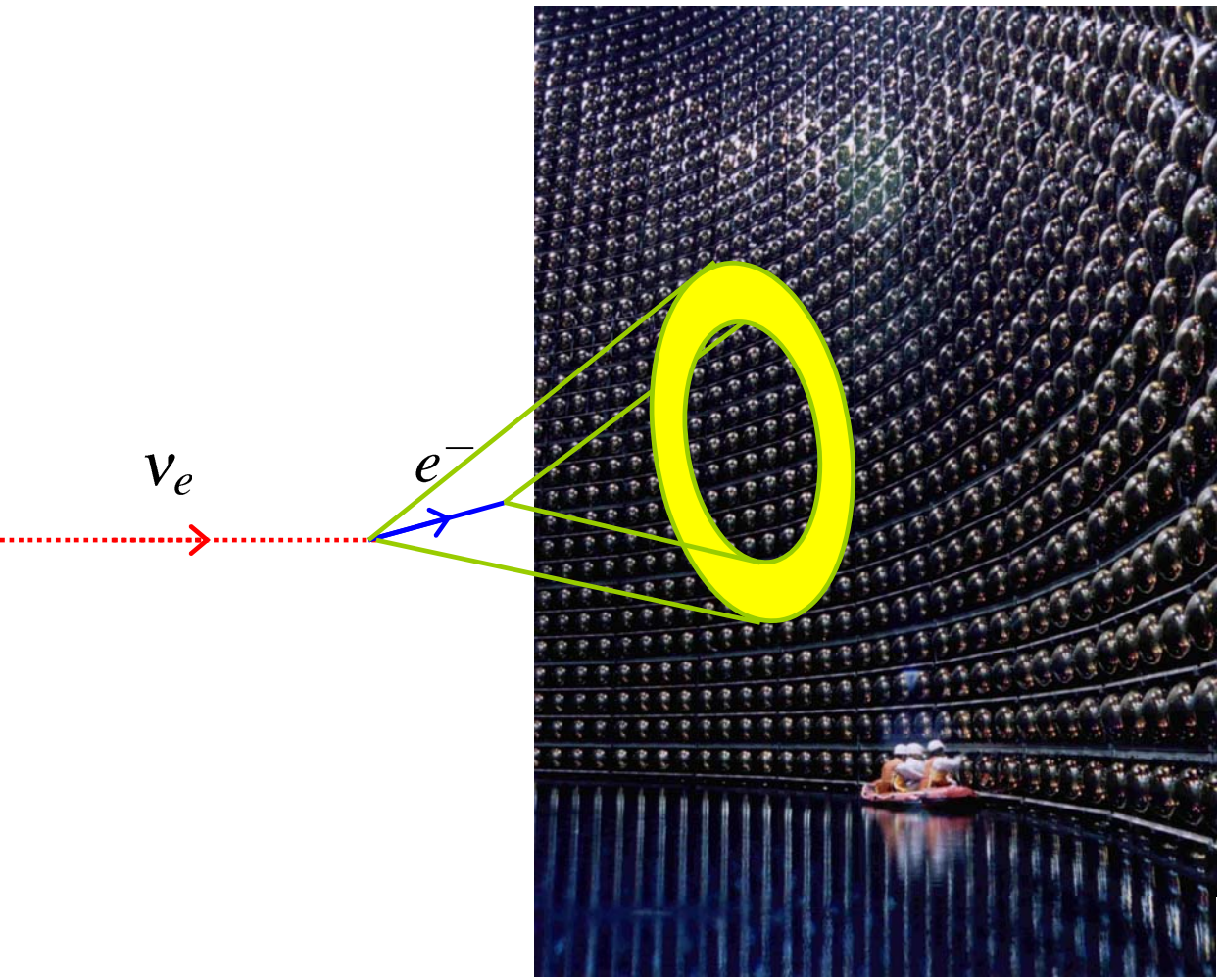
- 50000 ton water Čerenkov detector
- Water viewed by 11146 Photo-multiplier tubes
- Deep underground to filter out cosmic rays otherwise difficult to detect rare neutrino interactions

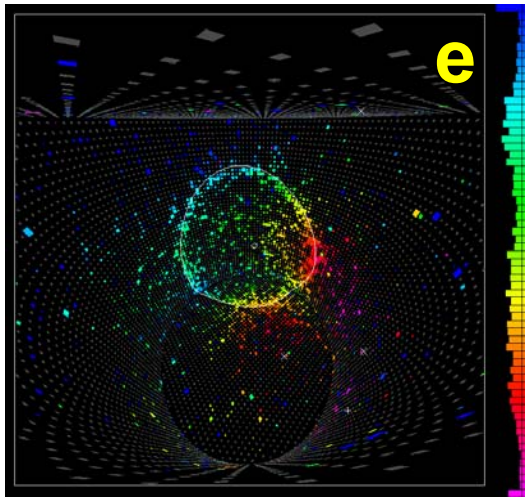
Mt. Ikenoyama, Japan

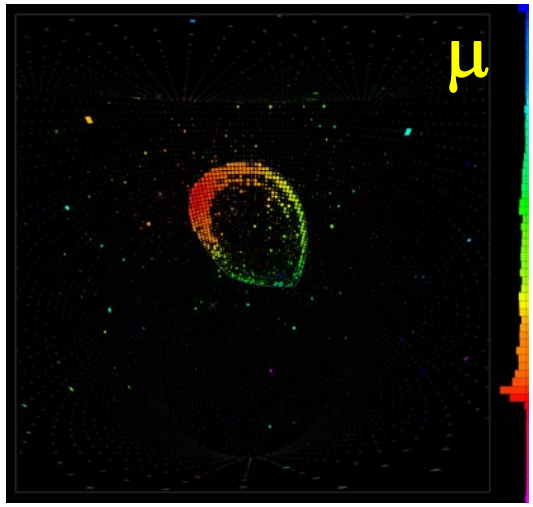




- **Detect neutrinos by observing Čerenkov radiation from charged particles which travel faster than speed of light in water *c/n***





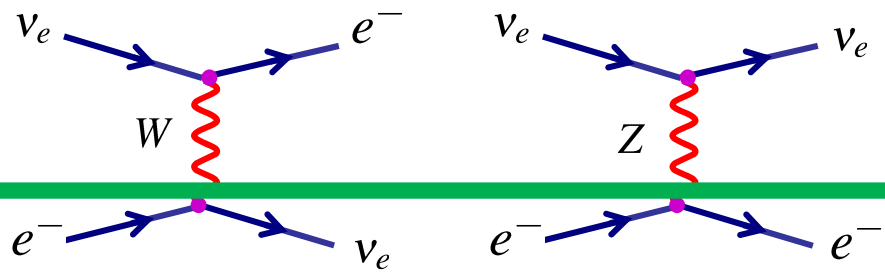


- **Can distinguish electrons from muons from pattern of light – muons produce clean rings whereas electrons produce more diffuse "fuzzy" rings**

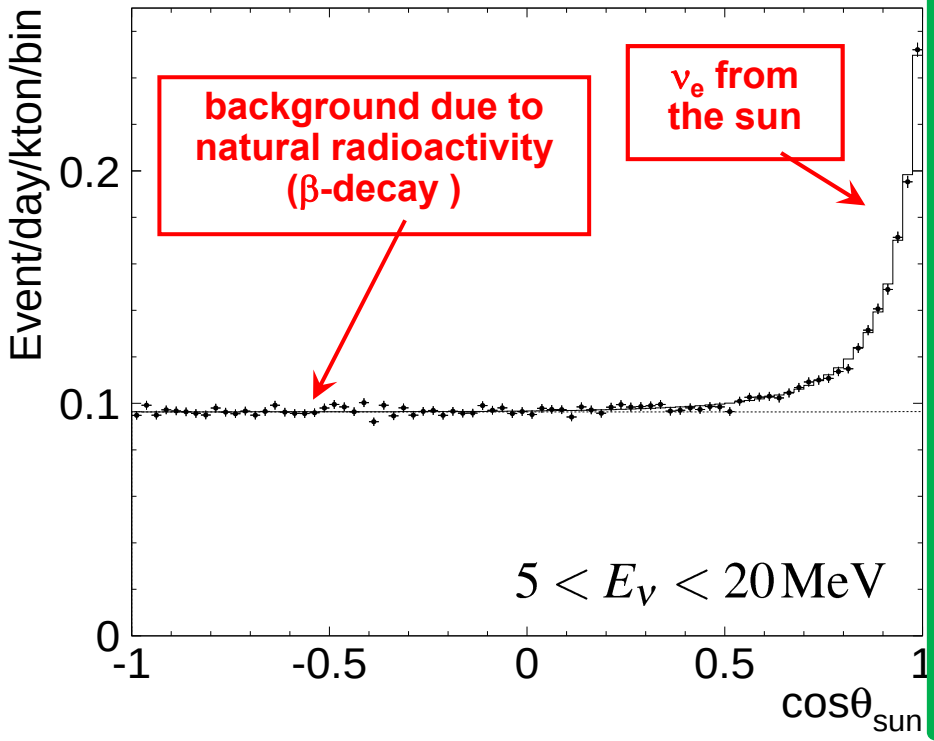
- Sensitive to solar neutrinos with $E_\nu > 5\,\mathrm{MeV}$
- For lower energies too much background from natural radioactivity (β-decays)
- Hence detect mostly neutrinos from $^8B \rightarrow {}^8Be^* + e^+ + \nu_e$
- Detect electron Čerenkov rings from

$$\nu_e + e^- \rightarrow \nu_e + e^-$$

- In LAB frame the electron is produced preferentially along the ν_e direction



S.Fukada et al., Phys. Rev. Lett. 86 5651-5655, 2001



Results:

- Clear signal of neutrinos from the sun
- However too few neutrinos

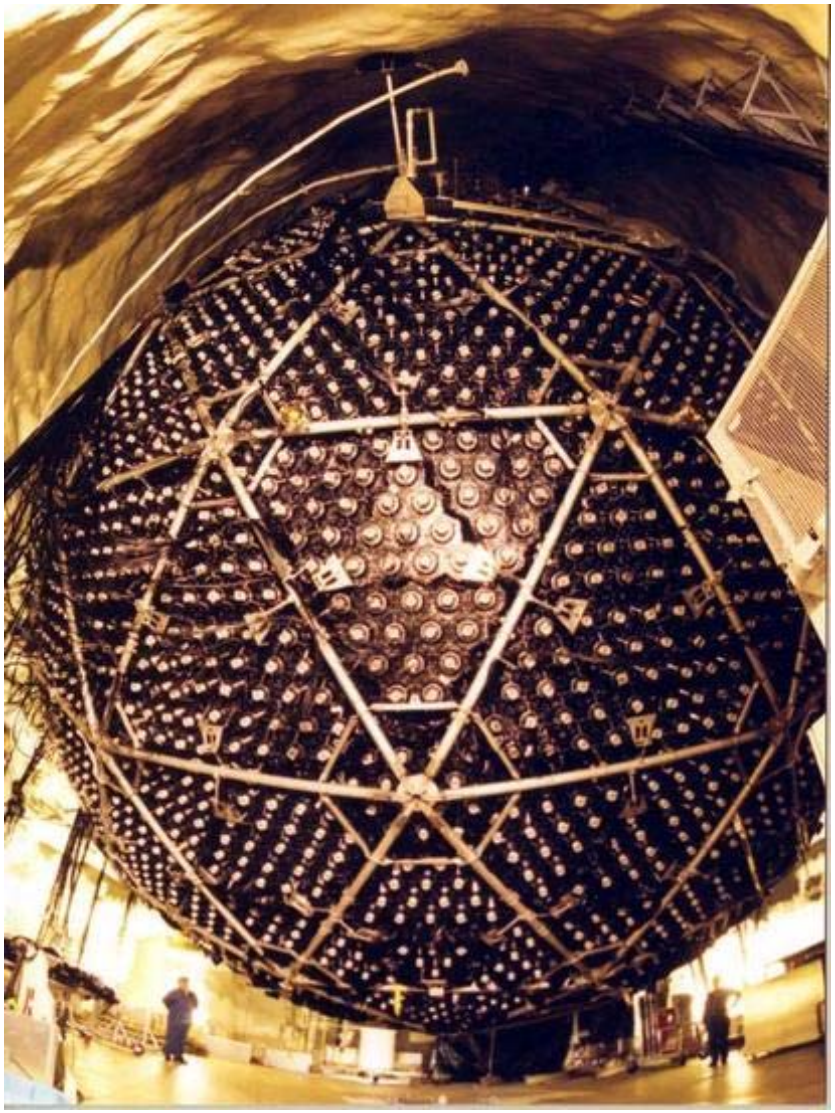
DATA/SSM = 0.45±0.02

SSM = "Standard Solar Model" Prediction

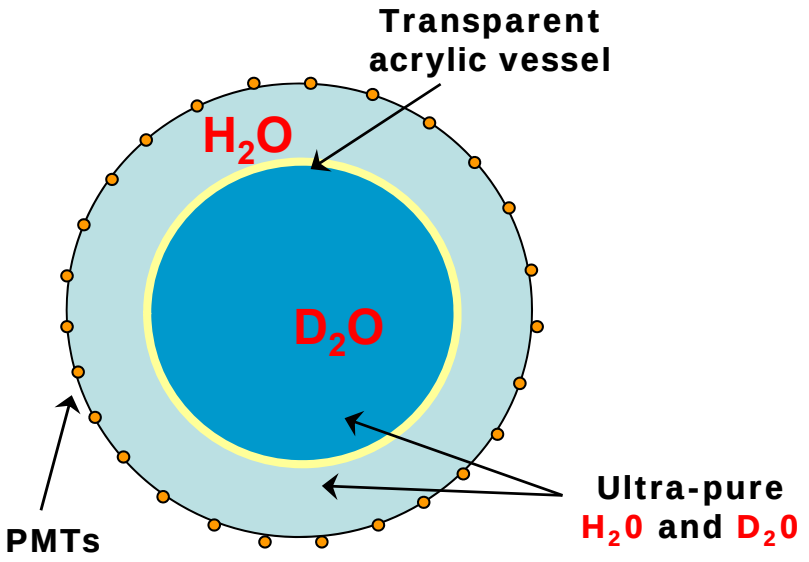
The Solar Neutrino "Problem"

Solar Neutrinos II: SNO

•Sudbury Neutrino Observatory located in a deep mine in Ontario, Canada



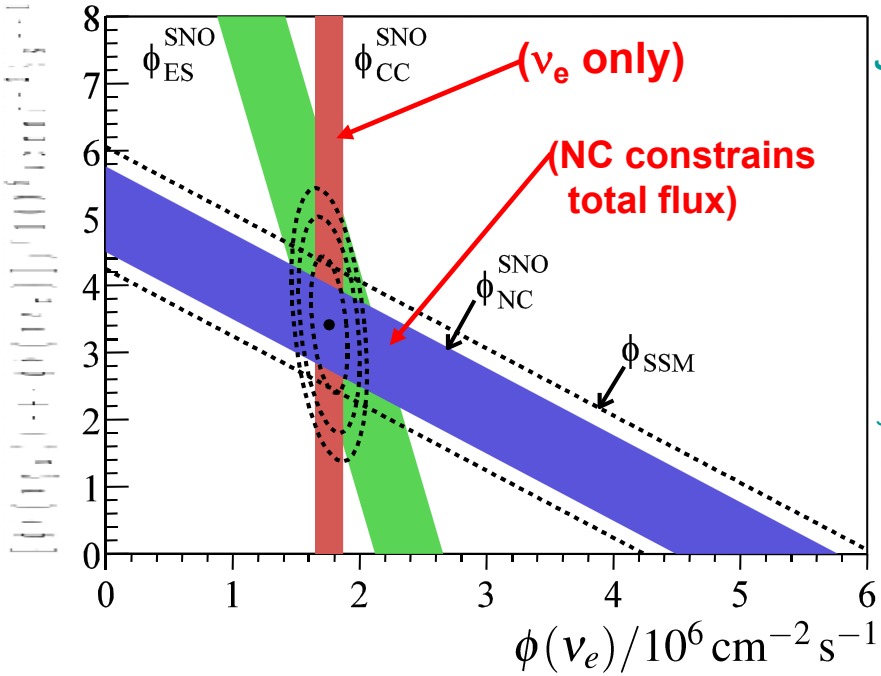
- 1000 ton heavy water (D_2O) Čerenkov detector
- D_2O inside a 12m diameter acrylic vessel
- Surrounded by 3000 tons of normal water
- Main experimental challenge is the need for very low background from radioactivity
- Ultra-pure H_2O and D_2O
- Surrounded by 9546 PMTs



★Using known cross sections can convert observed numbers of events into fluxes

★The different processes impose different constraints

★ Where constraints meet gives separate measurements of ν_e and $\nu_\mu + \nu_\tau$ fluxes



SNO Collaboration, Q.R. Ahmad et al., Phys. Rev. Lett. 89:011301, 2002

SNO Results:

$$\phi(\nu_e) = (1.8 \pm 0.1) \times 10^{-6}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$$

$$\phi(\nu_\mu) + \phi(\nu_\tau) = (3.4 \pm 0.6) \times 10^{-6}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$$

SSM Prediction:

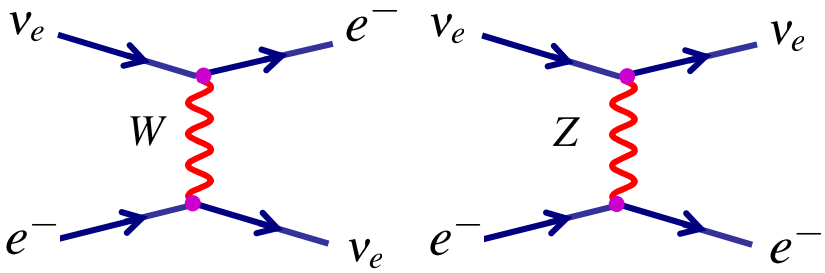
$$\phi(\nu_e) = 5.1 \times 10^{-6}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$$

- Clear evidence for a flux of ν_μ and/or ν_τ from the sun
- Total neutrino flux is consistent with expectation from SSM
- Clear evidence of $\nu_e \rightarrow \nu_\mu$ and/or $\nu_e \rightarrow \nu_\tau$ neutrino transitions

Interpretation of Solar Neutrino Data

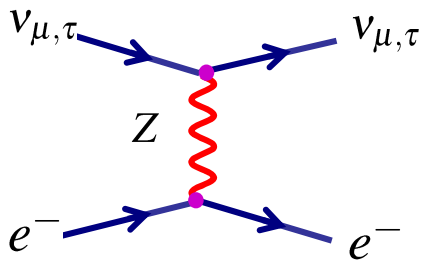
★ The interpretation of the solar neutrino data is complicated by **MATTER EFFECTS**

- The quantitative treatment is non-trivial and is not given here
- Basic idea is that as a neutrino leaves the sun it crosses a region of high electron density
- The coherent forward scattering process ($\nu_e \rightarrow \nu_e$) for an electron neutrino



CC + NC

is different to that for a muon or tau neutrino



NC

- Can enhance oscillations – "MSW effect"

★ A combined analysis of all solar neutrino data gives:

$$\Delta m^2_{\text{solar}} \approx 8 \times 10^{-5}\,\text{eV}^2,\ \sin^2 2\theta_{\text{solar}} \approx 0.85$$

1) Long Baseline Neutrino Experiments

- Initial studies of neutrino oscillations from atmospheric and solar neutrinos
  - atmospheric neutrinos discussed in **examinable** appendix
- Emphasis of neutrino research now on **neutrino beam** experiments
- Allows the physicist to take control – design experiment with specific goals
- In the last few years, long baseline neutrino oscillation experiments have started taking data: ~~K2K, MINOS, CNGS, T2K~~

DUNE next

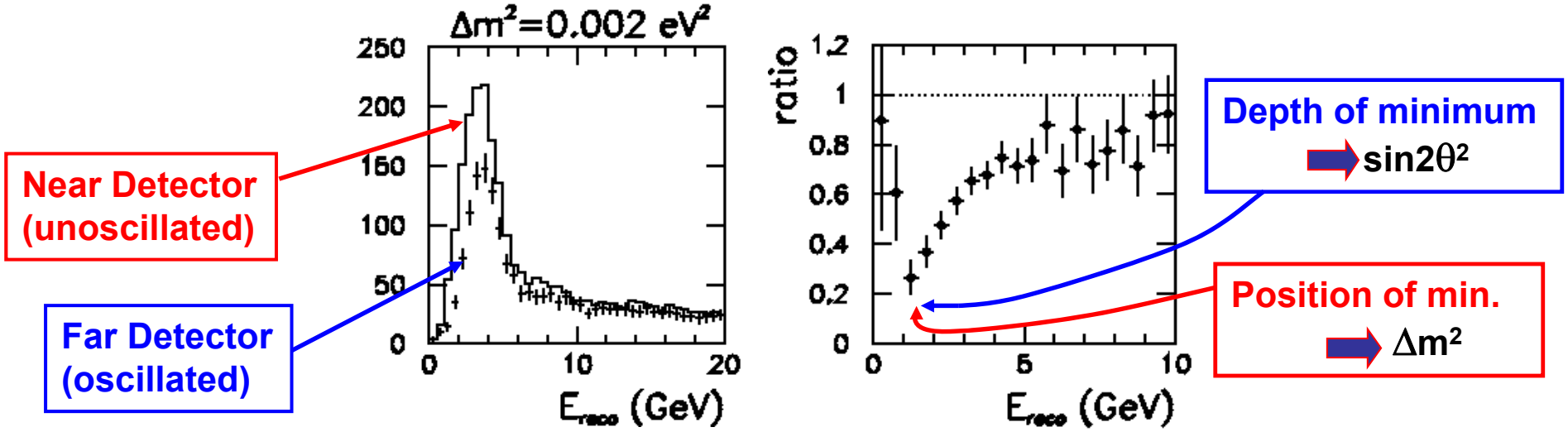
Basic Idea:

★ Intense neutrino beam

★ Two detectors: one close to beam the other hundreds of km away

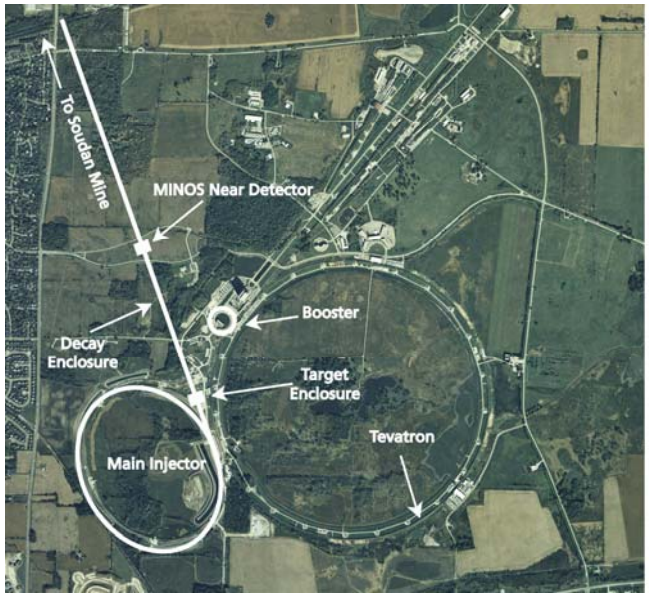
★ Measure ratio of the neutrino energy spectrum in far detector (oscillated) to that in the near detector (unoscillated)

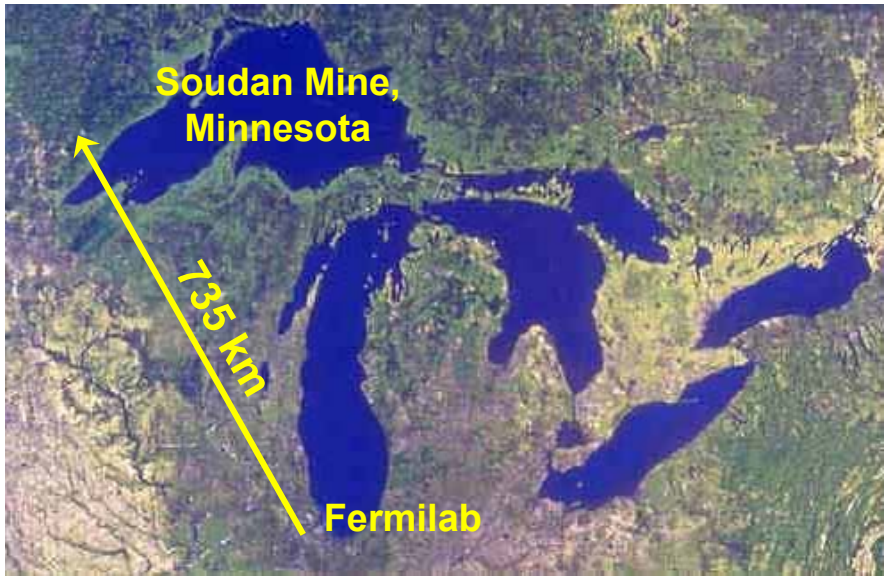
★ Partial cancellation of systematic biases



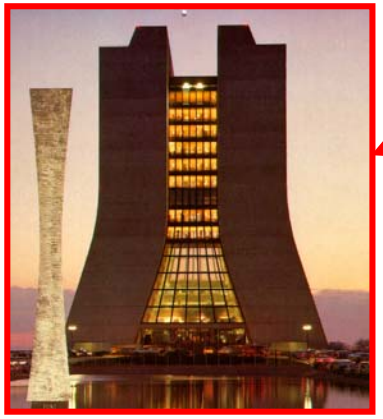
MINOS

- 120 GeV protons extracted from the MAIN INJECTOR at Fermilab (see p. 271)
- 2.5×10^{13} protons per pulse hit target ➡ very intense beam - 0.3 MW on target





Two detectors:



★ 1000 ton, NEAR Detector at Fermilab : **1 km** from beam

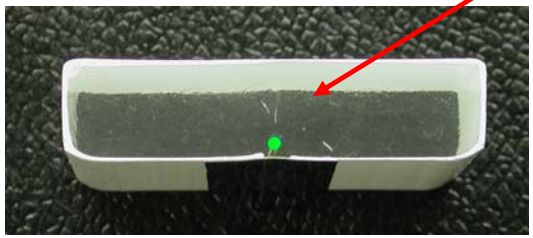
★ 5400 ton FAR Detector, 720m underground in Soudan mine, N. Minnesota: **735 km** from beam

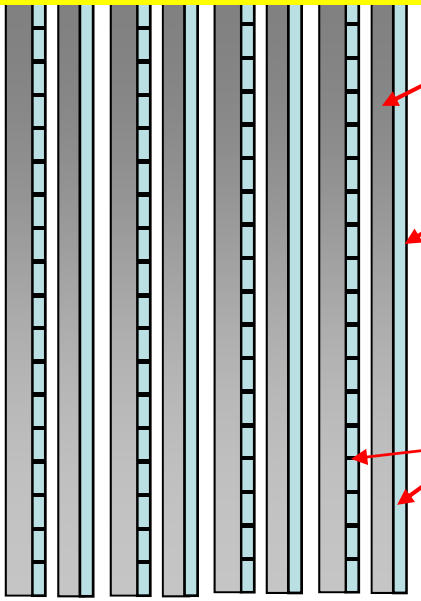


More like vN scattering experiments!!!

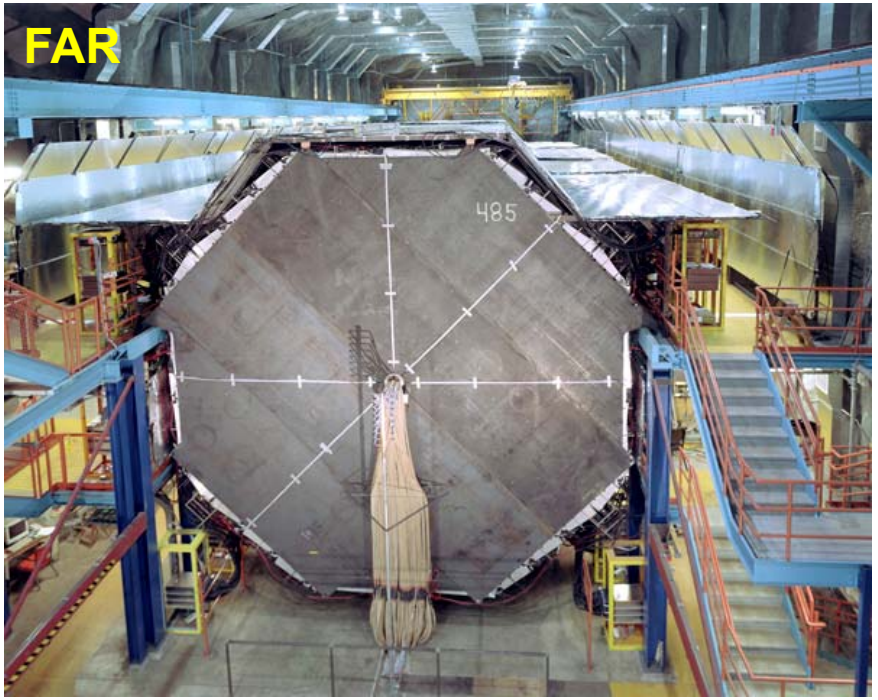
The MINOS Detectors:

- Dealing with high energy neutrinos $E_\nu > 1\,\mathrm{GeV}$
- The muons produced by ν_μ interactions travel several metres
- Steel-Scintillator sampling calorimeter
- Each plane: 2.54 cm steel +1 cm scintillator
- A charged particle crossing the scintillator produces light – detect with PMTs





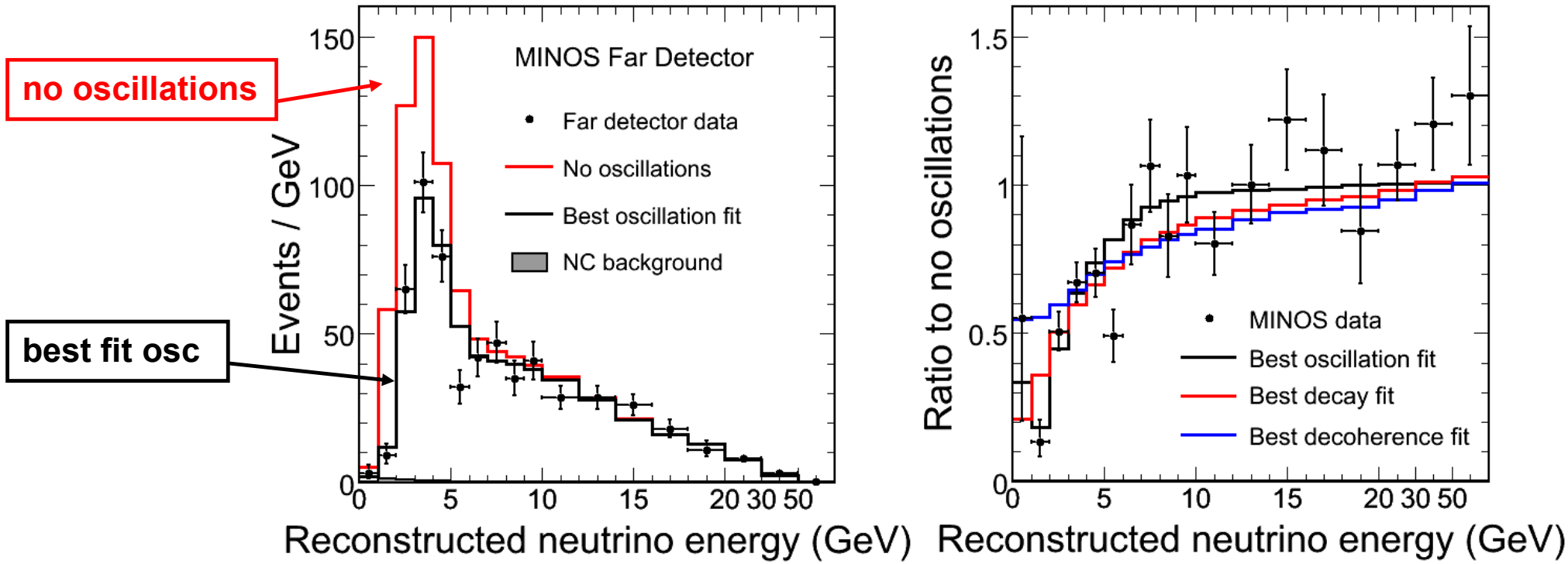




MINOS Results

- For the MINOS experiment L is fixed and observe oscillations as function of E_ν
- For $|\Delta m^2_{32}| \sim 2.5 \times 10^{-3}\,\mathrm{eV}^2$ first oscillation minimum at $E_\nu = 1.5\,\mathrm{GeV}$
- To a very good approximation can use two flavour formula as oscillations corresponding to $|\Delta m^2_{21}| \sim 8 \times 10^{-5}\,\mathrm{eV}^2$ occur at $E_\nu = 50\,\mathrm{MeV}$, beam contains very few neutrinos at this energy + well below detection threshold

MINOS Collaboration, Phys. Rev. Lett. 101, 131802, 2008



$$|\Delta m^2_{32}| = (2.43 \pm 0.12) \times 10^{-3}\,\mathrm{eV}^2$$

Exercise

- Using this formula:

- Calculate
  sin Δ21

$$\Delta_{21} = 1.27 \frac{\Delta m^2_{21}(\mathrm{eV}^2) L(\mathrm{km})}{E(\mathrm{GeV})}$$

For Dune L = 1300 km, E=10 GEV $\Delta m^2_{21} \sim 8 \times 10^{-5}\,\mathrm{eV}^2$

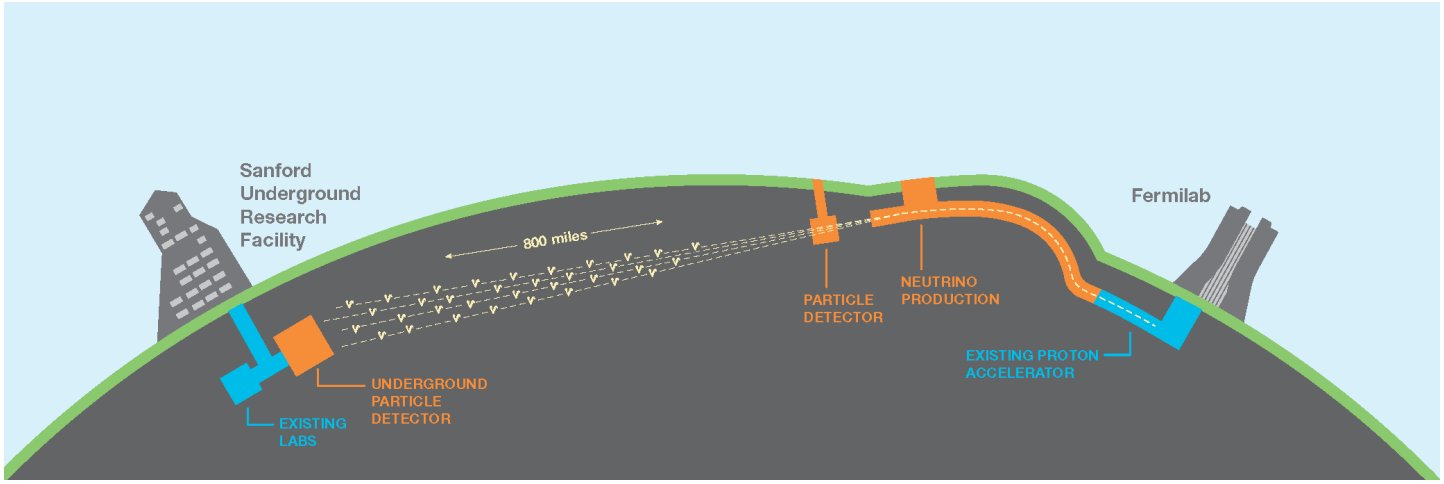
sin 1.27*8e-5*1300/10

~1.3%

Prob will be square of that ~1e-4 so very small

E will be a range but want most lower than 10→peak for MINOS like 2-3 GeV (*~10-25 x higher sin^2)

BUT Dune more interested in Δ23/Δ13 Δm^2 ~ 20 times higher



DUNE: The Deep Underground Neutrino Experiment

Mark Thomson

University of Cambridge & co-spokesperson of DUNE

The guy who wrote the textbook slides I'm using a lot from!
(I thought he was a theorist but I guess this also explains a lot!)

Heidelberg Physics Colloquium: 10th December 2015

UNIVERSITY OF CAMBRIDGE

DUNE

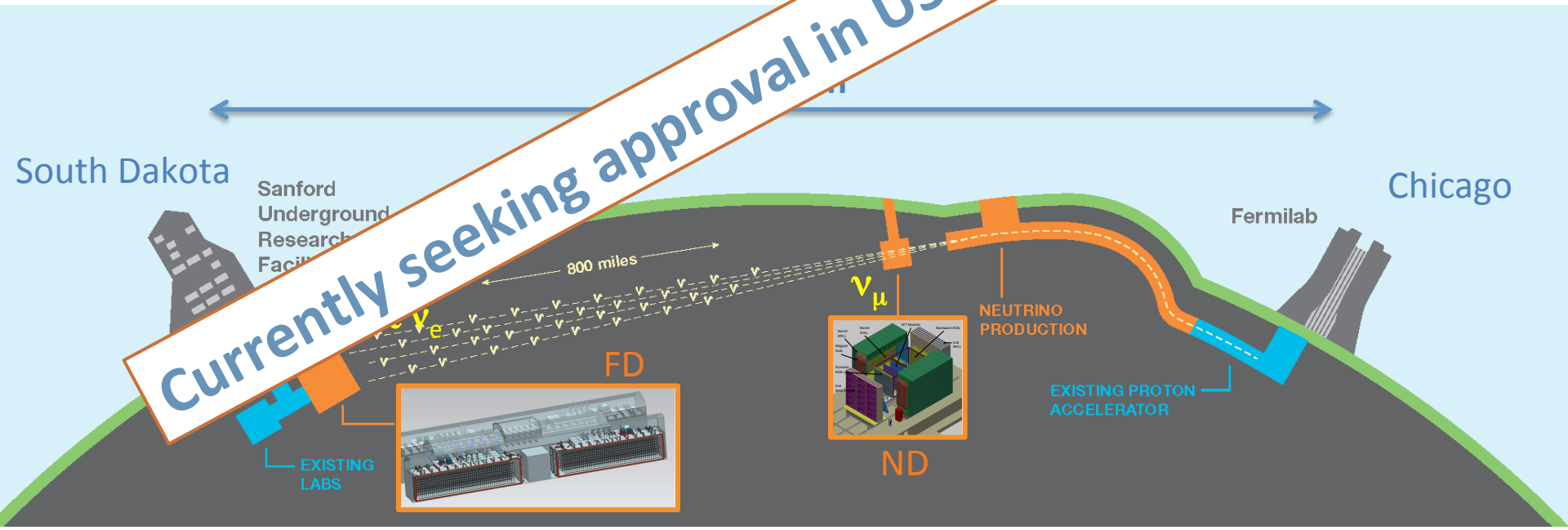
This talk

- 1. Introduction (1 slide)
- 2. Neutrino Basics: What are neutrinos? (5 slides)
  - and why they are so interesting
- 3. Neutrino oscillations – quantum mechanics in action (5 slides)
  - how we study neutrino properties
- 4. DUNE (4 slides)
- 5. DUNE Scientific Aims (6 slides)
  - what we intend to measure and why it matters
- 6. The LBNF/DUNE project (9 slides)
  - the accelerator/infrastructure (LBNF) and the detectors (DUNE)
- 7. The DUNE collaboration (3 slides)
- 8. Political Context (1 slide)
- 9. Opportunities (1 slide)
- 10. Summary (1 slide)

1. Introduction: What is DUNE?

- **DUNE will consist of**
  - **A powerful (MW) neutrino beam fired from Ferm**
  - **A massive (40000 t) underground detecto omestake mine, South Dakota**
  - **A large international collaborati ntly 800 scientists)**

Currently seeking approval in US: first beam 2025



The 2012 Revolution

★ **Two major discoveries in particle physics**

- A SM-like Higgs boson (ATLAS, CMS)
  - The key to EWSB and a possible window to the BSM world
- $\theta_{13} \sim 10^{o}$ (T2K, MINOS, Daya Bay, RENO)

  - about as large as it could have been !
  - The door to CP Violation in the leptonic sector

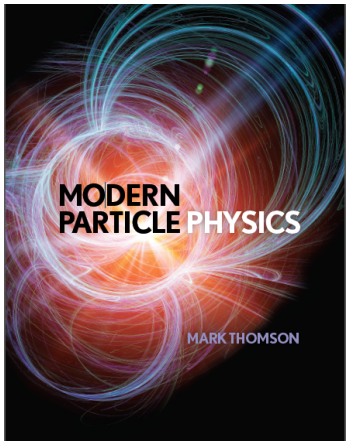
The 2012 Revolution

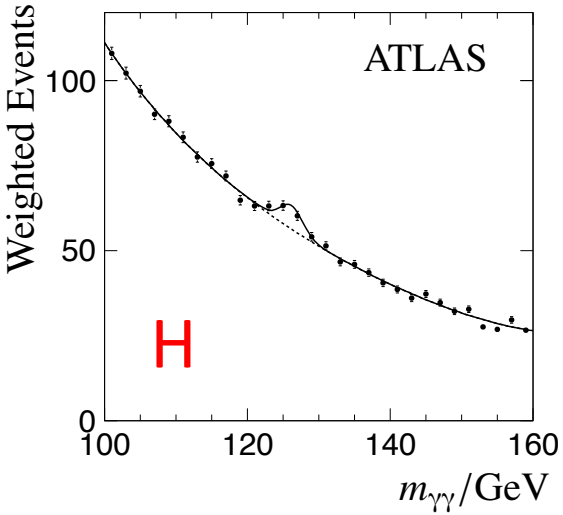
★ **Two major discoveries in particle physics**

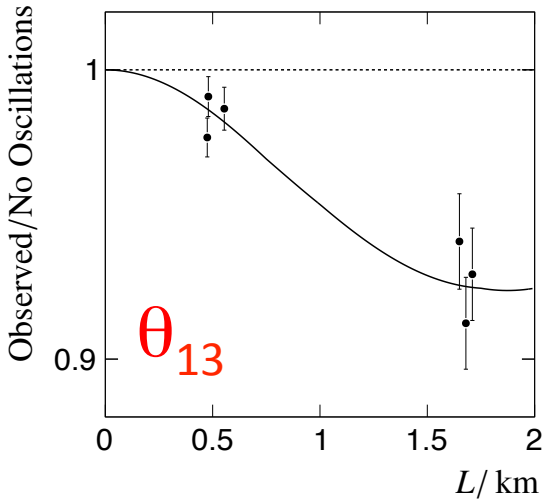
- A SM-like Higgs boson (ATLAS, CMS)
  - The key to EWSB and a possible window to the BSM world
- $\theta_{13} \sim 10^{o}$: determines rate of $P(\nu_\mu \rightarrow \nu_e)$
  - about as large as it could have been !
  - The door to CP Violation in neutrino oscillations

★ **Now standard textbook physics***

- **also** defines the next steps: ⇨ **DUNE**

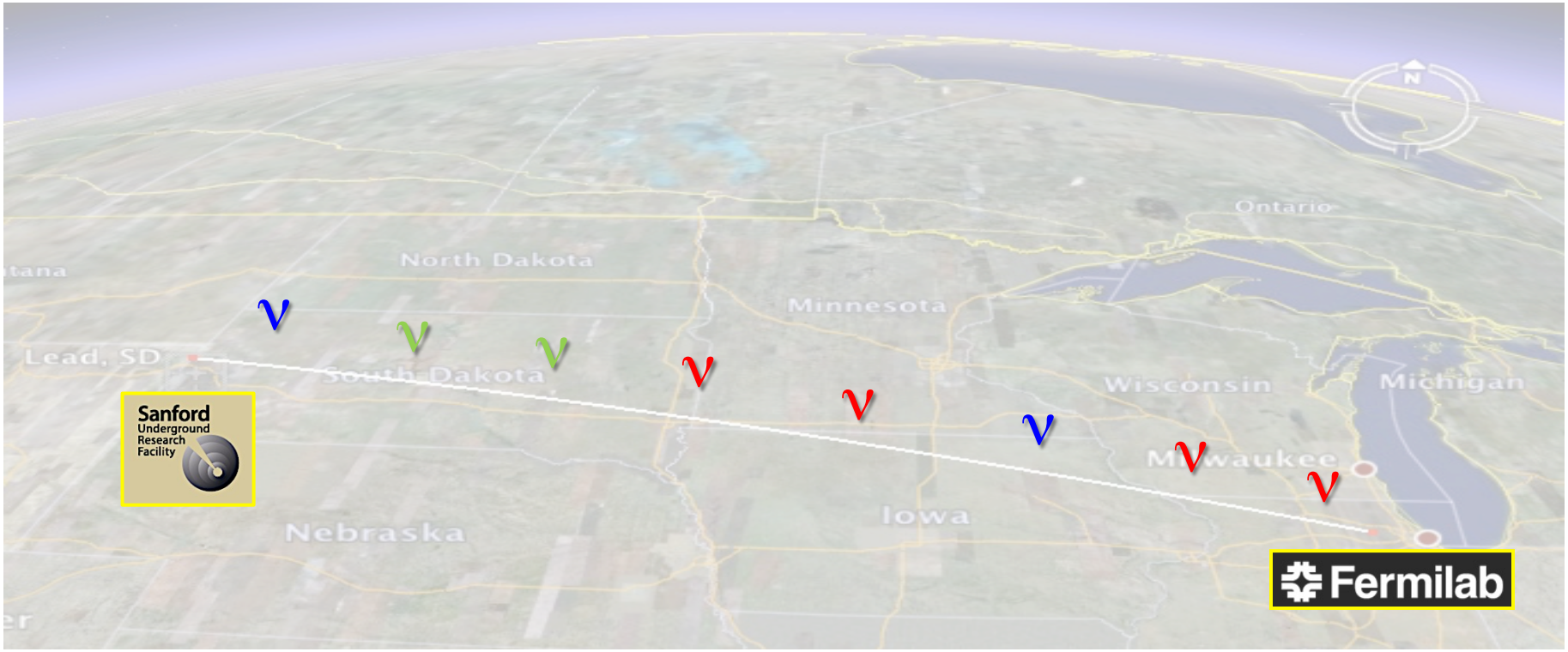






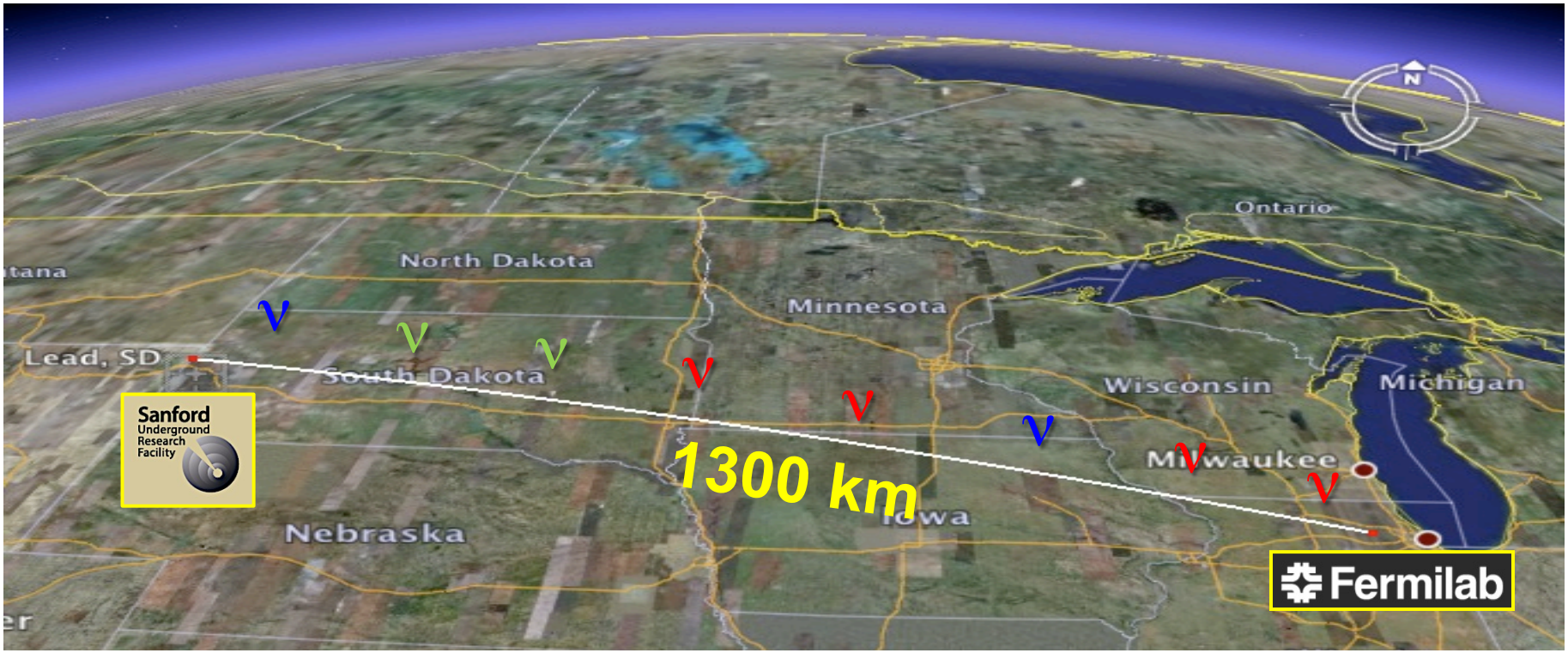
*apologies for gratuitous plug

4. DUNE


Ontario
North Dakota
Minnesota
Lead, SD
South Dakota
Wisconsin
Michigan
Milwaukee
Iowa
Nebraska
Sanford
Underground
Research
Facility
Fermilab

DUNE in a Nutshell

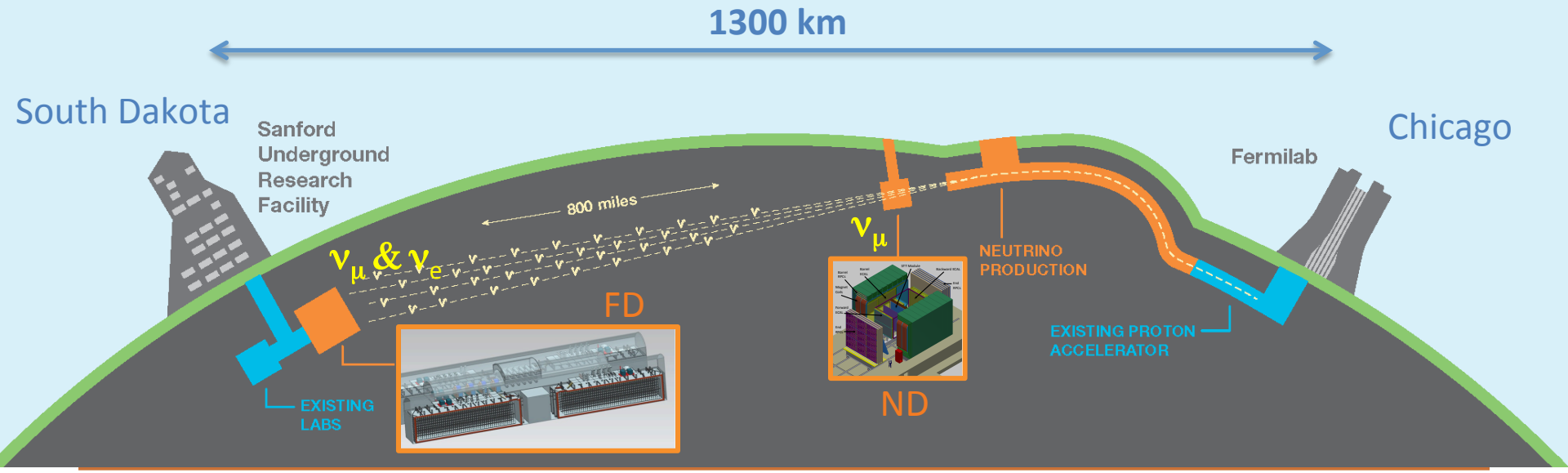
- ★ Intense beam of ν_μ or $\bar{\nu}_\mu$ fired 1300 km at a large detector
- ★ Compare $\nu_\mu \rightarrow \nu_e$ and $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ oscillations
- ★ Probe fundamental differences between matter & antimatter



DUNE in a Larger Nutshell

★LBNF/DUNE

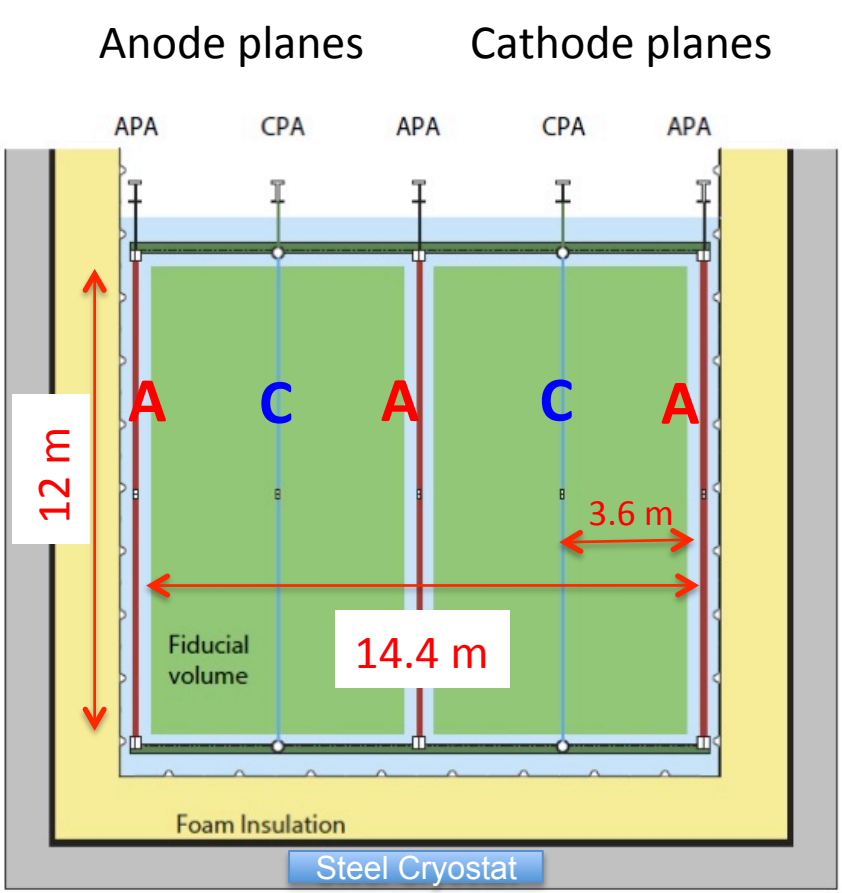
- Muon neutrinos/antineutrinos from high-power proton beam
  - **1.2 MW** from day one (upgradeable)
- Large underground **Liquid Argon Time Projection Chamber**
  - **4 x 17 kton ⇨ fiducial (useable) mass of >40 kton**
- Near detector to characterize the beam

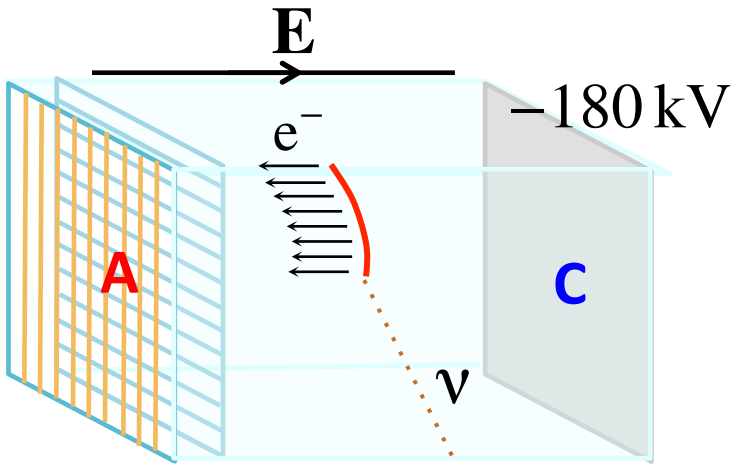


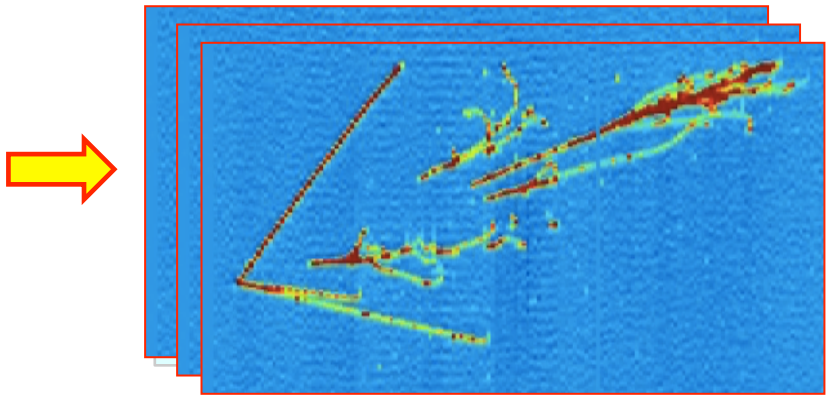
Liquid Argon TPC Basics

A **modular** implementation of Single-Phase TPC

- **Record ionization in LAr volume ⇨ 3D image**



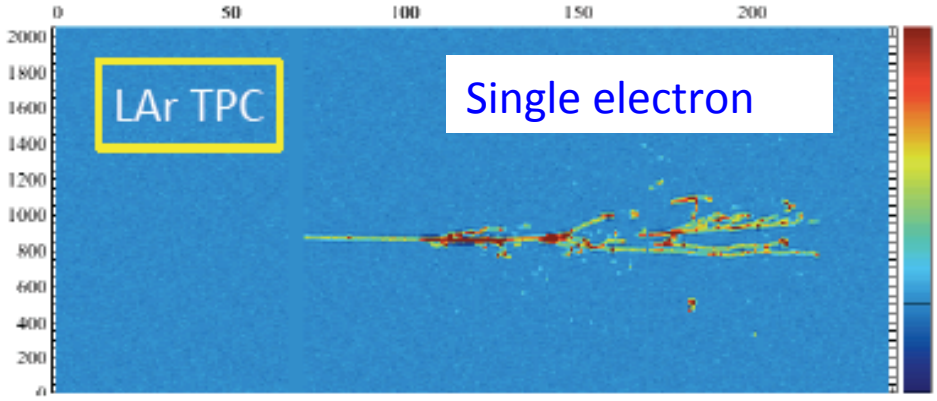


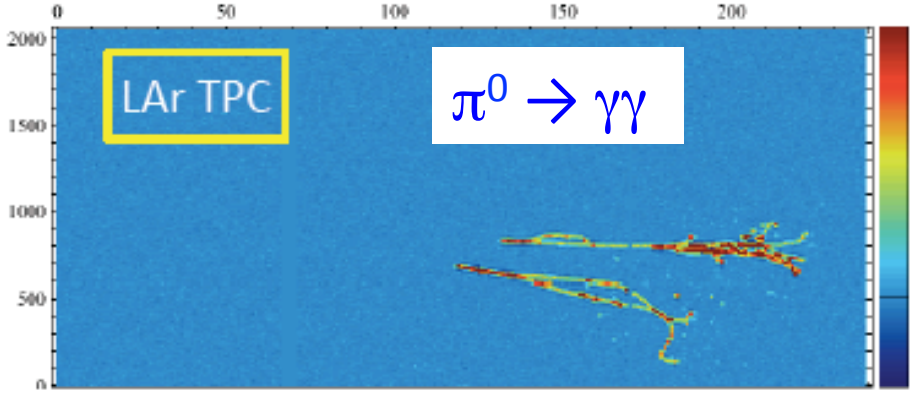


Benefits of an imaging detector

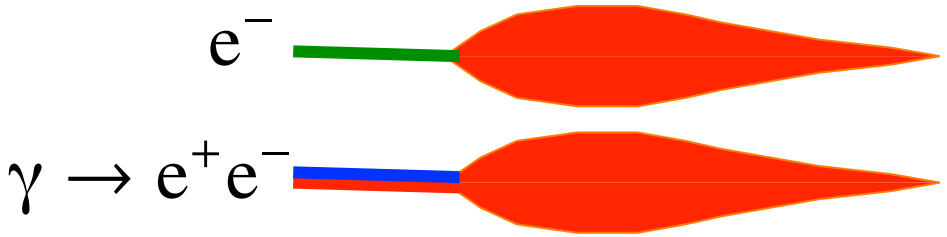
e.g. for electron neutrino appearance $e^{\pm} \leftrightarrow \gamma$ separation is vital

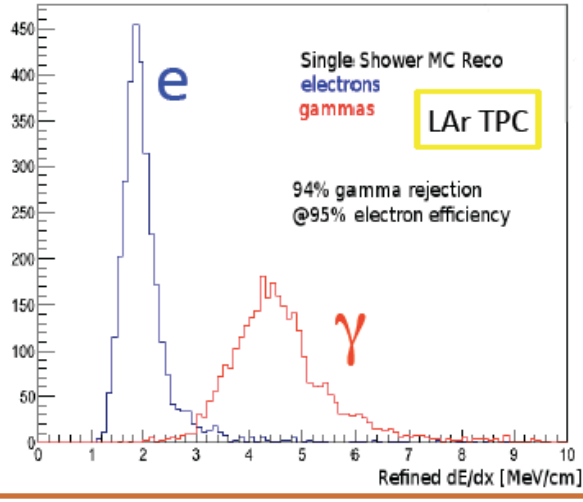
★ True for both photons from $\pi^0 \rightarrow \gamma\gamma$ or single photons





- Calorimetry to tag electrons/ gammas using dE/dx before EM shower evolves

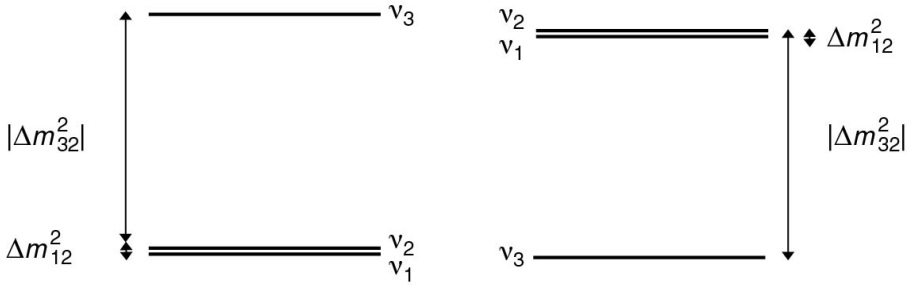




DUNE Primary Science Program

Focus on fundamental open questions in particle physics and astroparticle physics:

- **1) Neutrino Oscillation Physics**
  - **Discover CP Violation** in the leptonic sector
  - Mass Hierarchy
  - Precision Oscillation Physics:
    - e.g. parameter measurement, θ_{23} octant, testing the 3-flavor paradigm



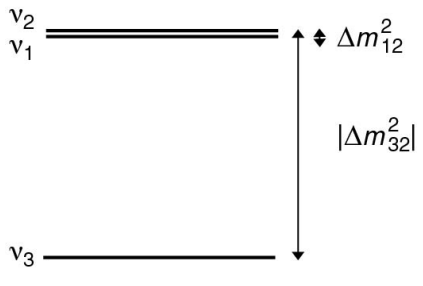
- **2) Nucleon Decay**
  - e.g. targeting SUSY-favored modes, $p \rightarrow K^+ \bar{\nu}$
- **3) Supernova burst physics & astrophysics**
  - Galactic core collapse supernova, sensitivity to ν_e

DUNE Primary Science Program

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  - Precision Oscill
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- **2) N cay**
  - eting SUSY-favored modes, $p \rightarrow K^{+}\bar{\nu}$
- **3) Supernova burst physics & astrophysics**
  - Galactic core collapse supernova, sensitivity to ν_e

All would be major discoveries

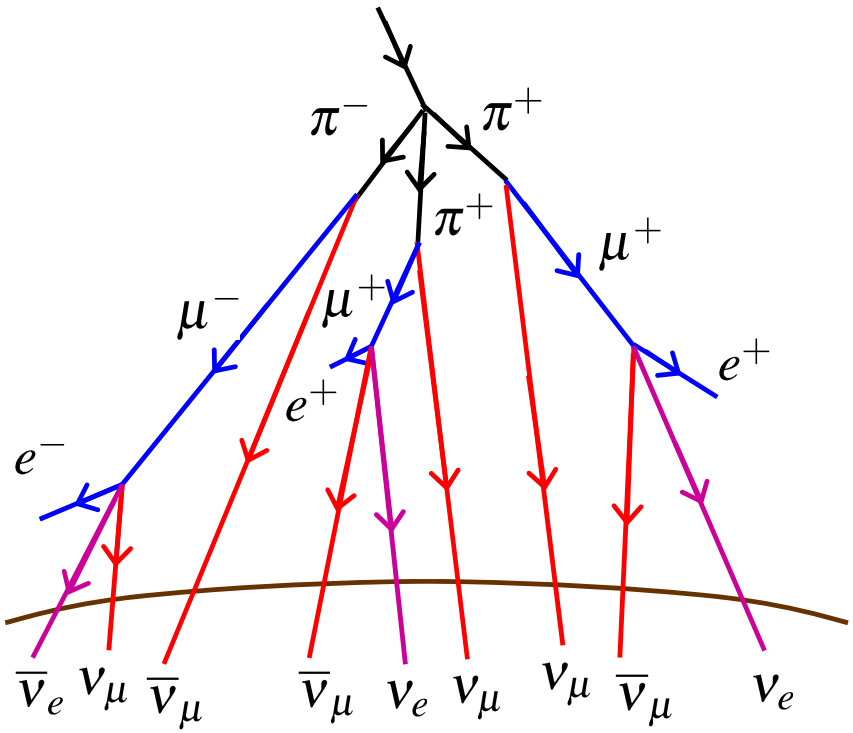


Addendums: Neutrino Oscillations

- Beam / "Long Baseline" experiments
  - Near detector
  - Far Detector
- Check slides on DuNE
  - → Biggest single item in DOE HEP budget!!! ($300 M)
- Oscillations only constrain mass differences…need other experiments for absolute mass
- Room for an extra generation ? (sterile neutrino – Beyond Standard Model) BSM

Examinable Appendix: Atmospheric Neutrinos

- **High energy cosmic rays (up to 10^{20} eV) interact in the upper part of the Earth's atmosphere**
- **The cosmic rays (~86% protons, 11% He Nuclei, ~1% heavier nuclei, 2% electrons) mostly interact hadronically giving showers of mesons (mainly pions)**



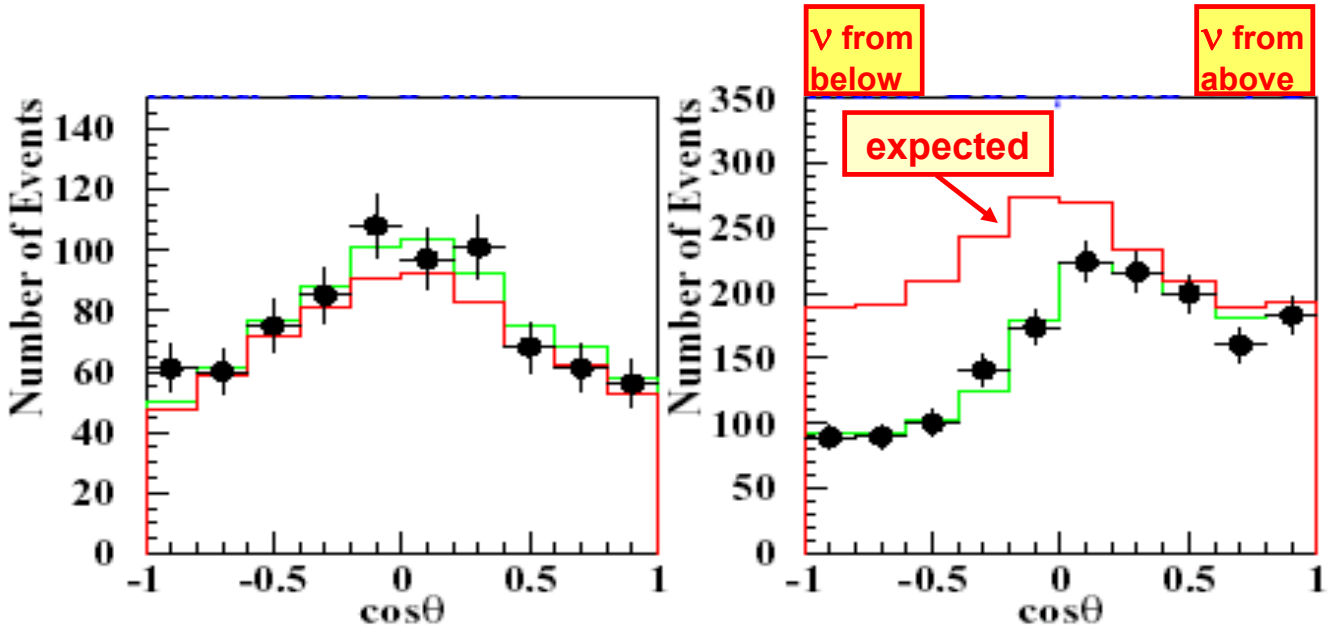
- **Neutrinos produced by:**

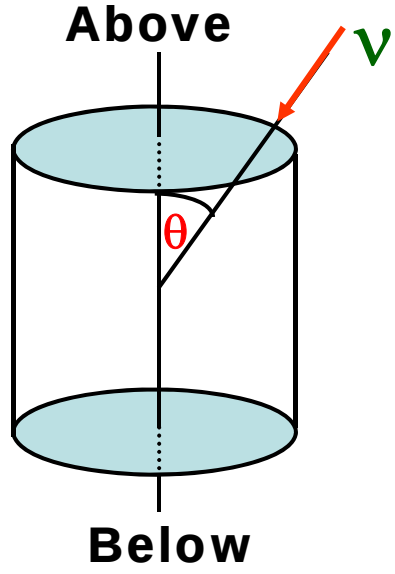
$$\pi^+ \to \mu^+ \nu_\mu \quad \hookrightarrow e^+ \nu_e \overline{\nu}_\mu \qquad \pi^- \to \mu^- \overline{\nu}_\mu \quad \hookrightarrow e^- \overline{\nu}_e \nu_\mu$$

- **Flux** $\sim 1\,\mathrm{cm}^{-2}\mathrm{sr}^{-1}\mathrm{s}^{-1}$
- **Typical energy :** $\mathrm{E}_\nu \sim 1\,\mathrm{GeV}$
- **Expect** $\dfrac{N(\nu_\mu + \overline{\nu}_\mu)}{N(\nu_e + \overline{\nu}_e)} \approx 2$
- **Observe a lower ratio with deficit of $\nu_\mu/\overline{\nu}_\mu$ coming from below the horizon, i.e. large distance from production point on other side of the Earth**

Super Kamiokande Atmospheric Results

- Typical energy: $E_\nu \sim 1\,\mathrm{GeV}$ (much greater than solar neutrinos – no confusion)
- Identify ν_e and ν_μ interactions from nature of Čerenkov rings
- Measure rate as a function of angle with respect to local vertical
- Neutrinos coming from above travel **~20 km**
- Neutrinos coming from below (i.e. other side of the Earth) travel **~12800 km**





★ Prediction for ν_e rate agrees with data

★ Strong evidence for disappearance of ν_μ for large distances

★ Consistent with $\nu_\mu \rightarrow \nu_\tau$ oscillations

★ Don't detect the oscillated ν_τ as typically below interaction threshold of **3.5 GeV**

3) Reactor Experiments

- To explain reactor neutrino experiments we need the full three neutrino expression for the electron neutrino survival probability (11) which depends on U_{e1}, U_{e2}, U_{e3}
- Substituting these PMNS matrix elements in Equation (11):

$$\begin{aligned} P(\nu_e \to \nu_e) &\approx 1 - 4U_{e1}^2 U_{e2}^2 \sin^2\Delta_{21} - 4(1-U_{e3}^2)U_{e3}^2 \sin^2\Delta_{32} \\ &= 1 - 4(c_{12}c_{13})^2 (s_{12}c_{13})^2 \sin^2\Delta_{21} - 4(1-s_{13}^2)s_{13}^2 \sin^2\Delta_{32} \\ &= 1 - c_{13}^4 (2s_{12}c_{12})^2 \sin^2\Delta_{21} - (2c_{13}s_{13})^2 \sin^2\Delta_{32} \\ &= 1 - \cos^4\theta_{13} \sin^2 2\theta_{12} \sin^2\Delta_{21} - \sin^2 2\theta_{13} \sin^2\Delta_{32} \end{aligned}$$

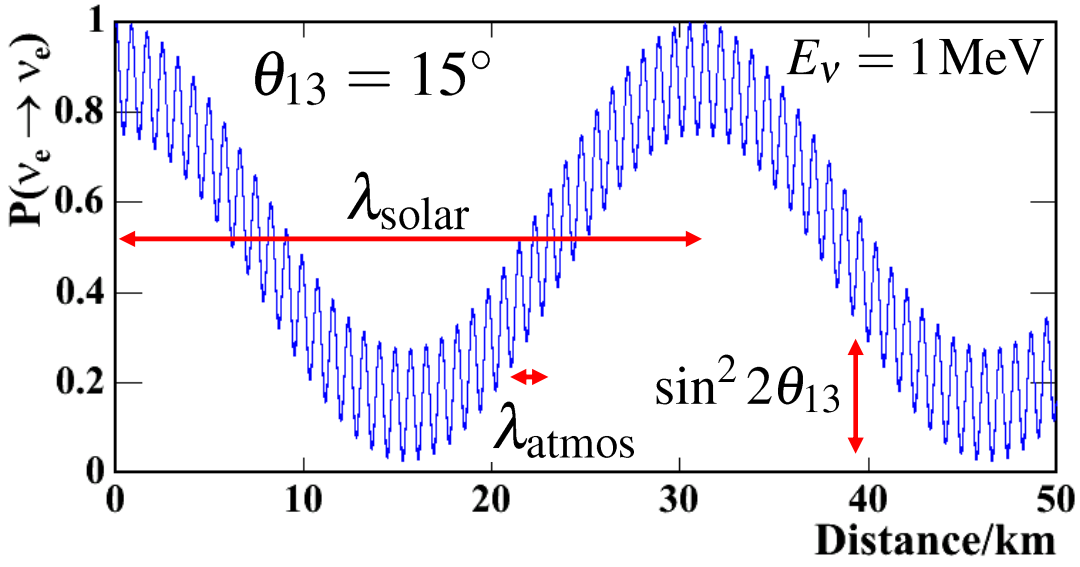
- Contributions with short wavelength (atmospheric) and long wavelength (solar)
- For a 1 MeV neutrino

$$\lambda_{\rm osc}({\rm km}) = 2.47 \frac{E({\rm GeV})}{\Delta m^2({\rm eV}^2)}$$

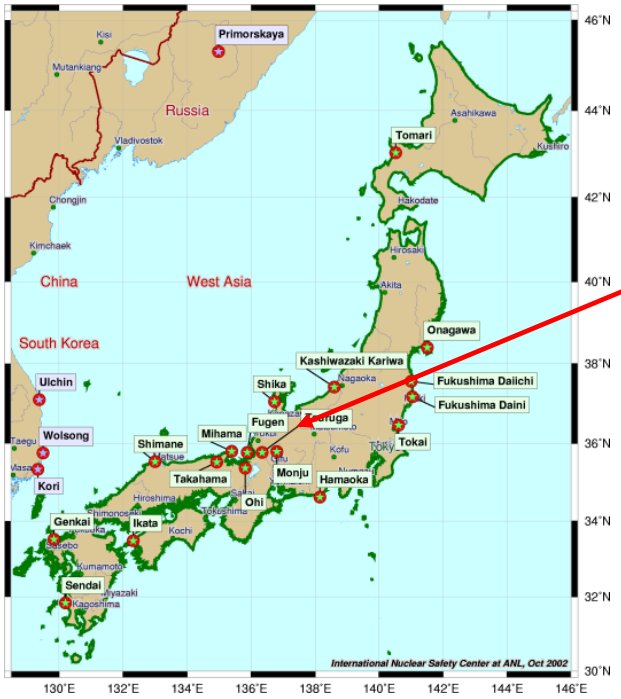
$$\Rightarrow \quad \lambda_{21} = 30.0\,{\rm km}$$

$$\lambda_{32} = 0.8\,{\rm km}$$

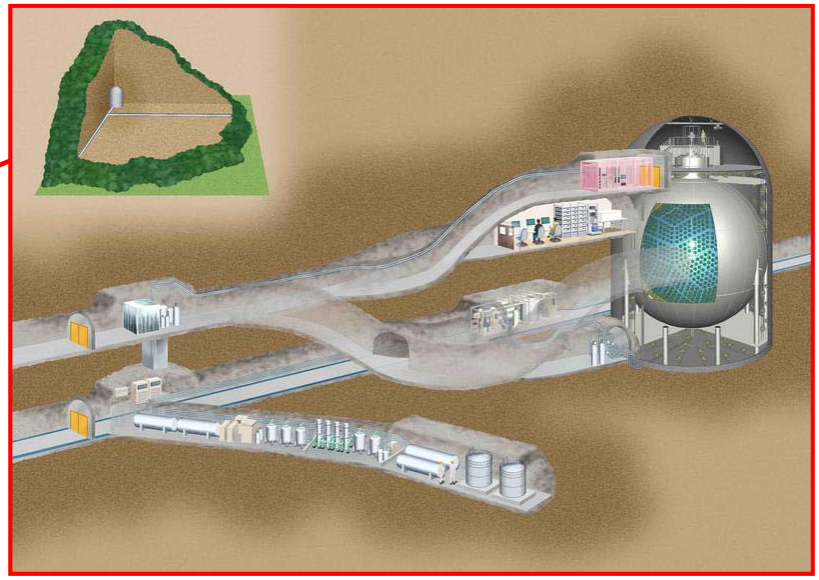
- Amplitude of short wavelength oscillations given by $\sin^2 2\theta_{13}$



Reactor Experiments II : KamLAND



- Detector located in same mine as Super Kamiokande



- 70 GW from nuclear power (7% of World total) from reactors within **130-240 km**
- Liquid scintillator detector, 1789 PMTs
- Detection via inverse beta decay: $\nu_e + p \rightarrow e^+ + n$

  Followed by $e^+ + e^- \rightarrow \gamma + \gamma$ prompt

  $n + p \rightarrow d + \gamma(2.2\,\mathrm{MeV})$ delayed

Next

- Weak Interaction
  - Already covered: Flavor structure and CKM Mixing (long ago)
  - Already covered: Basics of Feynman charge current scattering situations (F3) (last time)
  - (again) Back to CKM Neutrino Oscillations
  - **Weak CC (Charge Current) in DECAYS → Muon and Beta Decay**
  - Z boson "Neutral Current" Interactions
  - More on Chiral fermion states Weak Isospin & Electroweak Unification 1
  - Local Gauge Invariance, Spontaneous Symm Breaking and Higgs Mechanism Electroweak Unification 2

We've already mentionned several aspects of Weak Interactions charge current (W vertex) ...

More so lepton+lepton (ie ν + e or μ)

For q+q weak CC decays, we've only discussed the CKM aspect... look at Muon Decay / Beta Decay

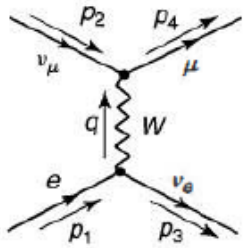
Muon Decay – Beta Decay

- Back to Weak Interaction
- MUON decay → Same diagram as mu+nu→ nu + e

Example 9.1 *Inverse Muon Decay* Consider the process

$$\nu_\mu + e^- \to \mu^- + \nu_e$$

represented (in lowest order) by the diagram



$$\Gamma(\mu^- \to e^- \bar{\nu}_e \nu_\mu) \equiv \frac{1}{\tau_\mu} = \frac{G_F^{(e)} G_F^{(\mu)} m_\mu^5}{192\pi^3}, \qquad (12.1)$$

Weak Interaction Feynman Rules

- Propagators

$$\frac{-i(g_{\mu\nu} - q_\mu q_\nu / M^2c^2)}{q^2 - M^2c^2} \quad \text{(propagator for } W \text{ and } Z\text{)}$$

$$\frac{ig_{\mu\nu}}{(Mc)^2} \quad \text{(propagator for } q^2 \ll (Mc)^2\text{)}$$

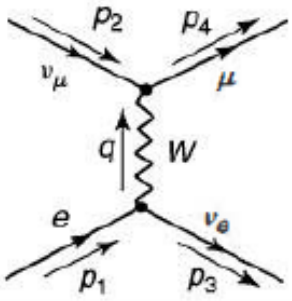
$$\frac{-ig_W}{2\sqrt{2}}\gamma^\mu(1-\gamma^5) \quad \text{(weak vertex factor)}$$

Only for CC, (non-quark) effectively

Example 9.1 *Inverse Muon Decay* Consider the process

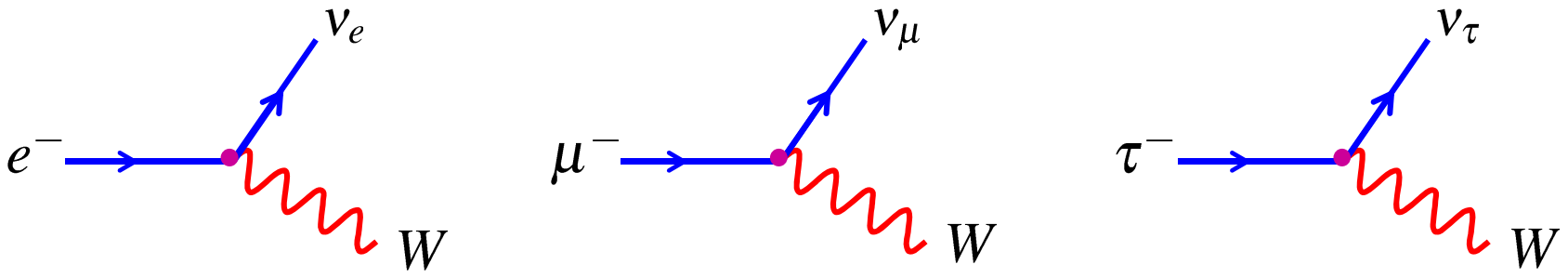
$$\nu_\mu + e^- \rightarrow \mu^- + \nu_e$$

represented (in lowest order) by the diagram

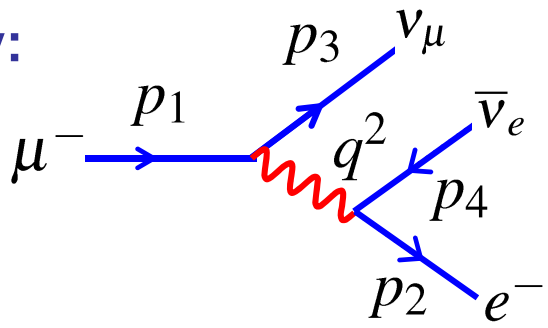


Muon Decay

★The leptonic charged current ($W^{\pm}$) interaction vertices are:



★Consider muon decay:



•It is straight-forward to write down the matrix element

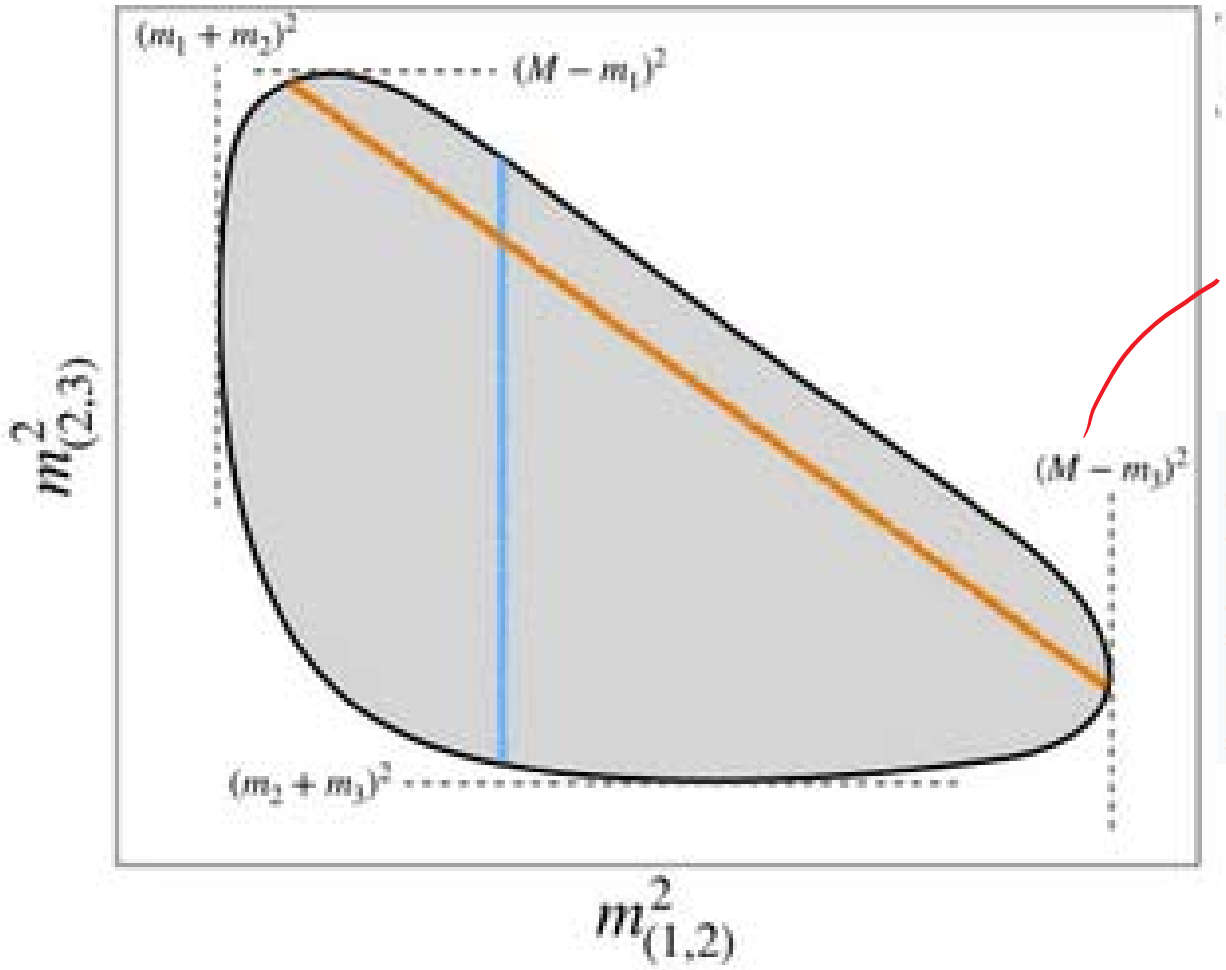
$$M_{fi} = \frac{g_W^{(e)} g_W^{(\mu)}}{8m_W^2}[\overline{u}(p_3)\gamma^\mu(1-\gamma^5)u(p_1)]g_{\mu\nu}[\overline{u}(p_2)\gamma^\nu(1-\gamma^5)v(p_4)]$$

Note: for lepton decay $q^2 \ll m_W^2$ so propagator is a constant $1/m_W^2$
i.e. in limit of Fermi theory

•Its evaluation and subsequent treatment of a three-body decay is rather tricky (and not particularly interesting). Here will simply quote the result

Muon and Neutron Beta DECAY 3=Boday

- 3-Body Decay - Relativistic case: Dalitz plots
- **Very important experimentally generally for n>3 decay products**



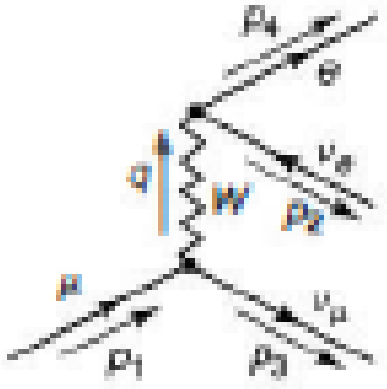
$$\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$$

M m_2 m_1 m_3

Dalitz plot for a three-body decay of a spin-0 particle of the mass M into three spin-0 particles of masses m_1, m_2, m_3. The grey area depicts the allowed kinematic region. The blue line shows a possible position of accumulation of events in case a spin-0 resonance is present as an intermediate state in this three-body decay, which then decays to particles 1 and 2. The orange line shows the position of accumulation of events in case another spin-0 resonance is present, decaying to particles 1

3-Body Decay

different golden rule



$$d\Gamma = \frac{(|\mathcal{M}|^2)}{2\hbar m_\mu}\left(\frac{d^3\mathbf{p}_2}{(2\pi)^3 2|\mathbf{p}_2|}\right)\left(\frac{d^3\mathbf{p}_3}{(2\pi)^3 2|\mathbf{p}_3|}\right)\left(\frac{d^3\mathbf{p}_4}{(2\pi)^3 2|\mathbf{p}_4|}\right)$$

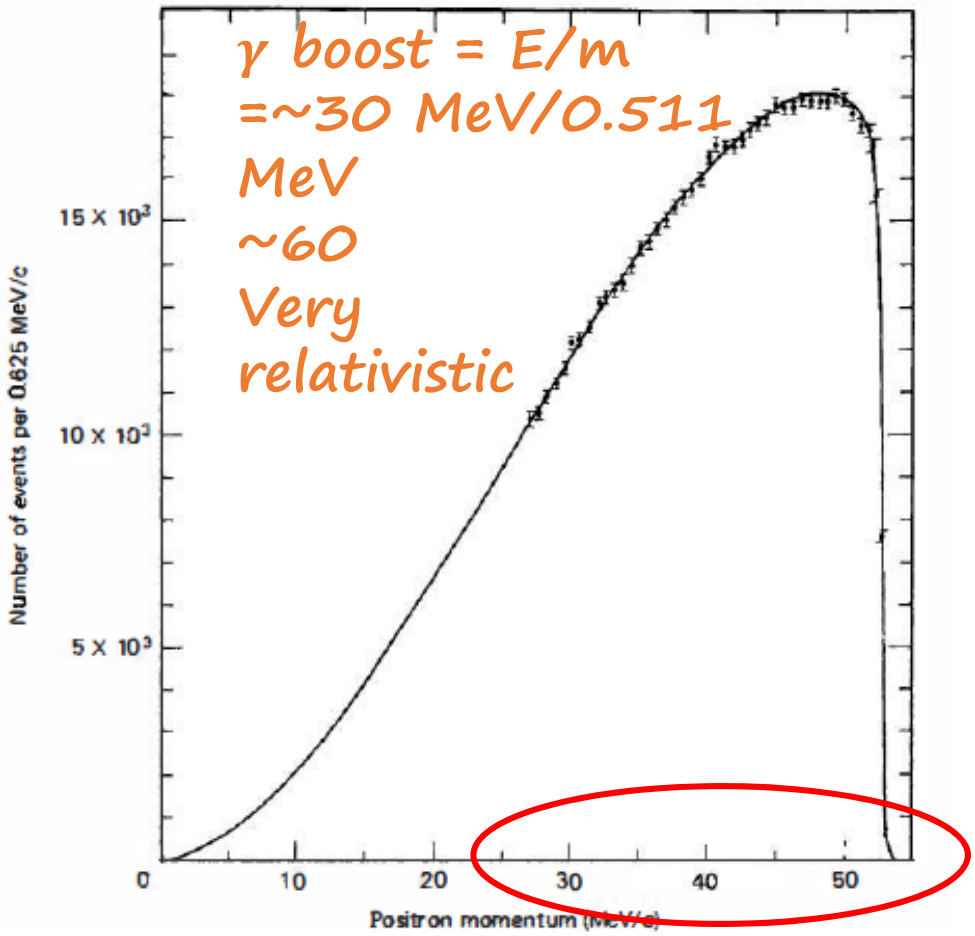
$$\times (2\pi)^4 \delta^4(p_1 - p_2 - p_3 - p_4)$$

- ~Still relativistic
- Projections

(and 0 otherwise) – which is to say (Problem 9.4),

$$\left\{\begin{array}{r} |\mathbf{p}_2| < \tfrac{1}{2} m_\mu c \\ |\mathbf{p}_4| < \tfrac{1}{2} m_\mu c \\ (|\mathbf{p}_2| + |\mathbf{p}_4|) > \tfrac{1}{2} m_\mu c \end{array}\right\} \qquad (9.30)$$

phase space defines boundaries e..g on Dalitz plots



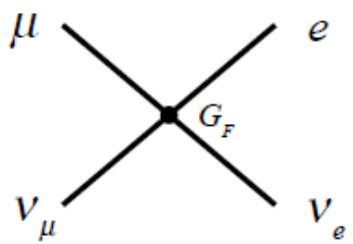
Griff Calc for Mu decay

Results of muon decay rate calculation:

$$\Gamma = \left(\frac{m_\mu g_w}{M_W}\right)^4 \frac{m_\mu c^2}{12\hbar(8\pi)^3} \quad \Rightarrow \quad \tau_\mu = \frac{1}{\Gamma} = \left(\frac{M_W}{m_\mu g_w}\right)^4 \frac{12\hbar(8\pi)^3}{m_\mu c^2}$$

Note that g_w and M_W do not appear separately, only as a ratio. The original β-decay theory of Fermi expressed this process as a "contact" interaction with strength:

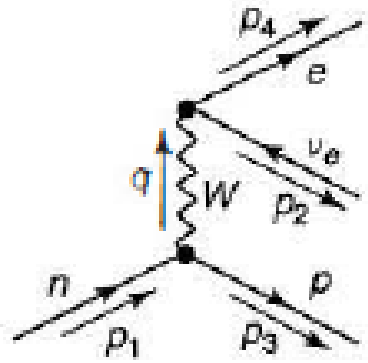
$$G_F = \frac{\sqrt{2}}{8}\left(\frac{g_w}{M_W c^2}\right)^2 (\hbar c)^3 \qquad \tau_\mu = \frac{192\pi^3\hbar^7}{G_F^2 m_\mu^5 c^4}$$



Putting in the observed muon mass and lifetime ➔ G_F = 1.166 x 10^{-5} GeV^{-2}.

neutron beta decay

- Griff same M
- ~s^2
- dsig ~s



really d to u quark

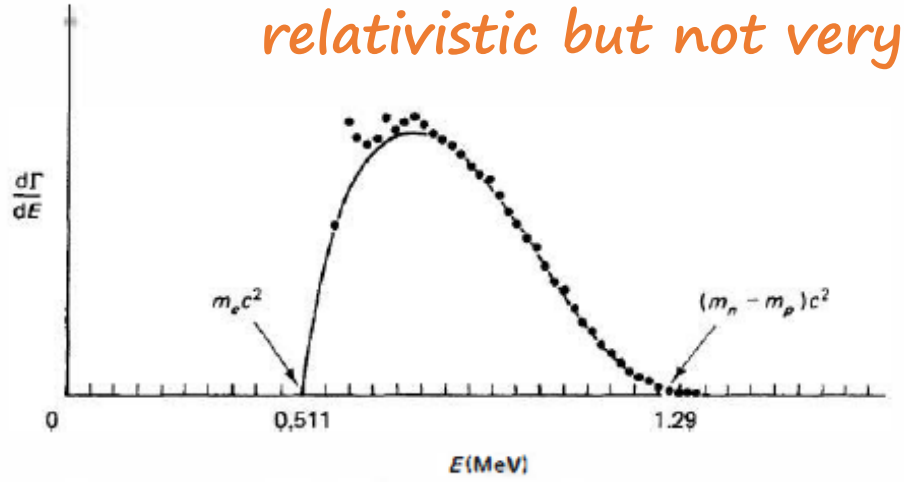
$$\langle |\mathscr{M}|^2 \rangle = 2\left(\frac{g_w}{M_w c}\right)^4 (p_1 \cdot p_2)(p_3 \cdot p_4) \qquad (9.41)$$

same as scattering s^2

$$\frac{d\Gamma}{dE} = \frac{1}{\pi^3 \hbar}\left(\frac{g_w}{2M_w c^2}\right)^4 E\sqrt{E^2 - m_e^2 c^4}\,[(m_n - m_p)c^2 - E]^2$$

derived in Griff! from scratch!

γ boost 1-3
relativistic but not very



$$\Gamma = \frac{1}{4\pi^3 \hbar}\left(\frac{g_w}{2M_w c^2}\right)^4 (m_e c^2)^5 \times \left[\frac{1}{15}(2a^4 - 9a^2 - 8)\sqrt{a^2 - 1} + a \ln(a + \sqrt{a^2 - 1})\right]$$

where

$$a \equiv \frac{m_n - m_p}{m_e}$$

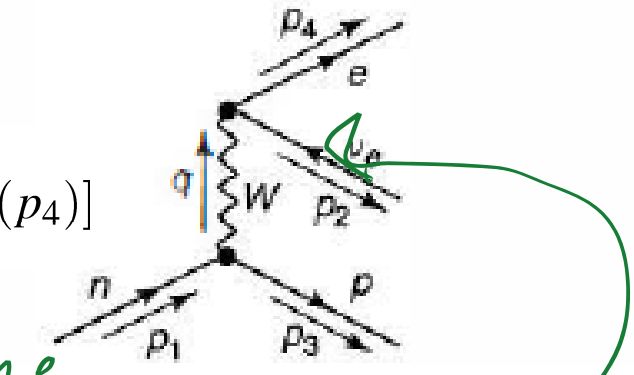
Putting in the numbers, I find (Problem 9.12)

$$\tau = \frac{1}{\Gamma} = 1318 \text{ s}$$

Accounting for u/d and nucleon structure effects

- Replace $1-\gamma^5$ in vertex

$$M_{fi} = \frac{g_W^{(e)} g_W^{(\mu)}}{8m_W^2}[\bar{u}(p_3)\gamma^\mu(1-\gamma^5)u(p_1)]g_{\mu\nu}[\bar{u}(p_2)\gamma^\nu(1-\gamma^5)v(p_4)]$$



This one

ok e, ν, W

→ $C_V \mathbb{1} - C_A \gamma 5$

$C_V^W = 1.000$ $\quad C_A^W = 1.27$

- Improves lifetime calc to 950s (real lifetime 880s)
  - (also added CKM factors)
- Note these c_A,cv factors are for CC W+/- diagram –later cA/cV Z boson

Real Nuclei- Mann

- nuclear beta decay

$$\mathrm{Cs}^{136} \longrightarrow \mathrm{Ba}^{136} + e^- + \overline{\nu}_e$$

mass diff ~2 MeV

$$\mathrm{C}^{10} \longrightarrow \mathrm{B}^{10} + e^+ + \nu_e$$

mass diff ~3 MeV

The matrix element $\mathcal{M}$ is determined by what we postulate our theory of β-decay to be. Fermi's original idea was to take the square of the matrix element to be proportional to the energies of the emitted particles, i.e.

$$|\mathcal{M}|^2 = E'_e E'_\nu F(Z, E'_e) \, |\mathcal{M}_0|^2 \tag{19.9}$$

where an additional factor $F(Z, E)$ – called the *Coulomb factor* – has been included. It takes account of the energy lost (by electrons) or gained (by positrons) as they escape from the Coulomb field of the nucleus. Non-relativistically it is [189]

$$F(Z, E_e) = \frac{2\pi\eta}{1 - \exp[-2\pi\eta]} \quad \text{where} \quad \eta = \pm\frac{Ze^2}{4\pi\epsilon_0\hbar v} \tag{19.10}$$

for β particles of speed v emitted from a nucleus of charge Ze. Relativistically it is complicated to calculate, but can be done exactly.

Hence we find

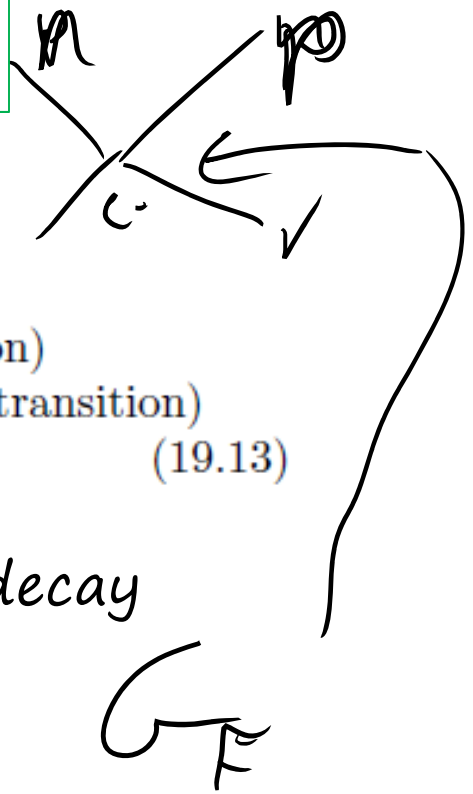
$$d\Gamma = \frac{c^5 \, |\mathcal{M}_0|^2}{(32\pi^3)\,\hbar M_N M'_N} p_e^2 \, (E_0 - E'_e) \, F(Z, E'_e) \sqrt{(E_0 - E'_e)^2 - m_\nu^2} \, dp'_e \tag{19.11}$$

Coloumb factor

part related to Feyman rules - M_0

Fermi's approach-ν mass

$$d\Gamma = \frac{c^5 |\mathcal{M}_0|^2}{(32\pi^3)\hbar M_N M'_N} p_e^2 (E_0 - E'_e) F(Z, E'_e) \sqrt{(E_0 - E'_e)^2 - m_\nu^2 c^4} dp'_e \quad (19.11)$$



- Before weak bosons were understood
- Direct interaction with this G_F

$$|\mathcal{M}_0|^2 = (M_p c^2)^2 \times \begin{cases} G_F^2 & \text{if } \Delta J = 0 \text{ (Fermi transition)} \\ 3G_F^2 & \text{if } \Delta J = 1 \text{ (Gamow-Teller transition)} \end{cases} \quad (19.13)$$

$$= \frac{g_W}{M_W}$$

- Alt way for neutrino mass → tritium beta decay

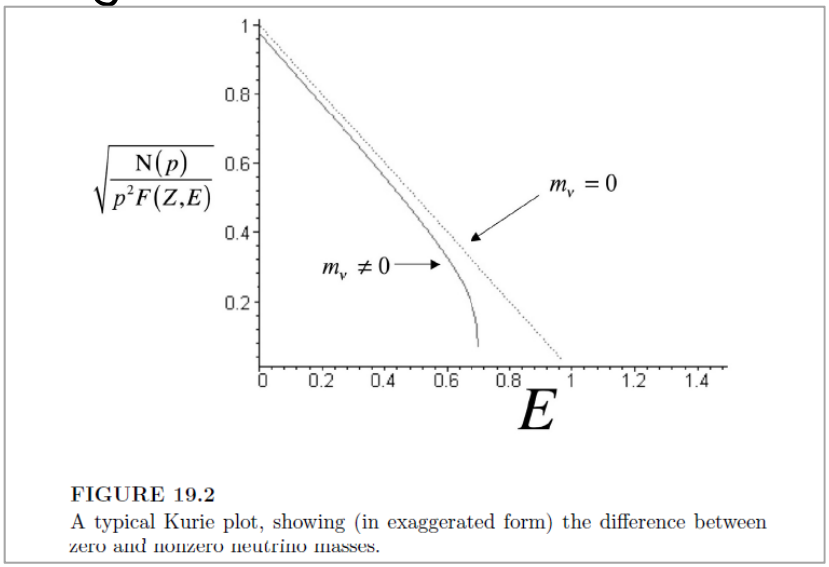


FIGURE 19.2
A typical Kurie plot, showing (in exaggerated form) the difference between zero and nonzero neutrino masses.

Role of Phase Space in Beta Decay

- Once embedded **into Nuclei** of course the neutron may NOT decay
- From a QFT perspective what causes this?
- Best Answer PHASE SPACE : → Qvalue determines available phase

$$\Gamma_{\text{free}} \propto \int |\mathcal{M}|^2 \, d\Phi_3$$

- $d\Phi_3$ = 3-body phase space

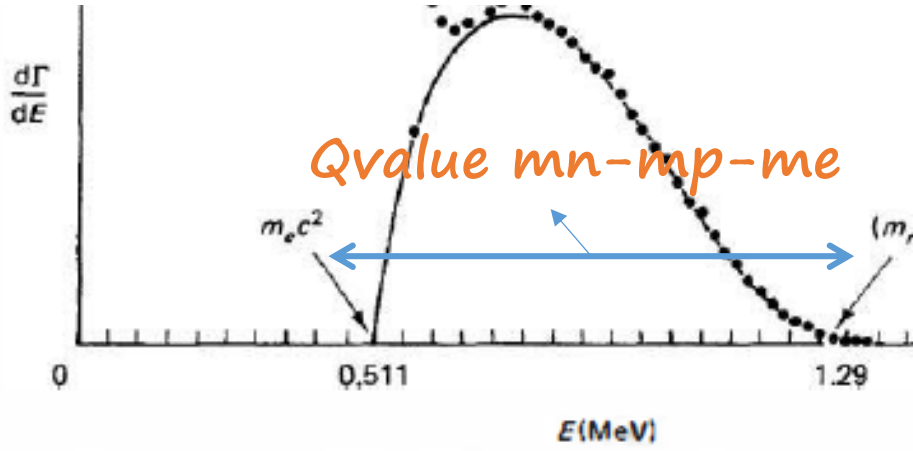
- Griff result (n→ p +e + ν):

$$\Gamma = \frac{1}{4\pi^3 \hbar} \left(\frac{g_w}{2M_W c^2} \right)^4 (m_e c^2)^5 \int \phi_3'$$

$$\Gamma_{\text{free}} \propto G_F^2 \, (Q_{\text{free}})^5$$

- →
- Nuclei: (N→ N-1 + e + ν)

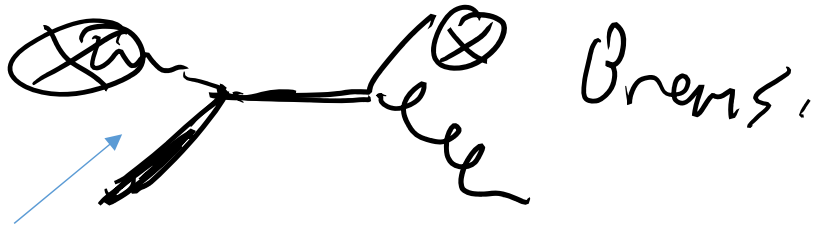
$$\Gamma_{\text{nucleus}} \propto G_F^2 \, |M_{\text{nuclear}}|^2 \, (Q_{\beta-})^5$$



So rate decreases as Q^5, if Q is negative it doesn't happen!

Independent "Fundamental" → Coherent QFT

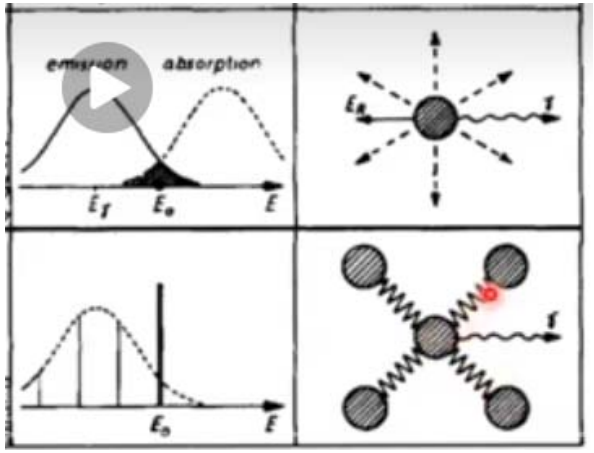
- Full understanding in Nuclei of Beta Decay:
- Needs Coupling of W boson to coherent collection of quarks/gluons in Nucleons & Nuclei
- Need different kind of Quantum Field Theory using multi particle Coherent states:
  - **General kinds CASES where need Experimental Inputs still !!!**



- Examples in this course so far:
  - Bremstrahlung ←→ Compton Feynman
  - Gluon Saturation – IN this case the Color Glass (CGC) Theory provides already derived theory example



- Ultimate example **Mossbauer Effect !**



- Nuclear Gamma Decay:
- Iron atomic crystal lattice coherently absorbs individual nucleus recoil !!!

Connects QED→Nuclear Structure→Atomic Physics

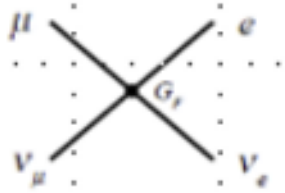
Exercise

- From leading order formula calculate the lifetime of the muon

$$\Gamma_\mu^{\mathrm{LO}} = \frac{G_F^2 m_\mu^5}{192\pi^3},$$

- Use GeV units
  - Try to motivate the value of GF

$$G_F = \frac{\sqrt{2}}{8}\left(\frac{g_w}{M_w c^2}\right)^2 (\hbar c)^3 \qquad \tau_\mu = \frac{192\pi^3\hbar^7}{G_F^2 m_\mu^5 c^4}$$



Putting in the observed muon mass and lifetime → G_F = 1.166 x 10^{-5} GeV^{-2}.

- You may need the right value for hbar

$$6.582119569 \times 10^{-25}\ \mathrm{GeV}\cdot\mathrm{s})$$

-Use uncert rel hbar = $\Delta E\ \Delta t$

Guide : Weak/Related Physics

- Weak Interaction
  - Already covered: Flavor structure and CKM Mixing (long ago)
  - Already covered: Basics of Feynman charge current scattering situations (F3) (last time)
  - (again) Back to CKM Neutrino Oscillations
  - Weak CC (Charge Current – W) in DECAYS → Muon and Beta Decay
  - Z boson "Neutral Current" Interactions
  - More on Chiral fermion states & Electroweak Unification Weak Isospin
- Local Gauge Invariance, Spontaneous Symm Breaking and Higgs Mechanism

cA, CV, gA gV gL, gR

- even with W one will in separate "parts" of the vertex factor

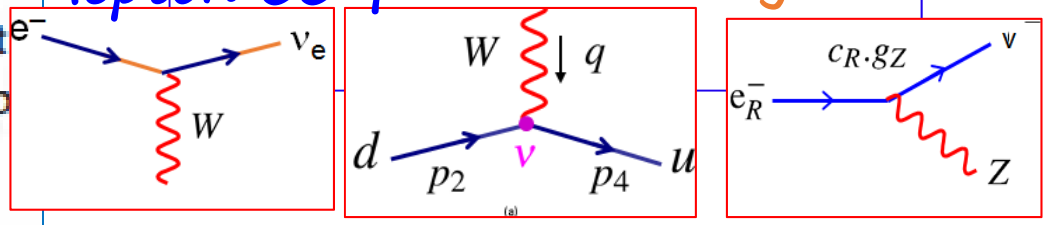
$$(1-\gamma^5) \rightarrow (c_V - c_A\gamma^5)$$

where c_V is the correction to the vector 'we
to the axial vector 'weak charge'. Luckily, the
occurs not only for the *free* neutron, but
have, in principle, many independent opp
experimental results are as follows:

$$c_V = 1.000, \quad c_A = 1.270 \pm 0.003$$

For W CA/CV= " ϵ "= 1,
but inside hadrons
"effectively" not for
quark-W vertices
For Z there are always NC

lepton CC quark CC all NC



never
$\epsilon \neq 1$

sometimes
(at lowE)
$\epsilon \neq 1$

ALWAYS

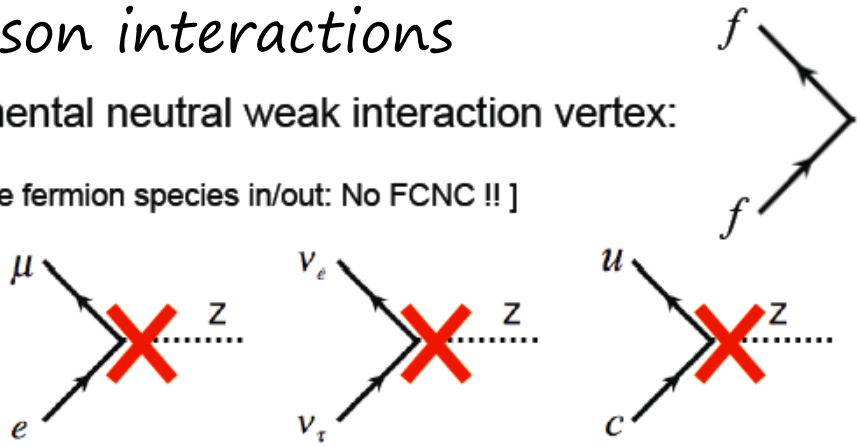
- These parameters are related to "Weak charge" or weak hypercharge (e.g. Qweak?)
  - in case of hadronic interactions: due to hadron structure/weak structure of hadrons (NonPerturb)
- **But for Z boson interactions this in herent in the fundamental vertex couplings**

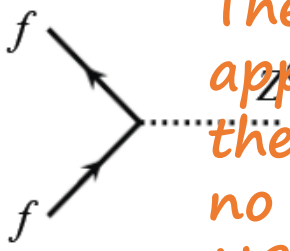
Z-boson Interactions

- Z-boson interactions

Fundamental neutral weak interaction vertex:

[Same fermion species in/out: No FCNC !!]



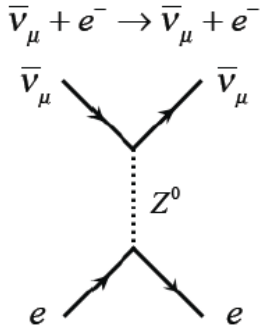


The CKM mixing doesn't apply to the vertex cause it's the same flavor for quarks → no FCNC- flavor changing NC

Note however there are box diagram higher order interactions w/ Z and W's involved that can be "FCNC"

- So it's coupling looks like photon for charge particles
- Annihilation e.g. can proceed through photon or Z
- But unique to Z's are when neutrinos are on the other end





CA, CV → DEPEND ON THE INTERACTING PARTICLE!!!

★In terms of **V** and **A** components for Z

$$j_\mu^Z = \frac{g_Z}{2}\bar{u}\gamma_\mu\left[c_V - c_A\gamma_5\right]u$$

★We will sometimes prefer a "chiral" breakdown::

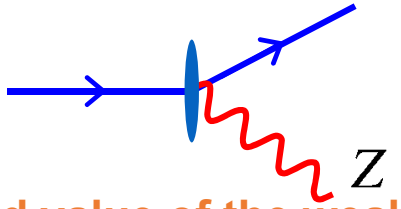
$$j_\mu^Z = \frac{g_Z}{2}\bar{u}\gamma_\mu\left[(c_L + c_R) + (c_R - c_L)\gamma_5)\right]u$$

related to "Weak Isopin" I_W also tbd next (cA is literally the spin up/down val of that)

$$c_V = c_L + c_R = I_W^3 - 2Q\sin^2\theta_W \qquad c_A = c_L - c_R = I_W^3$$

★ The vertex factor for the Z boson is:

$$-ig_Z\frac{1}{2}\gamma_\mu\left[c_V - c_A\gamma_5\right]$$



★ Using the experimentally determined value of the weak mixing angle (SEE GRIFF):

$$\sin^2\theta_W \approx 0.23$$

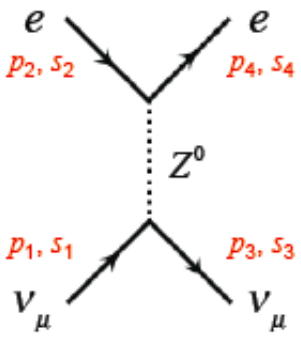
Fermion	Q	I_W^3	c_L	c_R	c_V	c_A
ν_e, ν_μ, ν_τ	0	$+\frac{1}{2}$	$+\frac{1}{2}$	0	$+\frac{1}{2}$	$+\frac{1}{2}$
e^-, μ^-, τ^-	-1	$-\frac{1}{2}$	-0.27	0.23	-0.04	$-\frac{1}{2}$
u, c, t	$+\frac{2}{3}$	$+\frac{1}{2}$	0.35	-0.15	$+0.19$	$+\frac{1}{2}$
d, s, b	$-\frac{1}{3}$	$-\frac{1}{2}$	-0.42	0.08	-0.35	$-\frac{1}{2}$

Example of Z Feynman rules

- (using the same large M_Z approx. for propagator)

The Z^0 propagator is $$-\frac{i}{q^2 - M_Z^2c^2}\left(g_{\mu\nu} - \frac{q_\mu q_\nu}{M_Z^2c^2}\right) = \frac{ig_{\mu\nu}}{M_Z^2c^2} \quad \text{for } q^2 << M_Z^2c^2$$

Look at $\nu_\mu e^- \to \nu_\mu e^-$ (we have already looked at the charged current process $\nu_\mu e^- \to \mu^- \nu_e$)



neutrino coupling

$$-i\mathcal{M} = \left[\bar{u}(3)\left(-\frac{ig_z}{2}\right)\gamma^\mu\left(\frac{1}{2} - \frac{1}{2}\gamma^5\right)u(1)\right]\left(\frac{ig_{\mu\nu}}{M_Z^2c^2}\right)\left[\bar{u}(4)\left(-\frac{ig_z}{2}\right)\gamma^\nu\underbrace{\left(c_V^e - c_A^e\gamma^5\right)}u(2)\right]$$

electron coupling

$$\mathcal{M} = \frac{g_z^2}{8(M_z c)^2}\left[\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1)\right]\left[\bar{u}(4)\gamma_\mu\left(c_V^e - c_A^e\gamma^5\right)u(2)\right]$$

includes a factor of 1/2 from the average over initial spin states

$$\left\langle|\mathcal{M}|^2\right\rangle = 2\left(\frac{g_z}{4M_z c}\right)^4 Tr\left[\gamma^\mu(1-\gamma^5)\not{p}_1\gamma^\nu(1-\gamma^5)\not{p}_3\right]Tr\left[\gamma_\mu\left(c_V^e - c_A^e\gamma^5\right)(\not{p}_2 + mc)\gamma_\nu\left(c_V^e - c_A^e\gamma^5\right)(\not{p}_4 + mc)\right]$$

$$= \frac{1}{2}\left(\frac{g_z}{M_z c}\right)^4\left[\left(c_V^e + c_A^e\right)^2(p_1\cdot p_2)(p_3\cdot p_4) + \left(c_V^e - c_A^e\right)^2(p_1\cdot p_4)(p_2\cdot p_3) - (mc)^2\left(\left(c_V^e\right)^2 - \left(c_A^e\right)^2\right)(p_1\cdot p_3)\right]$$

Chiral Structure of QED (Reminder)

★ Recall (Handout 4) introduced CHIRAL projections operators

$$P_R = \tfrac{1}{2}(1+\gamma^5); \qquad P_L = \tfrac{1}{2}(1-\gamma^5)$$

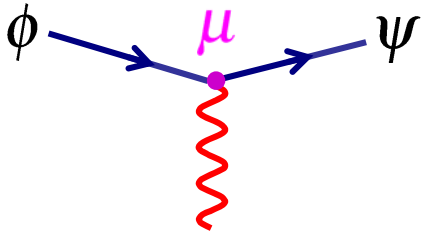
project out chiral right- and left- handed states

★ In the ultra-relativistic limit, chiral states correspond to helicity states

★ Any spinor can be expressed as:

$$\psi = \tfrac{1}{2}(1+\gamma^5)\psi + \tfrac{1}{2}(1-\gamma^5)\psi = P_R\psi + P_L\psi = \psi_R + \psi_L$$

•The QED vertex $\overline{\psi}\gamma^\mu\phi$ in terms of chiral states:

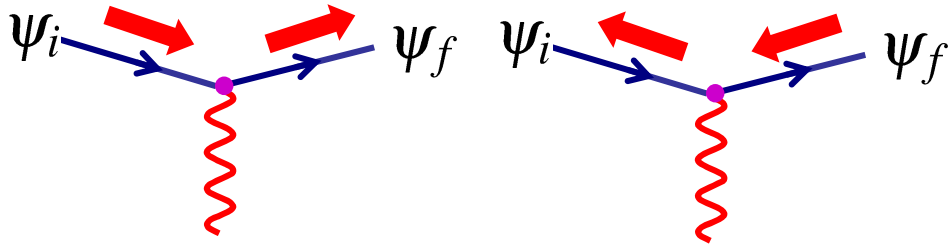


$$\overline{\psi}\gamma^\mu\phi = \overline{\psi}_R\gamma^\mu\phi_R + \overline{\psi}_R\gamma^\mu\phi_L + \overline{\psi}_L\gamma^\mu\phi_R + \overline{\psi}_L\gamma^\mu\phi_L$$

conserves chirality, e.g.

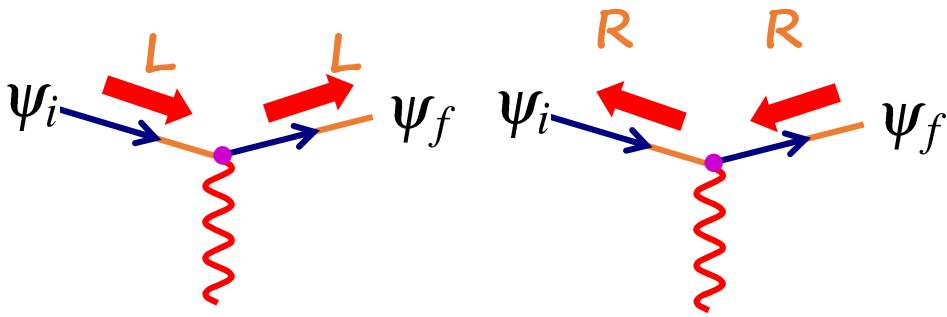
$$\begin{aligned}
\overline{\psi}_R\gamma^\mu\phi_L &= \tfrac{1}{2}\psi^\dagger(1+\gamma^5)\gamma^0\gamma^\mu\tfrac{1}{2}(1-\gamma^5)\phi \\
&= \tfrac{1}{4}\psi^\dagger\gamma^0(1-\gamma^5)\gamma^\mu(1-\gamma^5)\phi \\
&= \tfrac{1}{4}\overline{\psi}\gamma^\mu(1+\gamma^5)(1-\gamma^5)\phi = 0
\end{aligned}$$

★In the ultra-relativistic limit only two helicity combinations are non-zero



Chirality in "Currents"

- That means no cross terms for our original QED uγu current:



$$j_\mu^{em} = -\bar{e}\gamma_\mu e = -(\bar{e}_L + \bar{e}_R)\gamma_\mu(e_L + e_R) = -\bar{e}_L\gamma_\mu e_L - \bar{e}_R\gamma_\mu e_R$$

CONSERVES CHIRALITY (e.g. Helicity @ High E)

For the weak CC interaction same, but since the R handed no-exist-y neutrino, only the left handed term survives

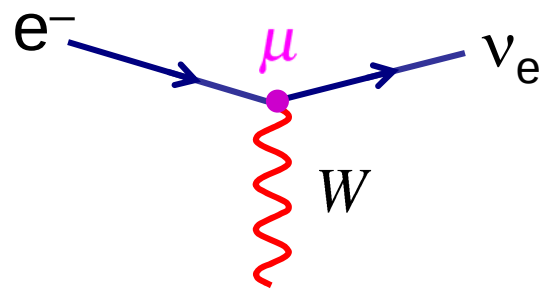
$$\left.\begin{aligned} j_\mu^- &= \bar{\nu}_L\gamma_\mu e_L \\ j_\mu^+ &= \bar{e}_L\gamma_\mu \nu_L \end{aligned}\right\} \text{ charged weak currents}$$

Mathematically enforced by "projection" view of weak vertex $1-\gamma^5$

Helicity Structure of the WEAK Interaction

★The charged current ($W^{\pm}$) weak vertex is:

$$\frac{-ig_w}{\sqrt{2}}\frac{1}{2}\gamma^{\mu}(1-\gamma^5)$$



★Since $\frac{1}{2}(1-\gamma^5)$ projects out left-handed chiral particle states:

$$\overline{\psi}\tfrac{1}{2}\gamma^{\mu}(1-\gamma^5)\phi = \overline{\psi}\gamma^{\mu}\phi_L$$

(question 16)

★Writing $\overline{\psi} = \overline{\psi}_R + \overline{\psi}_L$ and from discussion of QED, $\overline{\psi}_R\gamma^{\mu}\phi_L = 0$ gives

$$\overline{\psi}\tfrac{1}{2}\gamma^{\mu}(1-\gamma^5)\phi = \overline{\psi}_L\gamma^{\mu}\phi_L$$

⇒ **Only the left-handed chiral components of particle spinors and right-handed chiral components of anti-particle spinors participate in charged current weak interactions**

★At very high energy $(E \gg m)$, the left-handed chiral components are helicity eigenstates :

$\frac{1}{2}(1-\gamma^5)u \Rightarrow$ ⟵

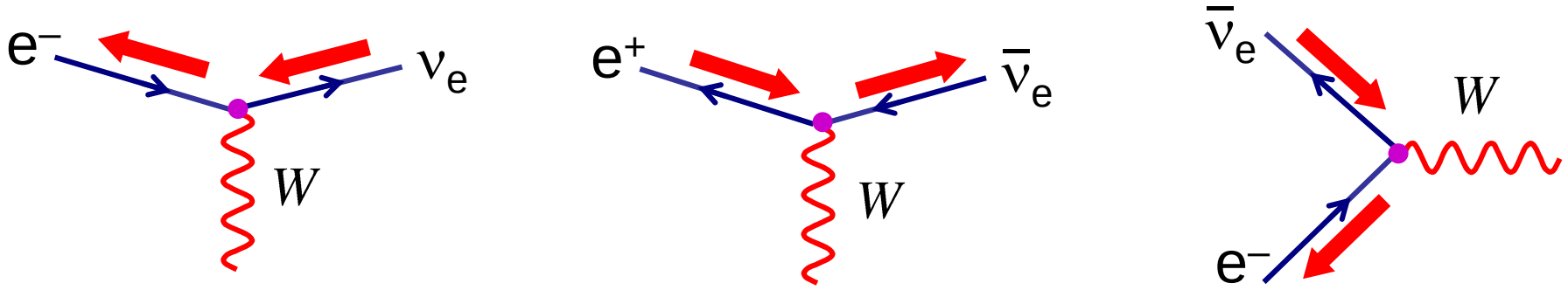
LEFT-HANDED PARTICLES
Helicity = -1

$\frac{1}{2}(1-\gamma^5)v \Rightarrow$ ⟶

RIGHT-HANDED ANTI-PARTICLES
Helicity = +1

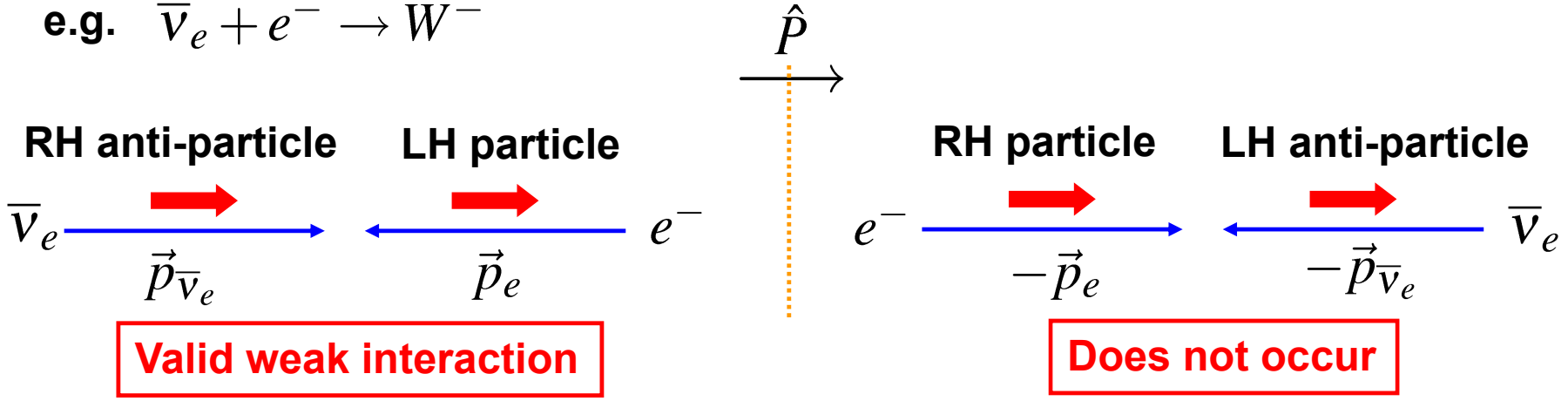
In the ultra-relativistic limit only left-handed particles and right-handed antiparticles participate in charged current weak interactions

e.g. In the relativistic limit, the only possible electron – neutrino interactions are:



★ The helicity dependence of the weak interaction ⟷ parity violation

e.g. $\overline{\nu}_e + e^- \rightarrow W^-$



Valid weak interaction

Does not occur

Z boson Chiral Currents

- The Z boson interactions CAN have right handed parts

$$J_Z \propto \bar{u}\gamma^\mu(1-\epsilon\gamma^5)u$$

$$\epsilon = C_A/C_V$$

Fermion	Q	I_W^3	c_L	c_R	c_V	c_A
ν_e,ν_μ,ν_τ	0	$+\frac{1}{2}$	$+\frac{1}{2}$	0	$+\frac{1}{2}$	$+\frac{1}{2}$
e^-,μ^-,τ^-	-1	$-\frac{1}{2}$	-0.27	0.23	-0.04	$-\frac{1}{2}$
u,c,t	$+\frac{2}{3}$	$+\frac{1}{2}$	0.35	-0.15	$+0.19$	$+\frac{1}{2}$
d,s,b	$-\frac{1}{3}$	$-\frac{1}{2}$	-0.42	0.08	-0.35	$-\frac{1}{2}$

- The extent to which this ϵ differs from 1 is the "EXTRA" beyond the LH Chirality projector → this much RH in its current
- But apparently only from its "EM" component
  - B.C. C_R is still ZERO for neutrinos!
- (from other references: →)

$$J_Z^\mu = J_{3L}^\mu - \sin^2\theta_W J_{em}^\mu$$

- Still looks different:

$$j_M^Z \propto \bar{\nu}_L\gamma^M\nu_L - k_1\bar{e}_R\gamma^M e_R - k_2\bar{e}_L\gamma^M e_L \ldots$$

(than jem b.c. ν's)

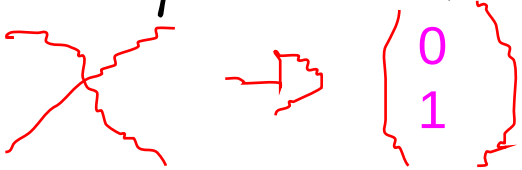
- Because of ν oscillations small #'s of ν_R must exist Z couple those too? current

Weak Isospin and Charge→ Electroweak Unif.

- All the factors discuss so far effectively modify the coupling. and thus the "charge" so there are many very different weak charges for each kind of particle and CC vs NC
  - gW and gZ are different but we will see they are fixed by electroweak unification
  - [this may be why the right-handed neutrinos coupling by Z are ruled out but I don't really know how much of a modification

- Driven by the desire to "unify" W+,W-, and Z (Weak) interactions (and eventually EM) we introduce weak isospin but it can only be with the LH-handed part of the Z interactions

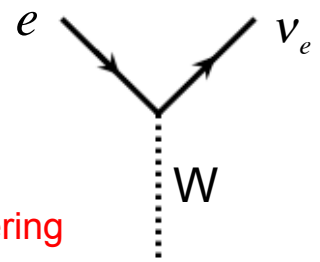
- This leads to a weak iso spin current and a weak "hyper charge" Y current which are thought of as the two main "kind" of weak charges
- The weak isospin "doublets" notation: (like for regular spin)



Follow some slides from Kreiger

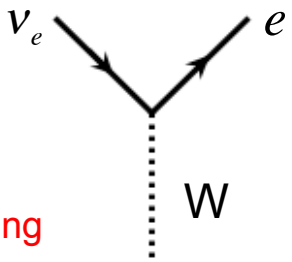
Neutral Weak Currents

We have expressions for the "positive" and "negative" weak charged current:



$$j_\mu^- = \bar{\nu}_L \gamma_\mu e_L$$

Weak-isospin lowering



$$j_\mu^+ = \bar{e}_L \gamma_\mu \nu_L$$

Weak-isospin raising

Can write this compactly by defining the left-handed (weak-isospin) doublet: $\chi_L = \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L$

Using the 2x2 matrices $\tau^+ \equiv \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\tau^- \equiv \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ we can then write:

$$\boxed{j_\mu^\pm = \bar{\chi}_L \gamma_\mu \tau^\pm \chi_L}$$

with $\tau^\pm = \frac{1}{2}(\tau_1 \pm i\tau_2)$ **Weak-isospin raising and lowering operators, with**

$$\tau_1 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \tau_2 = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

These are two of the Pauli spin matrices, written here as τ to avoid any confusion with ordinary spin (as we did when discussing isospin in our discussions of strong interactions). More accurately, these are two of the matrices that form a representation of the group SU(2) which describes systems in which two possible states are related by some symmetry transformation.

(e.g. the quantity in terms of which we are defining these doublets)

Refer to this as weak-isospin and anticipate full weak-isospin symmetry, which would imply the existence of a third "current" corresponding to $\frac{1}{2}\tau^3$ where τ^3 is the third Pauli spin matrix [of SU(2)]:

$$\tau^3 = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad j^3_\mu = \overline{\chi}_L \gamma_\mu \tfrac{1}{2}\tau^3 \chi_L = \frac{1}{2}\overline{\nu}_L \gamma_\mu \nu_L - \frac{1}{2}\overline{e}_L \gamma_\mu e_L$$

Here we have "predicted" the existence of a weak neutral current process. However, this still only couples LH particles to LH particles, while we know that the Z^0 has RH couplings as well (or the vertex factor would be just V-A like the charged weak interaction).

Consider the so-called weak-hypercharge current (which is a mixture of the two neutral currents j^3_μ and j^{em}_μ): [here the weak hypercharge Y is defined by $Q = I_3 + \frac{1}{2}Y$]

$$j^Y_\mu = 2j^{em}_\mu - 2j^3_\mu = 2\left(-\overline{e}_R \gamma_\mu e_R - \overline{e}_L \gamma_\mu e_L\right) - 2\left(\frac{1}{2}\overline{\nu}_L \gamma_\mu \nu_L - \frac{1}{2}\overline{e}_L \gamma_\mu e_L\right)$$

$$= -2\overline{e}_R \gamma_\mu e_R - \overline{e}_L \gamma_\mu e_L - \overline{\nu}_L \gamma_\mu \nu_L$$

$j^Y_\mu = -2\bar{e}_R\gamma_\mu e_R - \bar{e}_L\gamma_\mu e_L - \bar{\nu}_L\gamma_\mu \nu_L$ is invariant under a weak-isospin transformation

$$j^Y_\mu = -2\bar{e}_R\gamma_\mu e_R - \bar{\nu}_L\gamma_\mu \nu_L - \bar{e}_L\gamma_\mu e_L$$

$$e_R \rightarrow e_R$$
$$e_L \rightarrow \nu_L$$
$$\nu_L \rightarrow e_L$$

N.B. We have been discussing things in terms of electrons and electron-neutrinos, but the same applies to any of the weak-isospin doublets:

$$\chi_L = \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L \quad \begin{pmatrix} \nu_\mu \\ \mu \end{pmatrix}_L \quad \begin{pmatrix} \nu_\tau \\ \tau \end{pmatrix}_L \quad \begin{pmatrix} u \\ d' \end{pmatrix}_L \quad \begin{pmatrix} c \\ s' \end{pmatrix}_L \quad \begin{pmatrix} t \\ b' \end{pmatrix}_L$$

where primes on the down-type quarks denote the Cabibbo-rotated states

In each case one can construct three weak isospin currents, $\vec{j}_\mu = \frac{1}{2}\bar{\chi}_L\gamma_\mu\vec{\tau}\,\chi_L$ and one weak-hypercharge current: $j^Y_\mu = 2j^{em}_\mu - 2j^3_\mu$

Weak Hypercharge

$Y = 2Q - 2I_3$ Look at values for members of weak isospin doublet:

$$\chi_L = \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L \quad \begin{pmatrix} \nu_\mu \\ \mu \end{pmatrix}_L \quad \begin{pmatrix} \nu_\tau \\ \tau \end{pmatrix}_L \quad \begin{pmatrix} u \\ d' \end{pmatrix}_L \quad \begin{pmatrix} c \\ s' \end{pmatrix}_L \quad \begin{pmatrix} t \\ b' \end{pmatrix}_L \quad \begin{matrix} I_3 = +1/2 \\ I_3 = -1/2 \end{matrix}$$

$$\begin{pmatrix} \nu_e \\ e \end{pmatrix}_L \qquad 2Q - 2I_3 = \begin{pmatrix} 2(0) - 2(1/2) \\ 2(-1) - 2(-1/2) \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} u \\ d' \end{pmatrix}_L \qquad 2Q - 2I_3 = \begin{pmatrix} 2(2/3) - 2(1/2) \\ 2(-1/3) - 2(-1/2) \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1/3 \end{pmatrix}$$

e.g. value of Y is the same for the two members of a weak isospin doublet

GSW Model for Electroweak Mixing

The GWS model asserts that the three weak-isospin currents couple with strength g_w to a weak iso-triplet of vector bosons $\vec{W}$ while the weak hypercharge current couples with strength $g'/2$ to an iso-singlet vector boson B .

$$-i\left[g_w \vec{j}_\mu \cdot \vec{W}^{\,\mu} + \frac{g'}{2} j_\mu^Y B^\mu \right]$$ [where the vectors are vectors in weak-isospin space]

$$\vec{j}_\mu \cdot \vec{W}^{\,\mu} = j_\mu^1 W^{1\mu} + j_\mu^2 W^{2\mu} + j_\mu^3 W^{3\mu} = \frac{1}{\sqrt{2}} j_\mu^+ W^{+\mu} + \frac{1}{\sqrt{2}} j_\mu^- W^{-\mu} + j_\mu^3 W^{3\mu}$$

where $W_\mu^\pm \equiv \frac{1}{\sqrt{2}}\left(W_\mu^1 \mp W_\mu^2\right)$. We can now read off the couplings to the $W^\pm$:

e.g. for $e^- \to \nu_e + W^-$ $\quad j_\mu^- = \bar{\nu}_L \gamma_\mu e_L = \bar{\nu}\left[\gamma_\mu \left(\frac{1}{2}\right)\left(1-\gamma^5\right)\right] e$ which yields a term of

$$\left\{ \textit{i.e. recalling that } j_\mu^- = \bar{\nu}\gamma_\mu \left(\frac{1-\gamma^5}{2}\right) e = \bar{\nu}\left(\frac{1+\gamma^5}{2}\right)\gamma_\mu \left(\frac{1-\gamma^5}{2}\right) e = \bar{\nu}_L \gamma_\mu e_L \right\}$$

$$-ig_w \frac{1}{\sqrt{2}} j_\mu^- W^{-\mu} = -i\frac{g_w}{2\sqrt{2}}\left[\bar{\nu}\gamma_\mu \left(1-\gamma^5\right) e\right] W^{-\mu}$$ so the vertex factor is $\quad -i\frac{g_w}{2\sqrt{2}}\gamma_\mu\left(1-\gamma^5\right)$

Electroweak mixing

In the GWS model, the W^3 and B mix to produce one massless combination $\left(\gamma \equiv A^\mu\right)$ and one (orthogonal) massive combination, the Z^0:

$$\left.\begin{aligned} A_\mu &= B_\mu \cos\theta_w + W_\mu^3 \sin\theta_w \\ Z_\mu &= -B_\mu \sin\theta_w + W_\mu^3 \cos\theta_w \\ \\ W_\mu^3 &= Z_\mu \cos\theta_w + A_\mu \sin\theta_w \\ B_\mu &= -Z_\mu \sin\theta_w + A_\mu \cos\theta_w \end{aligned}\right\}$$

Write out neutral component of

$$-i\left[g_w \vec{j}_\mu \cdot \vec{W}^\mu + \frac{g'}{2} j_\mu^Y B^\mu \right]$$

In terms of the physical states A_μ and Z_μ

$$-i\left[g_w j_\mu^3 W^{3\mu} + \frac{g'}{2} j_\mu^Y B^\mu \right] = -i\left\{ \left[g_w \sin\theta_w j_\mu^3 + \frac{g'}{2} \cos\theta_w j_\mu^Y \right] A^\mu + \left[g_w \cos\theta_w j_\mu^3 - \frac{g'}{2} \sin\theta_w j_\mu^Y \right] Z^\mu \right\}$$

We know that the EM coupling is $-ig_e j_\mu^{em} A^\mu$. Recall that $j_\mu^Y = 2j_\mu^{em} - 2j_\mu^3 \Rightarrow j_\mu^{em} = j_\mu^3 + \frac{1}{2} j_\mu^Y$

This implies that we need $g_w \sin\theta_w = g' \cos\theta_w = g_e$. That is, the weak and electromagnetic coupling constants are NOT independent!

Griff Ch. 10 Local Gauge Invariance

- Idea. 1) Start w/ Free Particle Lagrangian
- 2) Demand Local Gauge Invariance of Free Lagrangian
  - ALSO ABOUT CERTAIN SPECIFIC GUAGE SYMMETRIES U(1)
- **1+2: Field/Boson (A^μ) terms automatically generated for QED/QCD/QFD !!!**
  - ….w/ proper properties (e.g. "non-Albelian" gluons)
- Some kind of "unification" but why is demanding that the key ? (still not known, don't think string theory provides an answer either).

$$\mathscr{L} = i\hbar c\bar{\psi}\gamma^\mu\partial_\mu\psi - mc^2\bar{\psi}\psi$$

$$\psi \to e^{i\theta(x)}\psi \qquad \text{(local phase transformation)}$$

$$\lambda(x) \equiv -\frac{\hbar c}{a}\theta(x)$$

$$\psi \to e^{-iq\lambda(x)/\hbar c}\psi$$

$$\mathscr{L} \to \mathscr{L} + (q\bar{\psi}\gamma^\mu\psi)\partial_\mu\lambda$$

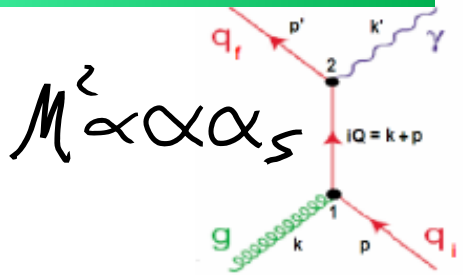
$$\mathcal{L}_{QED} = \bar{\psi}(i\gamma^\mu\partial_\mu\psi - m)\psi - q\bar{\psi}\gamma^\mu A_\mu\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

All fermion A and AA ("Proca" Eq) Self Interaction terms needed→ QFT

Follow steps of Canonical Quantz. L&B →Feyn. Rules etc

Griff Ch. 10 Spontanous Symmetry Breaking Higgs mechanism High Level

- Big Reason for Desiring **LGI Local Gauge Inv**:
  - Automatically theory is renormalizable
- Standard Model Lagrangian: L_{SM}, $L_{QED} + L_{QCD}$



Together➔ Diagrams like Gluon Compton Direct Photons

- **Problem: Local Gauge Invariance can't explain W Z MASSIVE BOSON FIELDS appearance**
- ALSO we know from the GWS Electroweak Unification "proven experimentally" Z/W/A unified through massless A, B
  - Need Mechanism to cause MASSLESS Aμ Bμ , … fields to be come massive
- Answer: The most studied NON-LGI **Renormalizable** theory: ϕ^4 QFT !!!!

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$$

- ADD THIS (Higg's field h(x) → ϕ(x) TO LagrangianPLUS SPONTANEOUS SYMMETRY BREAKING → W,Z get mass
- New massive scalar field exists → can give all particles "acquire" their Higg's mass

Higg's Mechanism

- Thomson Slides and Reading → VERY HIGH LEVEL SUMMARY:
- High points:
- Higg's fields related to most significant VACUUM STRUCTURE of our Universe → **Nonzero Vacuum Expectation Value (VEV)** defining the stable point of potential Field (Vint)
  - Our Gauge Theories work w/ all particles MASSLESS
  - Symmetries for massless → (very high energy?)
  - VEV v value (263 GeV we'll see later) a key parameter of our universe
- Spontaneous Symm Breaking:
- Lower energy we see particular value of VEV is actual reality

Dynamical generation of "mass"

★ **A simple analogy from classical electromagnetism**

- Consider EM radiation propagating through a plasma
- Because the plasma acts as a polarisable medium obtain the "dispersion relation"

$$n^2 = 1 - \frac{n_e e^2}{\varepsilon_0 m_e \omega^2} = 1 - \frac{\omega_p^2}{\omega^2}$$

n = refractive index
ω = angular frequency
ω_p = plasma frequency

★ **Quanta only propagate if have frequency/energy greater than some minimum**

$$E > E_0 = \hbar\omega_p$$

★ **Above this energy, waves propagate with group velocity** $v_g = \frac{c^2}{v_p} = nc$

- **Dropping the subscript and using the previous expression for** n

$$v^2 = c^2 n^2 = c^2\left(1 - \frac{\hbar^2\omega_p^2}{\hbar^2\omega^2}\right) = c^2\left(1 - \frac{E_0^2}{E^2}\right)$$

- **Rearranging gives**

$$\frac{E_0^2}{E^2} = 1 - \frac{v^2}{c^2} \quad \Rightarrow \quad E = E_0\left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}} = \gamma mc^2$$

with $m = E_0/c^2$

- **Massless photons propagating through a plasma behave as massive particles propagating in a vacuum**

Gauge Invariance Revisited

★ **Recall the Lagrangian corresponding to the free particle Dirac equation**

$$\mathcal{L} = i\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi - m\bar{\psi}\psi$$

★ **Now require invariance under a U(1) local gauge transformation**

$$\psi(x) \rightarrow \psi'(x) = e^{iq\chi(x)}\psi(x)$$

under this transformation

$$\mathcal{L} \rightarrow \mathcal{L}' = ie^{-iq\chi}\bar{\psi}\gamma^{\mu}\left[e^{iq\chi}\partial_{\mu}\psi + iq\left(\partial_{\mu}\chi\right)e^{iq\chi}\psi\right] - me^{-iq\chi}\bar{\psi}e^{iq\chi}\psi$$

$$= \mathcal{L} - \boxed{q\bar{\psi}\gamma^{\mu}\left(\partial_{\mu}\chi\right)\psi}$$

★ **Can achieve the desired gauge invariance by replacing ∂_{μ} with the covariant derivative D_{μ}**

$$\partial_{\mu} \rightarrow D_{\mu} = \partial_{\mu} + iqA_{\mu}$$

the desired cancellation of the term $q\bar{\psi}\gamma^{\mu}(\partial_{\mu}\chi)\psi$ can be achieved provided

$$A_{\mu} \rightarrow A'_{\mu} = A_{\mu} - \partial_{\mu}\chi$$

★ **Nothing new here, just reformulating in terms of the Lagrangian density and the full gauge-invariant Lagrangian for QED is**

$$\mathcal{L}_{QED} = \bar{\psi}(i\gamma^{\mu}\partial_{\mu}\psi - m)\psi - q\bar{\psi}\gamma^{\mu}A_{\mu}\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

where the normal field-strength tensor term is now included

The Problem with Mass

★ The corresponding "QED" Lagrangian with a massive photon would be

$$\mathcal{L}_{\mathrm{QED}'} \rightarrow \overline{\psi}(i\gamma^\mu\partial_\mu\psi - m_e)\psi + e\overline{\psi}\gamma^\mu A_\mu\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \boxed{\tfrac{1}{2}m_\gamma A_\mu A^\mu}$$

★ But the mass term, breaks the U(1) gauge invariance from $A_\mu \rightarrow A'_\mu = A_\mu - \partial_\mu\chi$

$$\tfrac{1}{2}m_\gamma^2 A_\mu A^\mu \rightarrow \tfrac{1}{2}m_\gamma^2\left(A_\mu - \partial_\mu\chi\right)(A^\mu - \partial^\mu\chi) \neq \tfrac{1}{2}m_\gamma^2 A_\mu A^\mu$$

★ This is not a problem for QED and QCD where the gauge bosons are massless, but breaks the $SU(2)_L$ gauge symmetry of the weak interaction with massive gauge bosons

★ There is also a problem with fermion masses…

e.g. consider the electron mass term in the QED Lagrangian

$$\begin{aligned} -m_e\bar{e}e &= -m_e\bar{e}\left[\tfrac{1}{2}(1-\gamma^5) + \tfrac{1}{2}(1+\gamma^5)\right]e \\ &= -m_e\bar{e}\left[\tfrac{1}{2}(1-\gamma^5)e_L + \tfrac{1}{2}(1+\gamma^5)e_R\right] \\ &= -m_e(\bar{e}_R e_L + \bar{e}_L e_R)\,. \end{aligned}$$

this also does not respect the SU(L) gauge symmetry, because LH particles transform as weak isospin doublets and RH particles transform as singlets

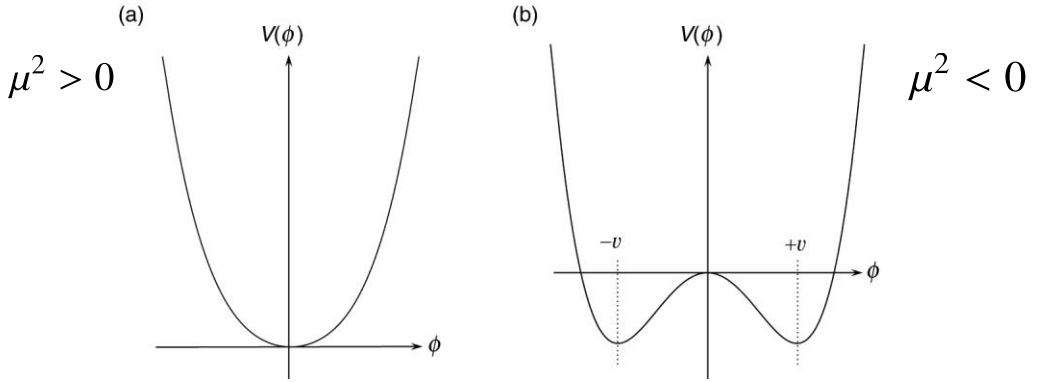
Towards the Higgs

★ The algebra of Higgs mechanism is quite fiddly to derive, mostly

★ Starting gently… a scalar field with a potential $V(\phi) = \frac{1}{2}\mu^2\phi^2 + \frac{1}{4}\lambda\phi^4$

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - V(\phi)$$
$$= \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}\mu^2\phi^2 - \frac{1}{4}\lambda\phi^4$$

★ For lambda positive, the potential can take two forms



Although the potential is symmetric, for $\mu^2 < 0$ the lowest energy "vacuum" states occur at either

$$\phi = \pm v = \pm\left|\sqrt{\frac{-\mu^2}{\lambda}}\right|$$

★ The choice of vacuum spontaneously breaks the symmetry of the Lagrangian

Towards the Higgs

★ **If the vacuum is chosen to be at $\phi = +v$, expanding about this point**

$$\phi(x) = v + \eta(x)$$

★ **The Lagrangian now can be written**

$$\begin{aligned}\mathcal{L}(\eta) &= \tfrac{1}{2}(\partial_\mu\eta)(\partial^\mu\eta) - V(\eta) \\ &= \tfrac{1}{2}(\partial_\mu\eta)(\partial^\mu\eta) - \tfrac{1}{2}\mu^2(v+\eta)^2 - \tfrac{1}{4}\lambda(v+\eta)^4\end{aligned}$$

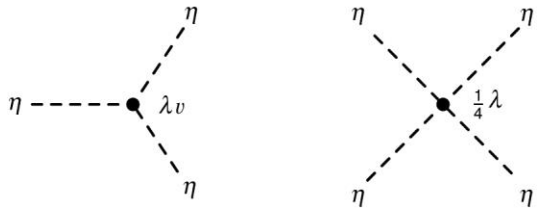
Since the minimum of the potential is given by $\mu^2 = \lambda v^2$ this is equivalent to

$$\mathcal{L}(\eta) = \tfrac{1}{2}(\partial_\mu\eta)(\partial^\mu\eta) - \lambda v^2\eta^2 - \lambda v\eta^3 - \tfrac{1}{4}\lambda\eta^4 + \tfrac{1}{4}\lambda v^4$$

★ **Dropping the constant**

$$\mathcal{L}(\eta) = \tfrac{1}{2}(\partial_\mu\eta)(\partial^\mu\eta) - \tfrac{1}{2}m_\eta^2\eta^2 - V(\eta)\,, \quad \text{with} \quad V(\eta) = \lambda v\eta^3 + \tfrac{1}{4}\lambda\eta^4$$

which is the Lagrangian for a massive scalar field $m_\eta = \sqrt{2\lambda v^2} = \sqrt{-2\mu^2}$ with triple and quartic self-interactions



One step closer

★ **Now consider a complex scalar field** $\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$

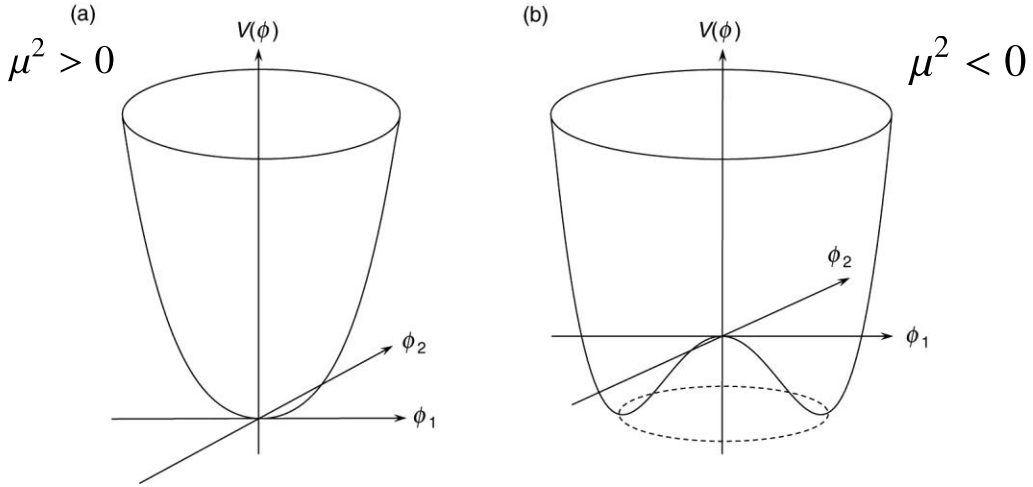
for which the Lagrangian is

$$\mathcal{L} = (\partial_\mu \phi)^*(\partial^\mu \phi) - V(\phi) \quad \text{with} \quad V(\phi) = \mu^2(\phi^*\phi) + \lambda(\phi^*\phi)^2$$

In terms of the two field components

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu \phi_1)(\partial^\mu \phi_1) + \tfrac{1}{2}(\partial_\mu \phi_2)(\partial^\mu \phi_2) - \tfrac{1}{2}\mu^2(\phi_1^2 + \phi_2^2) - \tfrac{1}{4}\lambda(\phi_1^2 + \phi_2^2)^2$$

★ **The shape of the potential takes two forms**



for $\mu^2 < 0$ **there are a continuous (infinite) set of minima**

$$\phi_1^2 + \phi_2^2 = \frac{-\mu^2}{\lambda} = v^2$$

Higg's Mechanism

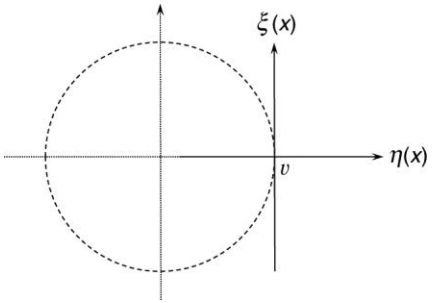
- Thomson Slides and Reading → VERY HIGH LEVEL SUMMARY:
- High points:

- ξ field → Massless→ Goldstone Boson of Symmetry
  - Massless because no interactions in direction of symmetry !!!
  - **Physical Higg's is not this**, "Ghost Particle" USE GLOBAL GAUGE TRANSFORMATION to "remove"
  - GHOST PARTICLES → Familiar in QFT : QCD Griffith's p. 288 mentions QCD needs these for gluon loop calculations

One step closer

★ **Break the symmetry: expand around the minimum** $(\phi_1, \phi_2) = (v, 0)$

$$\phi(x) = \frac{1}{\sqrt{2}}(\eta(x) + v + i\xi(x))$$



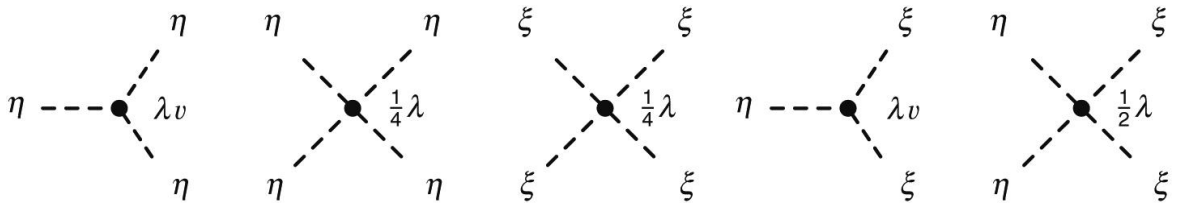
★ **Then express the original Lagrangian in terms of these new fields. After some simple algebra**

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \eta)(\partial^\mu \eta) - \frac{1}{2}m_\eta^2 \eta^2 + \frac{1}{2}(\partial_\mu \xi)(\partial^\mu \xi) - V_{int}(\eta, \xi)$$

with $m_\eta = \sqrt{2\lambda v^2}$ **and interactions**

$$V_{int}(\eta, \xi) = \lambda v \eta^3 + \frac{1}{4}\lambda \eta^4 + \frac{1}{4}\lambda \xi^4 + \lambda v \eta \xi^2 + \frac{1}{2}\lambda \eta^2 \xi^2$$

★ **This the Lagrangian for a massive scalar field** $\eta(x)$ **+ massless scalar** $\xi(x)$**, known as a Goldstone Boson, and interactions**



The Higgs Mechanism

★ Repeat the previous exercise for a complex scalar field, but now embedded in a theory with a U(1) local gauge symmetry and the same potential

$$V(\phi) = \mu^2\phi^2 + \lambda\phi^4$$

★ Because of the derivatives in

$$\mathcal{L} = (\partial_\mu\phi)^*(\partial^\mu\phi) - V(\phi)$$

the Lagrangian is not invariant under

$$\phi(x) \rightarrow \phi'(x) = e^{ig\chi(x)}\phi(x)$$

★ As before, the gauge invariance can be achieved using the covariant derivative

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + igB(x)$$

★ The resulting Lagrangian

$$\mathcal{L} = (D_\mu\phi)^*(D^\mu\phi) - V(\phi^2)$$

has the required local gauge invariance provided

$$B_\mu \rightarrow B'_\mu = B_\mu - \partial_\mu\chi(x)$$

★ The U(1) local gauge invariance requires a new field with well-defined gauge transformation properties – nothing new here

The Higgs Mechanism

★ **The combined Lagrangian for the complex scalar field and the new *massless vector* field *B is***

$$\mathcal{L} = -\tfrac{1}{4}F^{\mu\nu}F_{\mu\nu} + (D_\mu\phi)^*(D^\mu\phi) - \mu^2\phi^2 - \lambda\phi^4$$

where $F^{\mu\nu}F_{\mu\nu}$ is the kinetic term associated with the field-strength tensor

$$F^{\mu\nu} = \partial^\mu B^\nu - \partial^\nu B^\mu$$

★ **Again, adding an explicit mass term $\frac{1}{2}m_B B_\mu B^\mu$ would break the gauge invariance**

★ **The full expression for the Lagrangian (expanding the covariant derivative) is**

$$\mathcal{L} = -\tfrac{1}{4}F^{\mu\nu}F_{\mu\nu} + (\partial_\mu\phi)^*(\partial^\mu\phi) - \mu^2\phi^2 - \lambda\phi^4$$
$$- igB_\mu\phi^*(\partial^\mu\phi) + ig(\partial_\mu\phi^*)B^\mu\phi + g^2B_\mu B^\mu\phi^*\phi\,.$$

★ **Breaking the symmetry and repeating the previous expansion around the minimum**

$$\phi(x) = \tfrac{1}{\sqrt{2}}(v + \eta(x) + i\xi(x))$$

★ **After some straightforward algebra**

$$\mathcal{L} = \underbrace{\tfrac{1}{2}(\partial_\mu\eta)(\partial^\mu\eta) - \lambda v^2\eta^2}_{\text{massive } \eta} + \underbrace{\tfrac{1}{2}(\partial_\mu\xi)(\partial^\mu\xi)}_{\text{massless } \xi} - \underbrace{\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \tfrac{1}{2}g^2v^2B_\mu B^\mu}_{\text{massive gauge field}} - V_{int} + gvB_\mu(\partial^\mu\xi)$$

we now have a local gauge invariant mass term for the gauge boson!

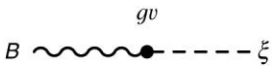
The Higgs Mechanism

★ **It all looks so good…**

$$\mathcal{L} = \underbrace{\tfrac{1}{2}(\partial_\mu\eta)(\partial^\mu\eta) - \lambda v^2\eta^2}_{\text{massive } \eta} + \underbrace{\tfrac{1}{2}(\partial_\mu\xi)(\partial^\mu\xi)}_{\text{massless } \xi} - \underbrace{\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \tfrac{1}{2}g^2v^2B_\mu B^\mu}_{\text{massive gauge field}} - V_{int} + \boxed{gvB_\mu(\partial^\mu\xi)}$$

but we seem to have two problems

- **We started with 4 degrees of freedom (two for the complex scalar field + two for the transverse polarisations of the massless vector gauge boson). We now appear to have 5. Two scalar fields and 3 polarisations – an additional degree of freedom associated with the longitudinal polarisation of the massive gauge boson**
- **There appears to be a coupling between a massless Goldstone scalar and massive spin-1 vector field**



★ **The solution is subtle. Both the above problems are indicative of the fact we are not working with fields that represent the physical particles. The $gvB_\mu(\partial^\mu\xi)$ term is reminiscent of an off-diagonal term in a mass matrix for a couple quantum system. We need to work in the diagonal basis if the fields are to represent the physical particles**

The Higgs Mechanism

★ **The Lagrangian**

$$\mathcal{L} = \underbrace{\tfrac{1}{2}(\partial_\mu\eta)(\partial^\mu\eta) - \lambda v^2\eta^2}_{\text{massive } \eta} + \underbrace{\tfrac{1}{2}(\partial_\mu\xi)(\partial^\mu\xi)}_{\text{massless } \xi} - \underbrace{\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \tfrac{1}{2}g^2v^2B_\mu B^\mu}_{\text{massive gauge field}} - V_{int} + \boxed{gvB_\mu(\partial^\mu\xi)}$$

is gauge invariant. Can we find a gauge in which the non-physical fields do not appear? Yes… the unitary gauge

★ **Noting that**

$$\tfrac{1}{2}(\partial_\mu\xi)(\partial^\mu\xi) + gvB_\mu(\partial^\mu\xi) + \tfrac{1}{2}g^2v^2B_\mu B^\mu = \tfrac{1}{2}g^2v^2\left[B_\mu + \frac{1}{gv}(\partial_\mu\xi)\right]^2$$

it is clear that the massless Goldstone field can be eliminated by making the gauge transformation

$$B_\mu(x) \rightarrow B'_\mu(x) = B_\mu(x) + \frac{1}{gv}\partial_\mu\xi(x)$$

★ **Because of the built in U(1) gauge invariance of the Lagrangian, nothing has physically changed. This is just equivalent of a U(1) transformation**

$$\phi(x) \rightarrow \phi'(x) = e^{-ig\frac{\xi(x)}{gv}}\phi(x) = e^{-i\xi(x)/v}\phi(x)$$

★ **To first order our original expansion about the SSB minimum can be written**

$$\phi(x) = \frac{1}{\sqrt{2}}(v + \eta(x) + i\xi(x)) \approx \frac{1}{\sqrt{2}}\left[v + \eta(x)\right]e^{i\xi(x)/v}$$

and it is clear that in the Unitary gauge the field $\xi(x)$ has been "gauged away"

The Higgs Mechanism

★ **The last step is equivalent to writing**

$$\phi(x) = \frac{1}{\sqrt{2}}(v + \eta(x)) \equiv \frac{1}{\sqrt{2}}(v + h(x))$$

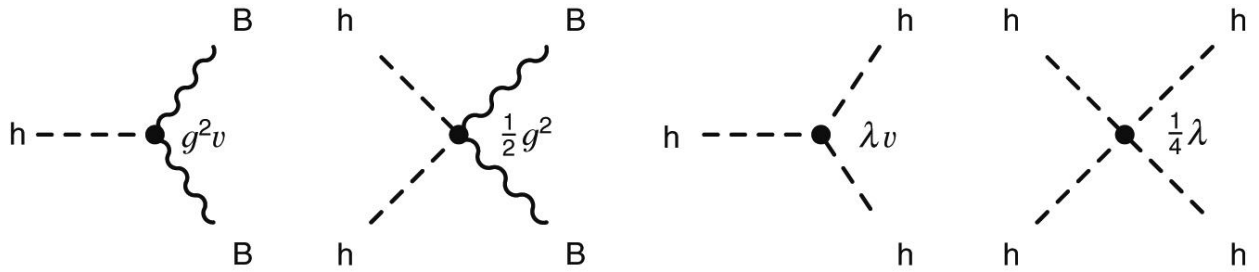
where the field η(x) has been rewritten as the Higgs field h(x) to emphasise the fact that this is the physical field

★ **In the Unitary gauge**

$$\mathcal{L} = \underbrace{\tfrac{1}{2}(\partial_\mu h)(\partial^\mu h) - \lambda v^2 h^2}_{\text{massive } h \text{ scalar}} \underbrace{- \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \tfrac{1}{2}g^2 v^2 B_\mu B^\mu}_{\text{massive gauge boson}}$$

$$+ \underbrace{g^2 v B_\mu B^\mu h + \tfrac{1}{2}g^2 B_\mu B^\mu h^2}_{h,B \text{ interactions}} \underbrace{- \lambda v h^3 - \tfrac{1}{4}\lambda h^4}_{h \text{ self-interactions}}$$

with $\boxed{m_B = g\,v}$ **and** $\boxed{m_{\mathrm{H}} = \sqrt{2\lambda}\,v}$ **and well-defined boson interactions**



Higg's Mechanism

- Thomson Slides and Reading → VERY HIGH LEVEL SUMMARY:
- High points:

- VEV v introduced as constant field
  - $\phi(x,y,z,t) = v$ [number]
- Simpler see B massive + v = VEV
- Specifically for GWS→ Use Weak Isospin Doublet structures
- (Zero , v+h(x))_L form

The Higgs and Electroweak Unification

★ **Electroweak unification is achieved by embedding the Higgs Mechanism in the local gauge symmetry** $\mathrm{U}(1)_Y \times \mathrm{SU}(2)_L$

★ **The scalar fields now need to be placed in a weak isospin doublet and the simplest model uses two complex scalar fields in a weak isospin doublet**

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1 + i\phi_2 \\ \phi_3 + i\phi_4 \end{pmatrix}$$

★ **The Lagrangian for the scalar fields can be written**

$$\mathcal{L} = (\partial_\mu \phi)^\dagger (\partial^\mu \phi) - V(\phi)$$

with $V(\phi) = \mu^2 \phi^\dagger \phi + \lambda (\phi^\dagger \phi)^2$

and a continuum of minima at $\phi^\dagger \phi = \frac{1}{2}(\phi_1^2 + \phi_2^2 + \phi_3^2 + \phi_4^2) = v^2 = -\frac{\mu^2}{2\lambda}$

★ **After symmetry breaking, we expect a neutral massless vector boson (the photon), so anticipating this expand the fields around the vacuum minimum**

$$\langle 0|\phi|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} \quad \text{i.e.} \quad \phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1(x) + i\phi_2(x) \\ v + \eta(x) + i\phi_4(x) \end{pmatrix}$$

The Higgs and Electroweak Unification

★ The local gauge symmetry can be imposed by replacing

$$\partial_\mu \rightarrow D_\mu = \partial_\mu + ig_{\rm W}\mathbf{T}\cdot\mathbf{W}_\mu + ig'\frac{Y}{2}B_\mu$$

where $\mathbf{T} = \frac{1}{2}\boldsymbol{\sigma}$ are the three SU(2) generators, represented by 2x2 matrices

★ The rest is a slog. However, anticipating that three Goldstone bosons will form the longitudinal polarisation states of three massive gauge bosons and the physical states will emerging after moving to the Unitary gauge we can cheat a bit and immediately write the Higgs doublet in the Unitary Gauge

$$\phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}$$

★ Putting this all together, get

$$(D_\mu\phi)^\dagger(D^\mu\phi) = \tfrac{1}{2}(\partial_\mu h)(\partial^\mu h) + \tfrac{1}{8}g_{\rm W}^2(W_\mu^{(1)} + iW_\mu^{(2)})(W^{(1)\mu} - iW^{(2)\mu})(v+h)^2$$
$$+ \tfrac{1}{8}(g_{\rm W}W_\mu^{(3)} - g'B_\mu)(g_{\rm W}W^{(3)\mu} - g'B^\mu)(v+h)^2$$

★ Gauge boson masses are quadratic in the fields, i.e. are contained in

$$\tfrac{1}{8}v^2 g_{\rm W}^2\left(W_\mu^{(1)}W^{(1)\mu} + W_\mu^{(2)}W^{(2)\mu}\right) + \tfrac{v^2}{8}\left(g_{\rm W}W_\mu^{(3)} - g'B_\mu\right)\left(g_{\rm W}W^{(3)\mu} - g'B^\mu\right)$$

therefore

$$\boxed{m_{\rm W} = \tfrac{1}{2}g_{\rm W}v}$$

The Neutral Boson Masses

★ the masses of the neutral bosons are obtained from

$$\frac{v^2}{8}\left(g_{\rm W}W_\mu^{(3)} - g'B_\mu\right)\left(g_{\rm W}W^{(3)\mu} - g'B^\mu\right) = \frac{v^2}{8}\begin{pmatrix} W_\mu^{(3)} & B_\mu \end{pmatrix}\begin{pmatrix} g_{\rm W}^2 & -g_{\rm W}g' \\ -g_{\rm W}g' & g'^2 \end{pmatrix}\begin{pmatrix} W^{(3)\mu} \\ B^\mu \end{pmatrix}$$

$$= \frac{v^2}{8}\begin{pmatrix} W_\mu^{(3)} & B_\mu \end{pmatrix}\mathbf{M}\begin{pmatrix} W^{(3)\mu} \\ B^\mu \end{pmatrix}$$

★ in the mass matrix there are terms that couple together the $W^{(3)}$ and B fields, which means that these are not the physical fields that are eigenstates of the free particle Hamiltonian. To obtain the physical fields, need to diagonalise the mass matrix using the characteristic equation $\det(\mathbf{M} - \lambda I) = 0$ giving eigenvalues from

$$(g_{\rm W}^2 - \lambda)(g'^2 - \lambda) - g_{\rm W}^2 g'^2 = 0$$

giving $\lambda = 0 \quad \text{or} \quad \lambda = g_{\rm W}^2 + g'^2$

★ In the diagonal basis the mass matrix is

$$\frac{1}{8}v^2\begin{pmatrix} A_\mu & Z_\mu \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & g_{\rm W}^2 + g'^2 \end{pmatrix}\begin{pmatrix} A^\mu \\ Z^\mu \end{pmatrix}$$

a massless field (the photon) and a massive neutral gauge boson the Z with

$$m_{\rm Z} = \frac{1}{2}\sqrt{(g_{\rm W}^2 + g'^2)}v$$

The Neutral Boson Masses

★ **The physical fields, that correspond to the normalised eigenvalues of the mass matrix are:**

$$A_\mu = \frac{g' W_\mu^{(3)} + g_{\rm W} B_\mu}{\sqrt{g_{\rm W}^2 + g'^2}} \quad \text{with} \quad m_A = 0\,,$$

$$Z_\mu = \frac{g_{\rm W} W_\mu^{(3)} - g' B_\mu}{\sqrt{g_{\rm W}^2 + g'^2}} \quad \text{with} \quad m_{\rm Z} = \tfrac{1}{2} v \sqrt{g_{\rm W}^2 + g'^2}$$

★ **Writing the ratio of the couplings of the underlying $U(1)_Y$ and $SU(2)_L$ local gauge symmetries as**

$$\frac{g'}{g_{\rm W}} = \tan\theta_{\rm W}$$

we obtain the relations

$$A_\mu = \cos\theta_{\rm W} B_\mu + \sin\theta_{\rm W} W_\mu^{(3)}\,,$$

$$Z_\mu = -\sin\theta_{\rm W} B_\mu + \cos\theta_{\rm W} W_\mu^{(3)}$$

★ **In GSW electroweak model everything is determined by the two couplings of the underlying gauge symmetries and the two free parameters of the Higgs potential**

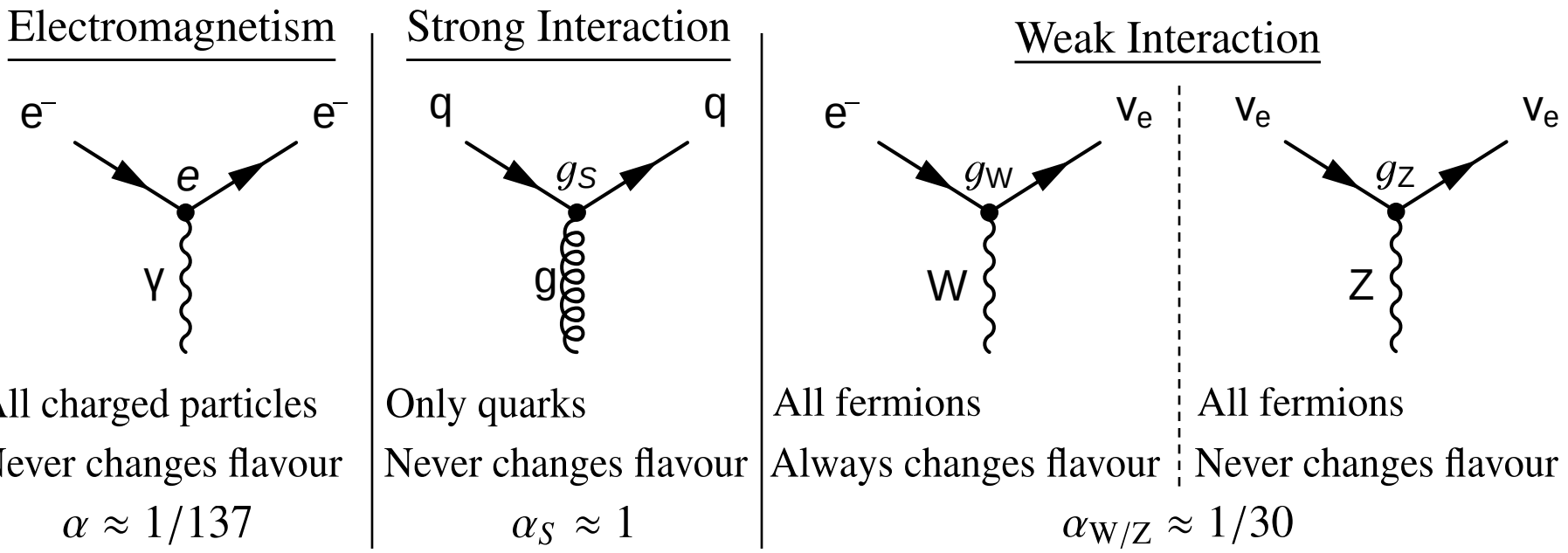
$$g_{\rm W},\ g',\ \mu,\ \lambda$$

Wrapping Up I

- ★ **The Higgs Mechanism is central to the electroweak unification in the standard model, fixing the properties of the W, Z and Higgs**
- ★ **Can also be used to write down gauge invariant terms in the Lagrangian for fermion masses – but has no power in determining the masses**

$$\overline{L}\phi_c R + \overline{R}\phi_c^\dagger L$$

- ★ **In the SM, in some sense all particles are massless, the mass terms in the Lagrangian are generated dynamically through symmetry breaking**



Higg's Mechanism

- Thomson Slides and Reading → VERY HIGH LEVEL SUMMARY:
- High points:
- **Fermion masses (incl Neutrino Mass!)**
- Terms like this break SU(2)L invariance:

$$-m\bar{\psi}\psi = -m\left(\bar{\psi}_R\psi_L + \bar{\psi}_L\psi_R\right),$$

- Recall Chiral structure for interactions:

★**Writing** $\bar{\psi} = \bar{\psi}_R + \bar{\psi}_L$ **and from discussion of QED,** $\bar{\psi}_R\gamma^\mu\phi_L = 0$ **gives**

$$\bar{\psi}\tfrac{1}{2}\gamma^\mu(1-\gamma^5)\phi = \bar{\psi}_L\gamma^\mu\phi_L$$

- Exact same Higg's solves this problem allowing the terms to exist, once it's embedded into the Stand.Model Lagrangian
- **Also in Thomson extra reading Majorana Mass Mechanism explained (motivation for Neutrinoless 2β Decay search)**

Selected Non-perturbative QCD Topics

- **There is a lot of freedom still to look for new understandings in the CLOUDED areas of non-perturbative QCD!!!**
- **Last thing in course: share a few of these with you**

- First – 2 Seconds about Lattice QCD

- Three topics from Extra Reading:

- 1)Non-perturbative String Model (Very soft hadron formation)
- 2)MIT Bag Model -- Simplest Possible Interpretation for hadron structure (softest momenta)→ still useful!
- 3) Diffractive Scattering – Pomerons – glueball like structures very important for total cross sections at low q^2 but high energy

- One extra example just in slides today of how uncertain things can be!:
- Simple question: How is Baryon number carried in hadrons?
  - "Gluon Junctions" Talks

Lattice QCD!

- What is Lattice QCD and what are its limitations !?!?
- Problem w/ pQCD: coupling constant from renorm "diverges" E→0
- "lattice approach" naturally regulates ALL divergences
  - e.g. no true Δr = 0 → no UltraViolet E→∞ diverge
- **USES Feynman Path Integrals**

  Wick Rotation $e^{iS} \rightarrow e^{-S}$ 4D Euclidean Space

$$\left(\frac{m}{2\pi i\Delta t}\right)^{n/2} dx_1 \dots dx_{n-1} \longrightarrow \text{const.} \prod_t dx(t) \equiv \mathcal{D}x$$

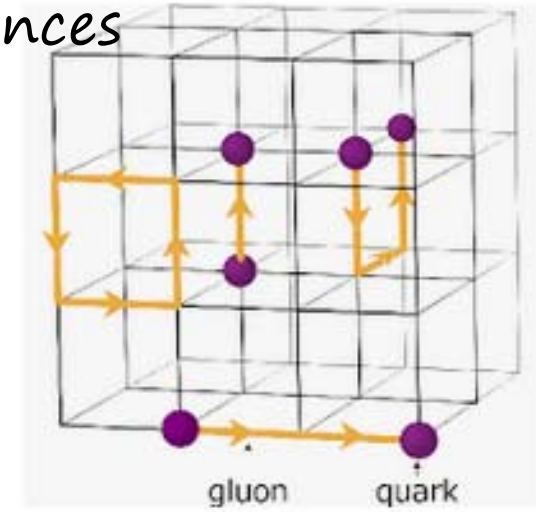
e at the path integral representation of the quantum me
e:

$$\langle x'|e^{-iHT}|x\rangle = \int \mathcal{D}x\, e^{iS}.$$

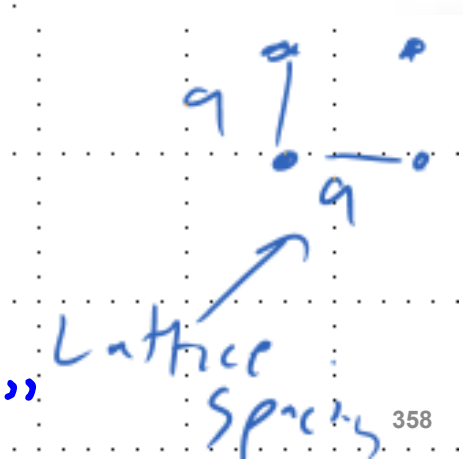
rticle in 3 dimensional space we generalize to paths $x_i(t$
and

$$\mathcal{D}x = \prod_t \prod_i dx_i(t).$$

$$U(i,f) = \int \mathcal{D}\Psi\mathcal{D}\Psi\mathcal{D}A \exp -i \int d^4x\mathcal{L}[\Psi,\Psi]$$



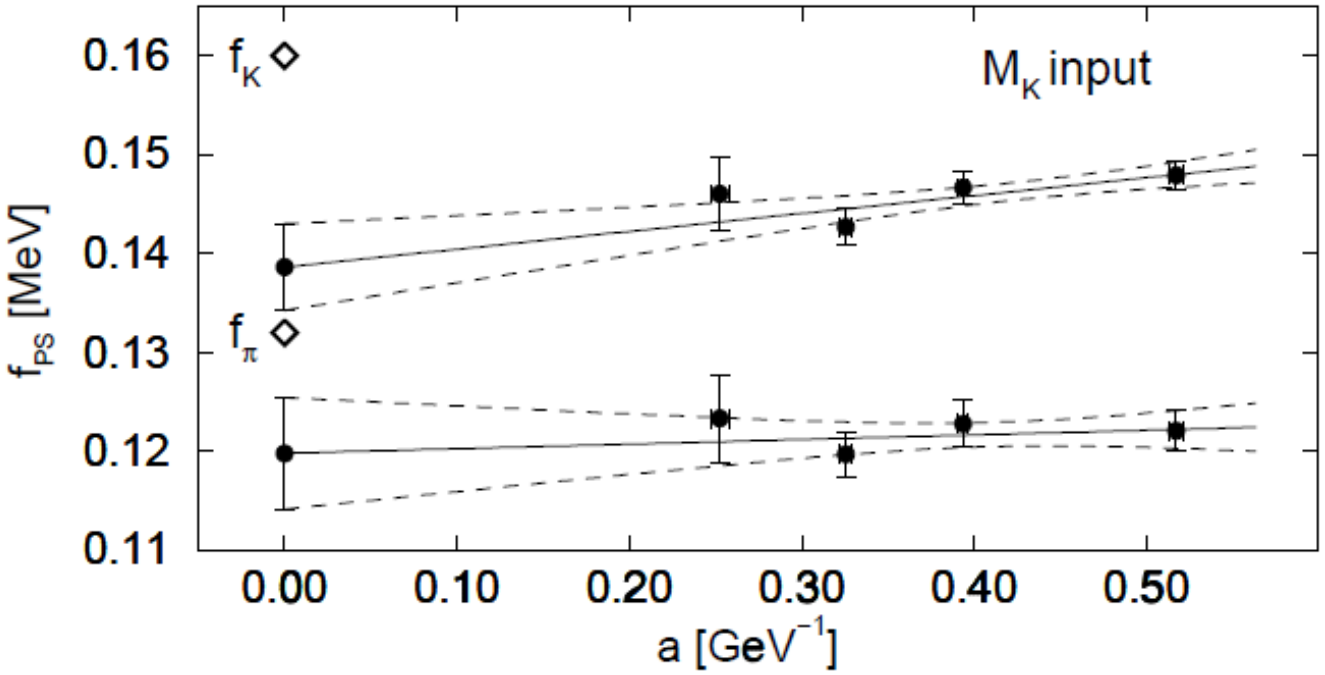
Lattice QCD discretization illus...
researchgate.net



Lattice size characterized by lattice spacing "a"

Paths/Lattice Spacings/Extrapolations

- Feynman Path Integrals integrating quantities over literal paths on lattice of space time points
  - Not limited by divergence problems can calculate where pQCD fails ✓
- (Below pion decay constant same one discussed in Griff Ch 9)
- **OBSERVABLES computed w/ decreasing lattice spacing a's extrapolated to a→ 0 sometimes need multiple lattices!**



To decrease convergence time, also use higher coupling → "higher pion mass" extrapolate to real pion mass

Fig. 7: CP-PACS Collaboration's results for the quenched K meson and pion decay constants in the continuum limit [15]. Experimental values are denoted by diamonds.

Lattice QCD – Computing Limitations

- Computing limited by size of lattice N^3 scaling of computing

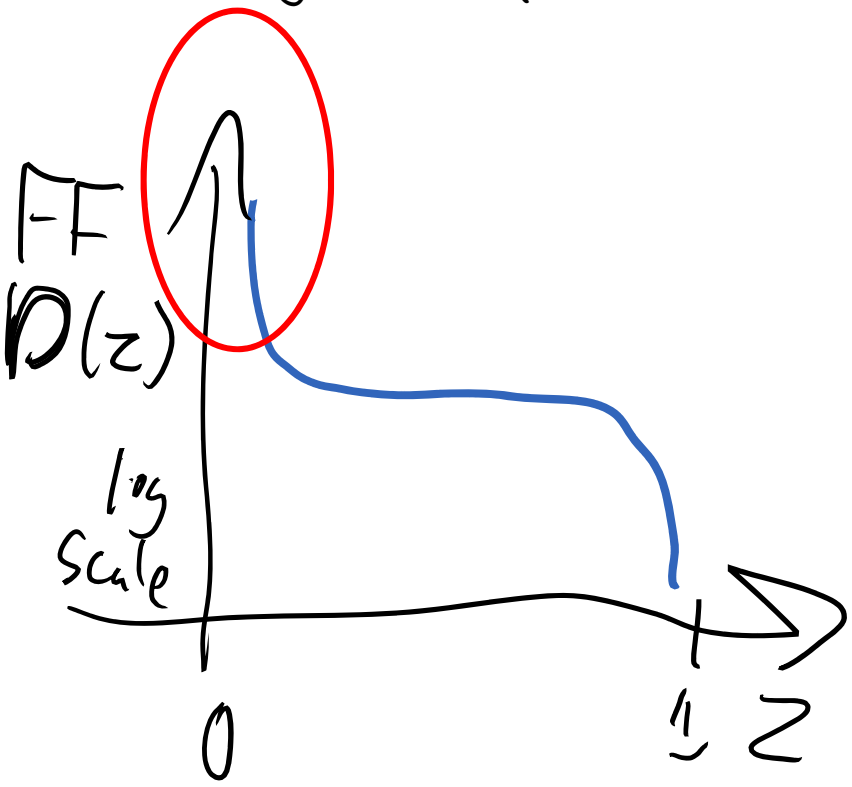
T-values	lattice size	configuration generation [TFlop/s-years]	configuration analysis [TFlop/s-years]
5	$32^3 \times 8$	21	630
5	$48^3 \times 12$	49	1990
5	$64^3 \times 16$	144	1010 (sixth order)

TABLE II. Requirements for the generation of 10^5 gauge-field configurations used for the calculation of conserved charge fluctuations up to eighth order with physical values of the quark masses at five values of the temperature between 150-170 MeV and for three values of the lattice spacings. Also given are the resources needed to calculate up to eighth order cumulants on a sample of 10^4 configurations using $2 \cdot 10^3$ random source vectors for the inversions of fermion matrices. Limiting the analysis on the $N_\tau = 12$ lattices to sixth order only would reduce the resource requirement by a factor 2.

- Lately some advances → predict many hadron masses very accurately
- But for practical purposes still not capable of calculating many practically needed quantites let alone "explore-able" for interesting phenomena!!!
- **Still need experimental data and models!!!**

Hadronization Models

- Fragmentation Functions have limits
  - Not measured well for ALL hadron types ! (too many!)
  - Very small z → perturbative methods (evolution) become inaccurate
- Need non-perturbative models for fairly numerous very soft particles → "underlying event" coming along w/ e.g. q+g->jets
  - String Model (Also Cluster model)



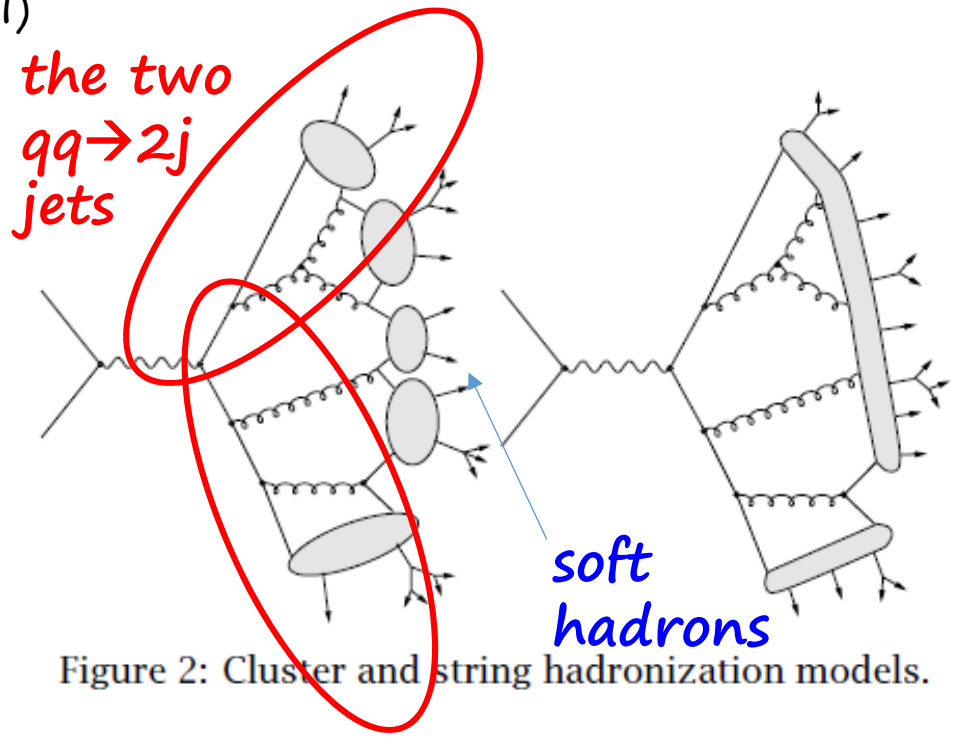


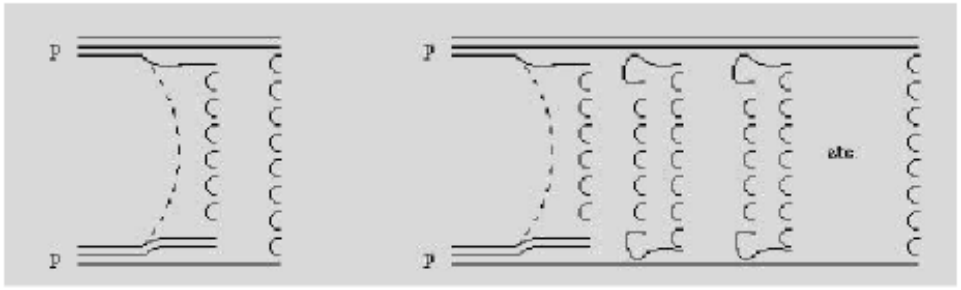
Figure 2: Cluster and string hadronization models.

String Model of Hadronization

- Used by PYTHIA also GEANT for some cases

Quark Gluon String Model

- Two or more strings may be stretched between partons within hadrons
  - Strings from cut cylindrical Pomerons
- Parton interaction leads to color coupling of valence quarks
  - sea quarks included too
- Partons connected by quark gluon strings, which hadronize



Strings ?!!?

- String Model → Simple literal tension modeling!
- Very successful in matching data!
- Define space time verticies (rapidity is used)
- Fractal like cycles

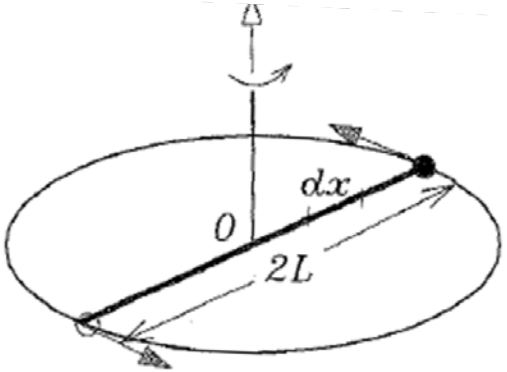


Fig. 7.3 A rotating relativistic string.

On the other hand, the angular momentum content of the segment dx is its energy content multiplied by vx. Hence, the total angular momentum of the string is

$$\begin{aligned} J &= 2\int_0^L \gamma\kappa v x\,dx \\ &= 2\int_0^L \frac{\kappa v x\,dx}{\sqrt{1-(x/L)^2}} \\ &= \kappa L^2\pi/2 \end{aligned}.$$

and

$$y = \frac{1}{2}\ln\frac{p_+}{p_-} = \frac{1}{2}\ln\frac{u}{v}. \tag{7.25b}$$

There is thus one more way to label the location of the vertex V by the proper momentum and the rapidity, (Γ, y). From Eq. (7.25a), the proper momentum variable Γ is related to the proper time coordinate, $\tau = \sqrt{t^2 - x^2}$, by

$$\sqrt{\Gamma} = \kappa\tau. \tag{7.26}$$

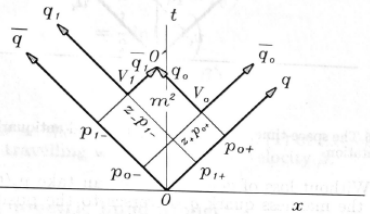


Fig. 7.7 The space-time trajectories of quarks and antiquarks from a string fragmentation at vertices V_0 and V_1.

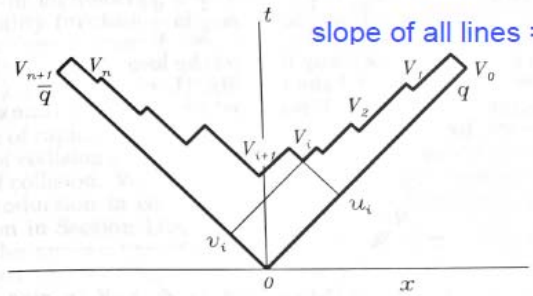
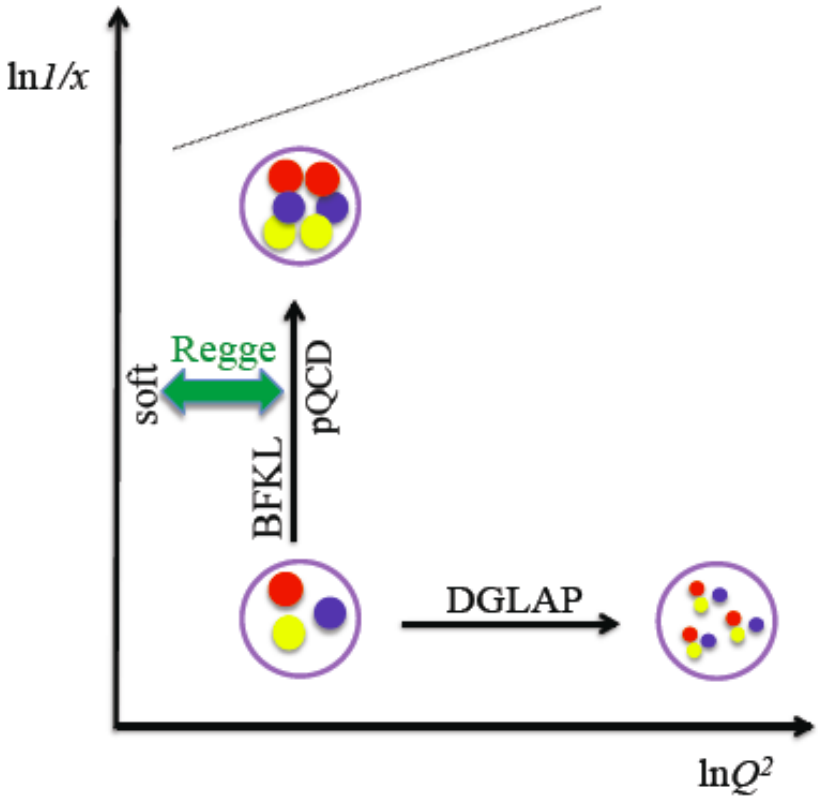


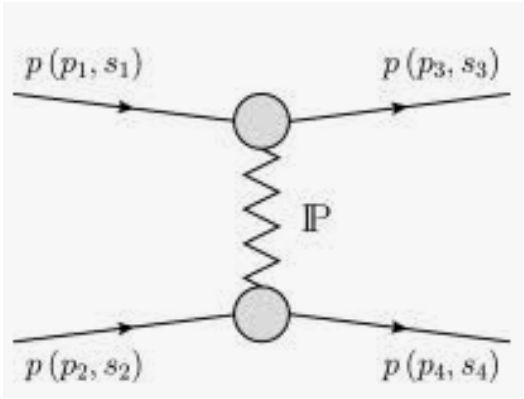
Fig. 7.6 The space-time trajectories of quarks and antiquarks from a string fragmentation.

Non-Perturbative QCD TOPICS- Pomeron

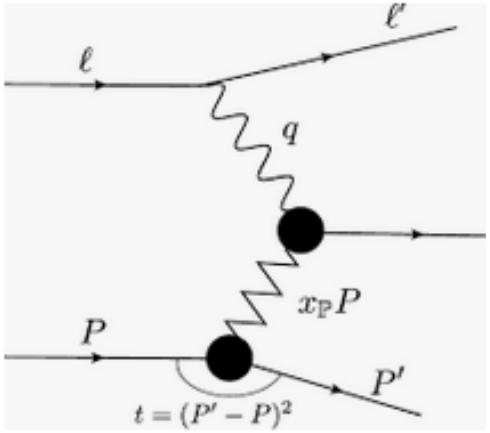
- "QCD" Diffraction -Pomeron Regge Theory

Proton stays intact
Interactions w/
"GLUEBALL" like object(s)





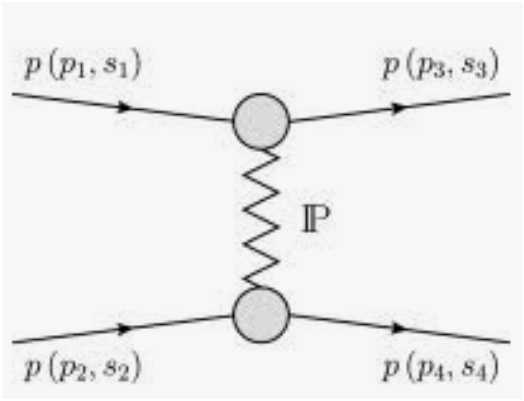
DIS-like



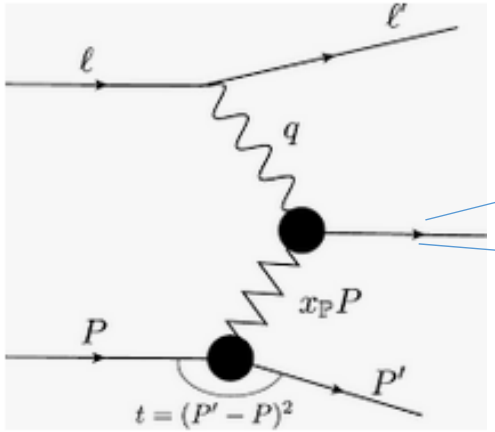
Relativistic Heavy Ions
Ze much larger "photon equivalent flux"
See Halzen Martin Weissacker Williams discussion

"Diffractive" Scattering - Pomeron

- More importantly these processes are "elastic" or pseudo elastic → despite high energy, all incoming hadrons are left intact by the scattering.
  - But ~usually, additional particles are produced
- The pomeron attains seemingly very real nature in some cases, because some of these processes involve e.g. jet production where it is thought that the quarks emerge from the breakup of the pomerons themselves.
- The EIC will have this kind of scattering as a major focus



Elastic most common

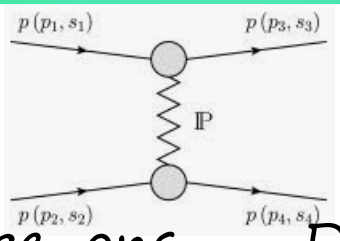


breakup- but not of incoming particles- Pom is dissected

Inelastic-more rare

Case Study: Other Particles - Pomeron





- Pomeron / Odderon /Regge-ons - Diffractive Physics - very odd "states"
- These are multigluon "states" that ~only act as force mediators of a sort, somewhat like pion is an effective force mediator for nucleons
- The scattering process involving pomeron exchange are responsible for a large percentage (majority at some energies) of the total high energy nuclear scattering **total cross section (x.s.) - IMPORTANT!**
  - But a little like low E photon minor scat/Bremst. leading to ∞ x.s. for in QED
- An early (<=70's) general classification of hadrons in scattering grouped them in a classifier type called Regge-ons → a theory that successfully explained the mysteriously rising .x.s. energy dependence through exchange of a Reggeon not observed elsewhere that had "vacuum quantum numbers" e.g. spin 0
  - **This x.s. rise is a non-perturbative QCD phenomenon**

High Points Regge Theory Slides/More about Pomeron scattering

- Regge Trajectories
- Pomeron is one == due to dependence on total h+h cross section w/ increasing energy

- Single pomeron exchange h+h → Total cross section influencer
- e + pomeron γ+pom → inelastic events (several/many hadrons produced) in e+p, e+A where nucleon/nuclei stay intact
- Doubly diffractive h+h → Pomeron+Pomeron both h's intact but inelastic extra hadron productions

- One Current Focus: Can we define PDF's for the Pomeron / Scattering?

An introduction to Regge Field Theory

Martin Poghosyan
(CERN)

Wilhelm und Else Heraeus Physics Summer School
"Diffractive and electromagnetic processes at high energies"
Heidelberg, Germany, September 2-6, 2013

Pomeranchuk theorem

$$\mathrm{Im}\,T_{ab}(s,t=0) = s\sigma_{tot}(ab)$$
$$\mathrm{Im}\,T_{ab}(u,t=0) = u\sigma_{tot}(a\bar{b})$$

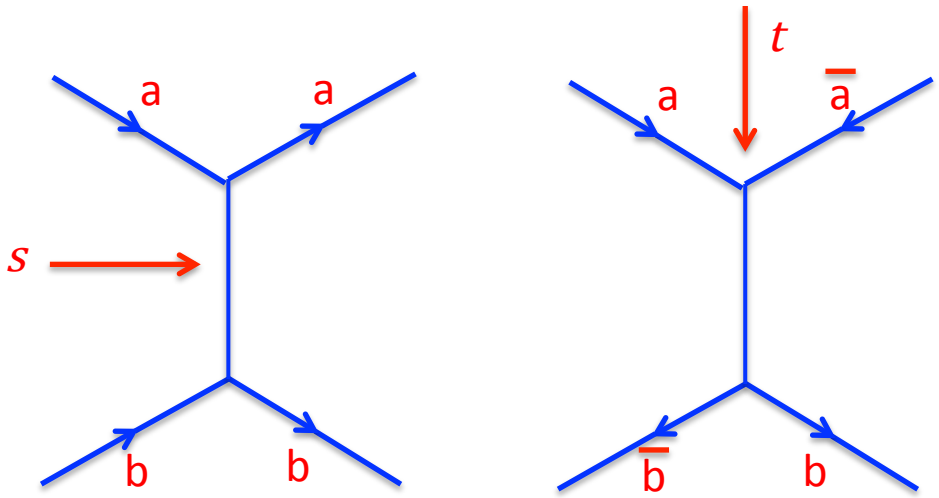
If T is not an oscillating function and $\dfrac{1}{\ln s}\dfrac{\mathrm{Re}\,T(s,0)}{\mathrm{Im}\,T(s,0)} \to 0$ at $s \to \infty$



$$\sigma_{tot}(ab) = \sigma_{tot}(a\bar{b}) \text{ at } s \to \infty$$

Pomeranchuk theorem may be violated. See O. Nachtmann's talk.

t-channel exchange picture



Suppose the $f_l(t)$ has a singularity of form

$$f_l(t) = \frac{r(t)}{l - \alpha(t)}$$

$$\frac{1}{2i}\left[f_l(t) - f_l^*(t)\right] \sim f_l(t) f_l^*(t)$$

$$f_l(t) - f_l^*(t) = \frac{r}{l-\alpha(t)} - \frac{r^*}{l-\alpha^*(t)} = \frac{r\left(l-\alpha^*(t)\right) - r^*\left(l-\alpha(t)\right)}{\left(l-\alpha(t)\right)\left(l-\alpha^*(t)\right)} \sim 2i\frac{rr^*}{\left(l-\alpha(t)\right)\left(l-\alpha^*(t)\right)}$$

$$r = r^*$$

$$\mathrm{Im}\,\alpha(t) \sim r \neq 0$$

No poles in the real axis in the l plane for $t > 4m^2$

Resonances

Assume for some $t = t_R \equiv M_R^2 > 4m^2$, $\mathrm{Re}\,\alpha(t_R) = l_R$ $\Rightarrow$ $\alpha(t) \approx l_R + i\,\mathrm{Im}\,\alpha(t_R) + \alpha'(t_R)(t - t_R)$

Taylor expansion

$$f_{l_R}(t) \approx \frac{r(t_1)}{l_R - [l_R + i\,\mathrm{Im}\,\alpha(t_R) + \alpha'(t_R)(t - t_R)]} = -\frac{\mathrm{Im}\,\alpha(t_1)}{\alpha'(t_R)(t - t_R) + i\,\mathrm{Im}\,\alpha(t_R)} = \frac{\mathrm{Im}\,\alpha(t_R)/\alpha'(t_R)}{t_R - t - i\,\mathrm{Im}\,\alpha(t_R)/\alpha'(t_R)}$$

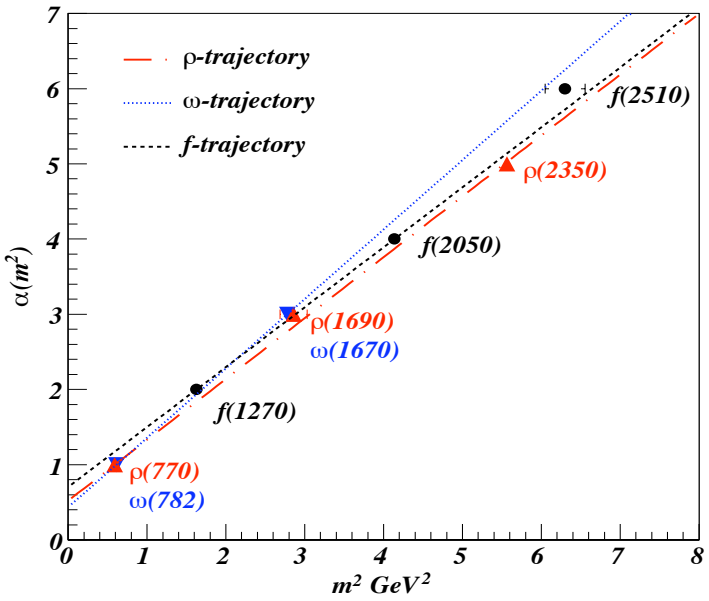
$$\mathrm{Im}\,\alpha(t_R)/\alpha'(t_R) \equiv M_R\Gamma$$

$$f_{l_R}(t) \sim \frac{1}{M_R^2 + t - iM_R\Gamma}$$

Breit-Wigner

Regge pole in the physical region of the t-channel ($t > 4m^2$) corresponds to a Briet-Wigner resonance with $\mathrm{Re}\,\alpha(t_1) = l_R$ (=spin of R)

Reggeon trajectory



Chew-Frautschi plot

Regge trajectories are almost straight lines and in standard Regge theory they are parameterized by $\alpha(t) = \alpha_0 + \alpha' \cdot t$

Regge pole gives a generalization of a particle exchange in the t-channel. It corresponds to an exchange in the t-channel by a state of noninteger spin $\alpha(t)$ (reggeon trajectory), which coincides with particles of spin J for $t = M_J^2$

Regge pole exchange amplitude

"Physical" region in the t−channel corresponds to $t > 4m^2, s < 0$. Analytically continue the amplitude to $s > 4m^2, t < 0$ (s−channel).

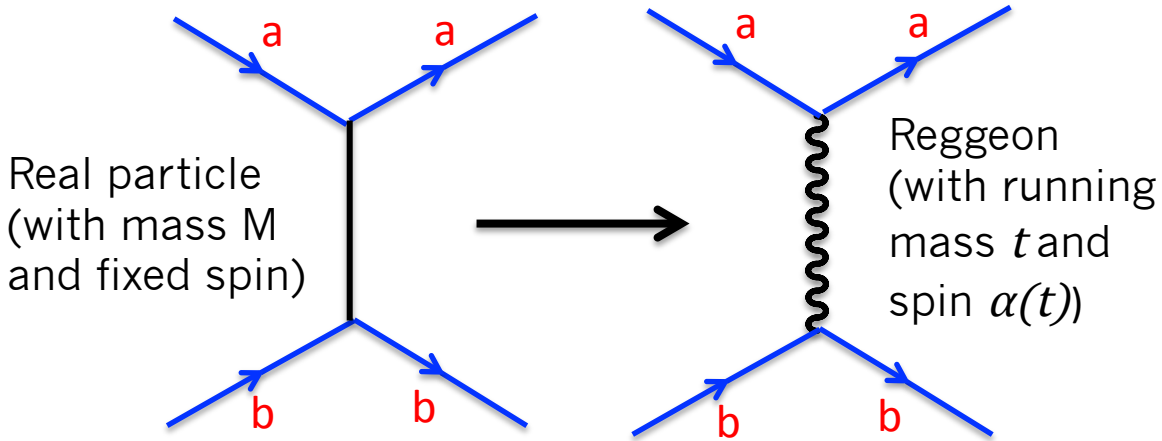
For $s >> 4m^2 > |t|$, $\cos\theta_t \sim \frac{s}{4m^2} >> 1$

$P_l(z) \sim z^l$ for $z >> 1$

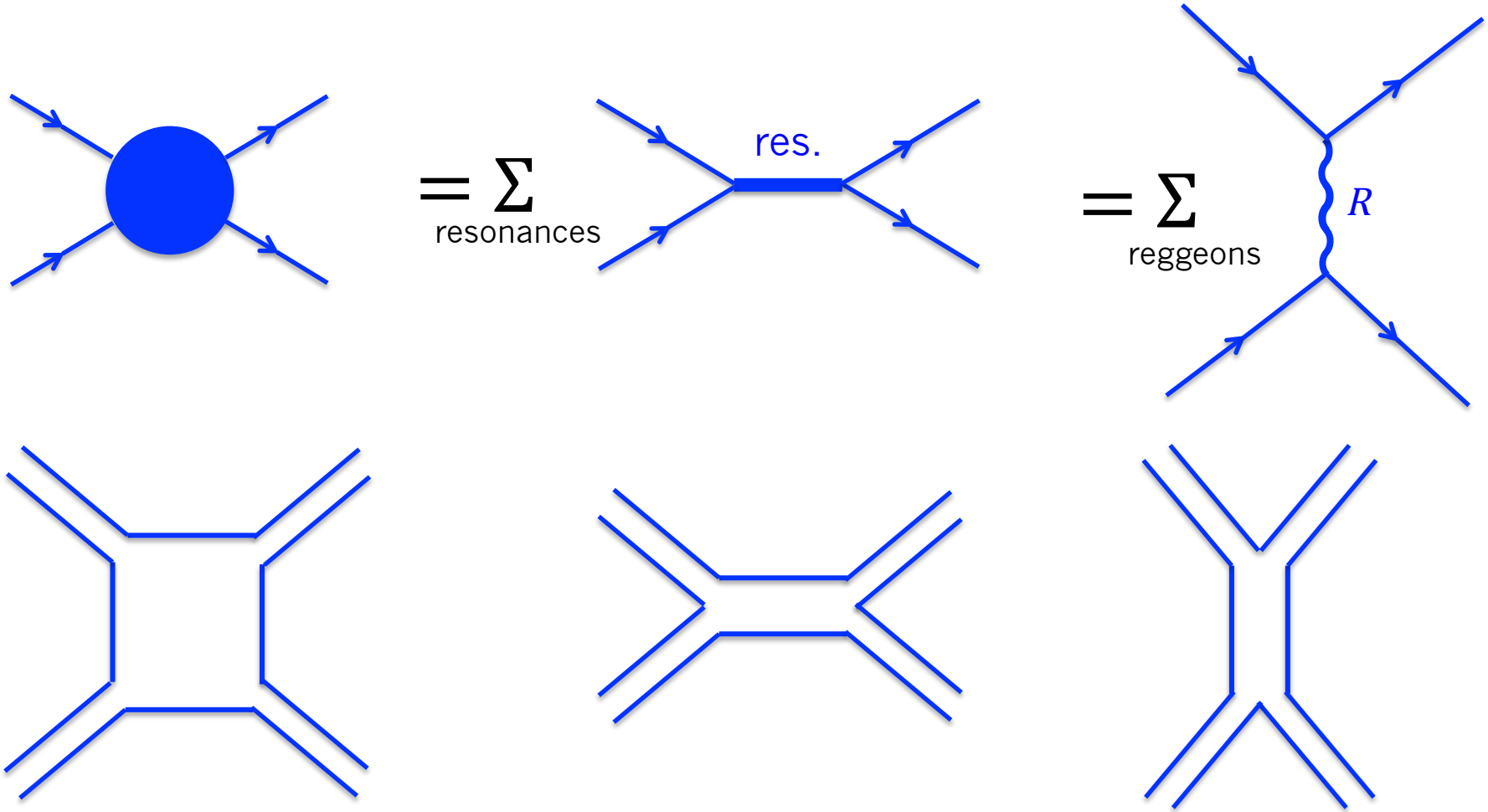
$$T(t,s) = \sum_{\sigma=\pm 1} \sum_{poles} \eta_\sigma \left(\alpha_i^\sigma(t)\right) \gamma_i^\sigma(t) \left(\frac{s}{s_0}\right)^{\alpha_i^\sigma(t)}$$

$$\sigma_{tot}(s) = \frac{1}{s} \mathrm{Im}\, T(s,0) \sim s^{\alpha(0)-1}$$

s_0 is a constant scale factor, usually chosen to be $s_0 = 1\ \mathrm{GeV}^2$.



Duality

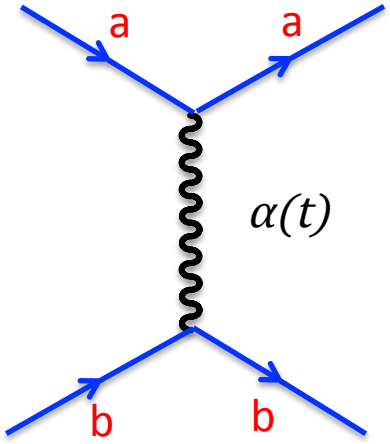


Factorization

What is the meaning of $\gamma(t)$?

In fact, all information about incoming and outgoing particles (baryon number, strangeness, etc.) are absorbed in $\gamma(t)$ and it does not depend on s.

$\gamma(t)$ should be related to Reggeon-hadron interaction vertex!



One can assume the initial state does not know anything about the final state: in the cross-channel the initial particles first transform into an intermediate state, which then gets converted into the final particles, with the amplitude independent of the properties of the initial state.

$$\gamma(t) = g_{aa}(t) g_{bb}(t)$$

It is not possible to predict the explicit form of $g_{aa}(t)$ from the analytical properties of the matrix element (model dependent).

Regge pole approximation

At fixed t, with $s >> t$

- Amplitude for a process governed by the exchange of a trajectory $\alpha(t)$ is

$$T(s,t) \propto \left(s / s_0\right)^{\alpha(t)}$$

- No prediction for t dependence

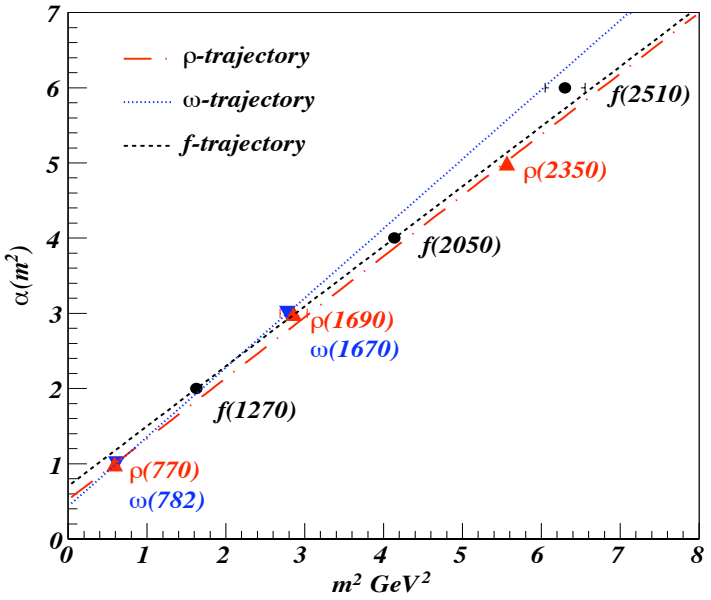
- Elastic cross section

$$\frac{d\sigma_{el}}{dt} \approx \frac{1}{s^2}\left|T(s,t)\right|^2 \propto s^{2(\alpha(t)-1)}$$

- Total cross section considering the optical theorem

$$\sigma_{tot} \approx \frac{1}{s}\,\mathrm{Im}\,T(s,0) \propto s^{\alpha(0)-1}$$

Reggeons



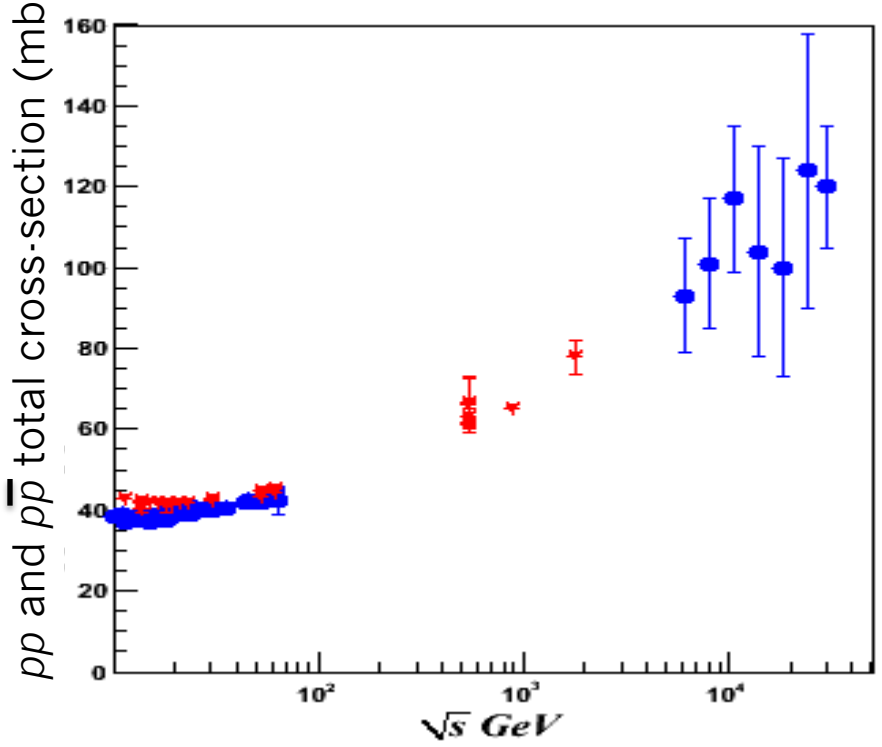
$$\alpha_i(t) = \alpha_i(0) + \alpha_i' \cdot t, \quad i = f, \rho, \omega.$$

$$\alpha_f(0) = 0.703 \pm 0.023 \quad \alpha_f' = 0.797 \pm 0.014 GeV^{-2}$$
$$\alpha_\rho(0) = 0.522 \pm 0.009 \quad \alpha_\rho' = 0.809 \pm 0.015 GeV^{-2}$$
$$\alpha_\omega(0) = 0.435 \pm 0.033 \quad \alpha_\omega' = 0.923 \pm 0.054 GeV^{-2}$$

$$\sigma_{tot} \propto \left(\frac{s}{s_0}\right)^{\alpha_i(0)-1} \Longrightarrow \quad \sigma_{tot} \to 0 \quad at \quad s \to \infty$$

Unexpected Reggeon?

Experiment: $\sigma_{tot} \not\to 0 \quad at \quad s \to \infty$

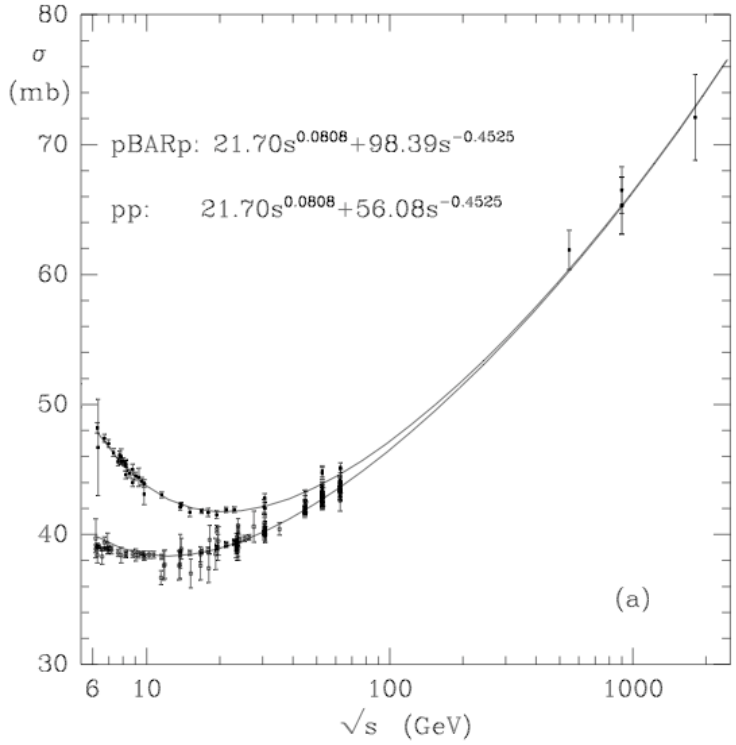


$$\sigma_{tot} \sim \left(\frac{s}{s_0}\right)^{\alpha_i(0)-1}$$



An object with α(0)= 1+ Δ >1 is needed

Pomeron



Donnachie and Landshoff (1992)

$$\sigma_{tot} = As^{0.0808}$$

grows as a power function of s

Unitarity requires that the total cross section at very high energies should not grow faster than $\ln^2 s$ (Froissart bound).

For describing DIS data

$$F_2(x,Q^2) = f(Q^2)x^{\Delta(Q^2)}$$ (CKMT 1992)

$$F_2(x,Q^2) = f_1(Q^2)x^{-0.08} + f_2(Q^2)x^{-0.42}$$ (DL 1998)

soft Pomeron hard Pomeron

Pomeron



It is usually assumed that the Pomeron in QCD is related to gluonic exchanges in the t–channel.

Δ_{eff} determined from fits to data are in general different from Δ.

See talks by
L. Jenkovszky
A. Martin
O. Nachtmann
W. Schäfer

DIFFRACTION:
In HEP any process involving Pomeron exchange

A simple parameterization of Regge residues

$g_{aa}(t) = g_{aa}\exp\left\{R_{aa}^2 t\right\}$ $\quad R_{aa}$ – Regge radius of hadron *a*

$\alpha(t) = \alpha_0 + \alpha' t$

$$\eta_\sigma(\alpha) = -\frac{\sigma + e^{-i\pi\alpha}}{\sin(\pi\alpha)} = e^{-i\pi\alpha/2}\begin{cases} \dfrac{-1}{\sin(\pi\alpha/2)} \text{ for } \sigma = +1 \\ \dfrac{i}{\cos(\pi\alpha/2)} \text{ for } \sigma = -1 \end{cases}$$

$$T(s,t) = g_{aa}g_{bb}\eta(\alpha(0))\exp\left\{\lambda t\right\}\left(\frac{s}{s_0}\right)^{\alpha_0}$$

$$\lambda \equiv R_{aa}^2 + R_{bb}^2 + \alpha'\left(\ln\left(s/s_0\right) - i\pi/2\right)$$

Impact parameter representation

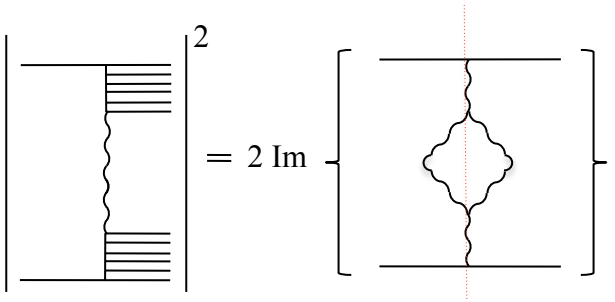
$$f_{ab}(s,\boldsymbol{b}) \sim \int d^2q_\perp \exp\left\{-i\mathbf{b}\mathbf{q}_\perp\right\}T(s,q_\perp^2) \sim \frac{\left(s/s_0\right)^{\alpha_0-1}}{\lambda}\exp\left\{-\frac{b^2}{4\lambda}\right\}$$

$$\sqrt{\overline{b}^2} = 2\,|\,\lambda\,| \approx 2\sqrt{R_a^2 + R_b^2 + \alpha'\ln\left(s/s_0\right)}$$

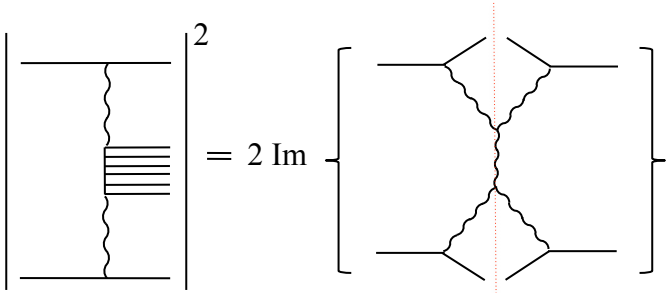
Radius of interaction increases with increasing *s*

$$\frac{d\sigma}{dt} \sim \left(\frac{s}{s_0}\right)^{2\alpha-2} \times \exp\left\{-2\left(R_{aa}^2 + R_{bb}^2 + \alpha'\ln\left(s/s_0\right)\right)|\,t\,|\right\}$$

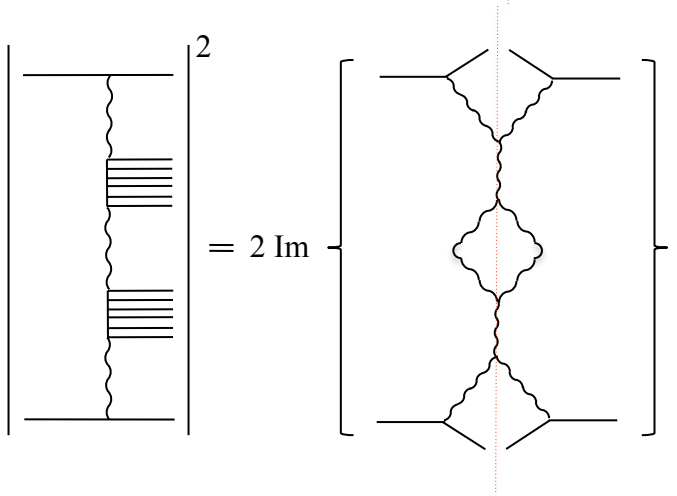
Increases with increasing *s*. Diffraction peak shrinkage.



Double diffraction



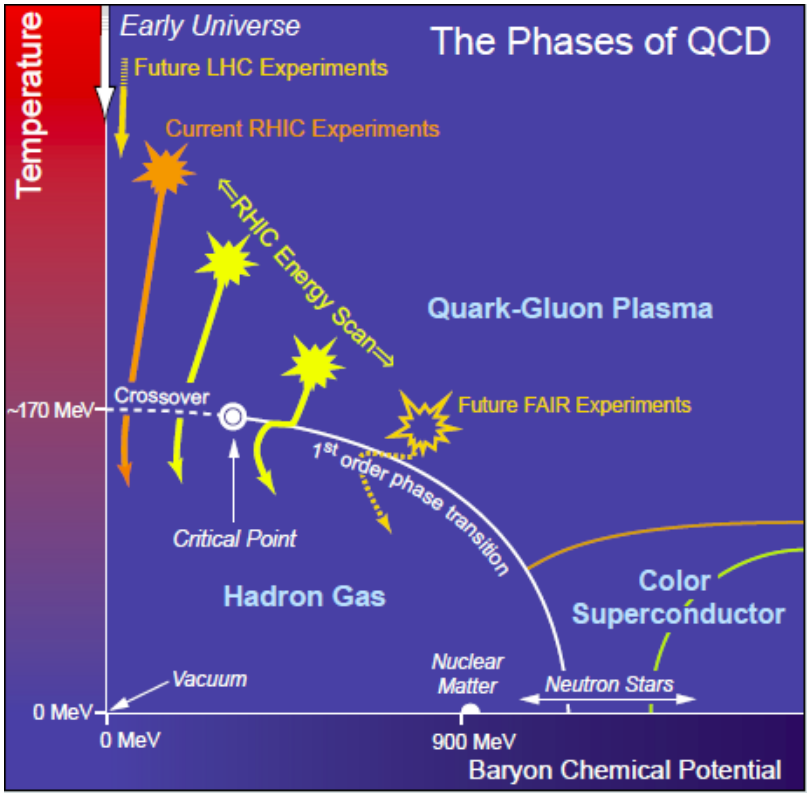
Central production



Double gap topology

Bag Model- Thermal degrees of Freedom

- Thermal / high density QCD are other regions where completely new QCD-related phenomena (e.g. yet even more different degrees of freedom) may exist
- Only broad description of very softest quark/gluon phenomena can describe "bulk" large numbers of low energy (temperature ~100's of MeV) QCD interactions
- Simplest possible model: MIT Bag Model



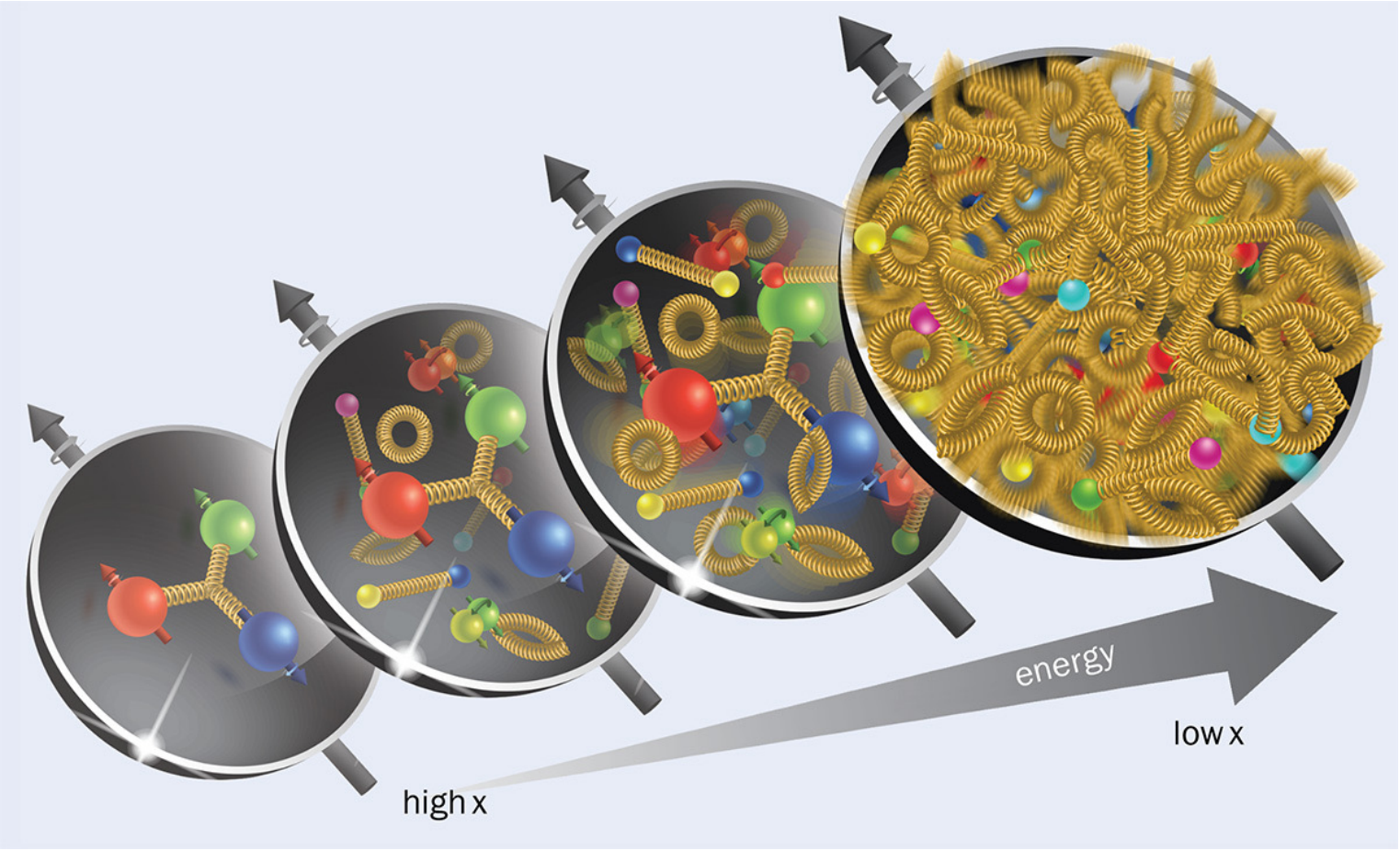
Density →

Gluon Junctions

- Example of types of degrees of freedoms we still don't know whether they may exist
- In this case, could carry Baryon number

- Talk by Kent State faculty Zhangbu Xu

Tracking the origin of baryon number with electron-ion collisions



Zhangbu Xu
(Brookhaven National Lab)

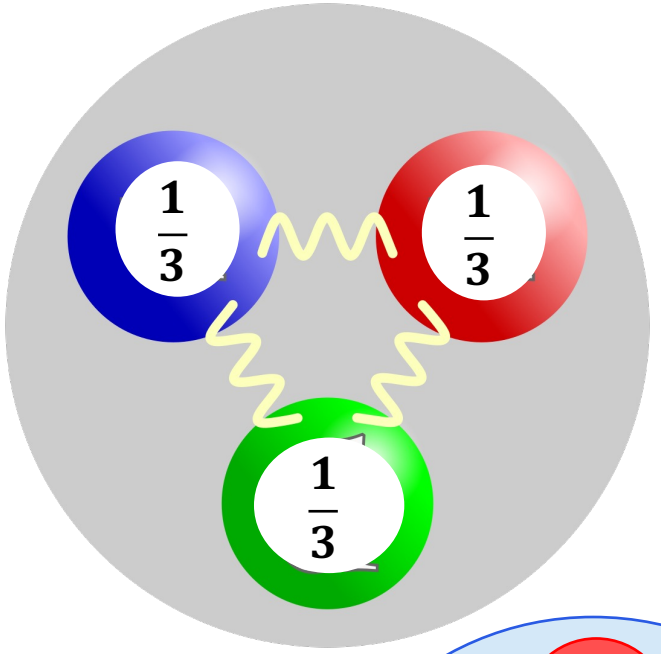
- baryon number carrier
- Three experimental approaches at RHIC
- Earlier theory and experiment work on pp and ep
- EIC perspectives and requirements

N. Lewis, T. Tsang, Y. Li, H. Klest, W.B. Zhao, N. Magdy, R.R. Ma, P. Tribedy, J.D. Brandenburg, Z.B. Tang, Z.W. Lin, C. Shen, B. Schenke, D. Kharzeev, *et al.*

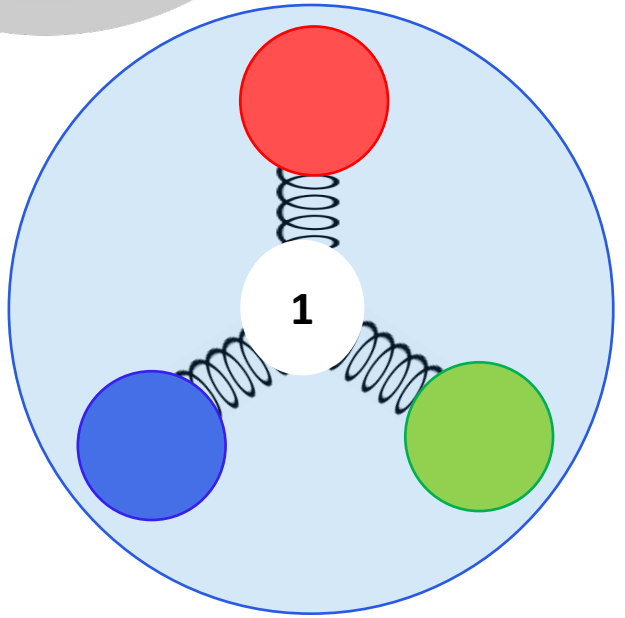
In part supported by
Brookhaven National Laboratory
U.S. DEPARTMENT OF ENERGY | Office of Science

Baryon Number (B) Carrier

https://en.wikipedia.org/wiki/Quark



- Textbook picture of a proton
  - Lightest baryon with strictly conserved baryon number
  - Each valence quark carries 1/3 of baryon number
  - Proton lifetime >10^{34} years
  - Quarks are connected by gluons
- Alternative picture of a proton
  - Proposed at the Dawn of QCD in 1970s
  - A Y-shaped gluon junction topology carries baryon number (B=1)
  - The topology number is the strictly conserved number
  - Quarks do not carry baryon number
  - Valence quarks are connected to the end of the junction always



[1]: Artru, X.; String Model with Baryons: Topology, Classical Motion. Nucl. Phys. B 85, 442–460 (**1975**).
[2]: Rossi, G. C. & Veneziano, G. A; Possible Description of Baryon Dynamics in Dual and Gauge Theories. Nucl. Phys. B 123, 507–545 (**1977**)

Model implementations of baryons at RHIC

- Many of the models used for heavy-ion collisions at RHIC (HIJING, AMPT, UrQMD) have implemented a nonperturbative baryon stopping mechanism

  V. Topor Pop, *et al,* Phys. Rev. C **70**, 064906 (2004)

  Zi-Wei Lin, *et al*, Phys. Rev. C **72**, 064901 (2005)

  M. Bleicher, *et al,* J.Phys.G **25,** 1859-1896 (1999)

- Baryon Stopping
  - Theorized to be an effective mechanism of stopping baryons in pp and AA

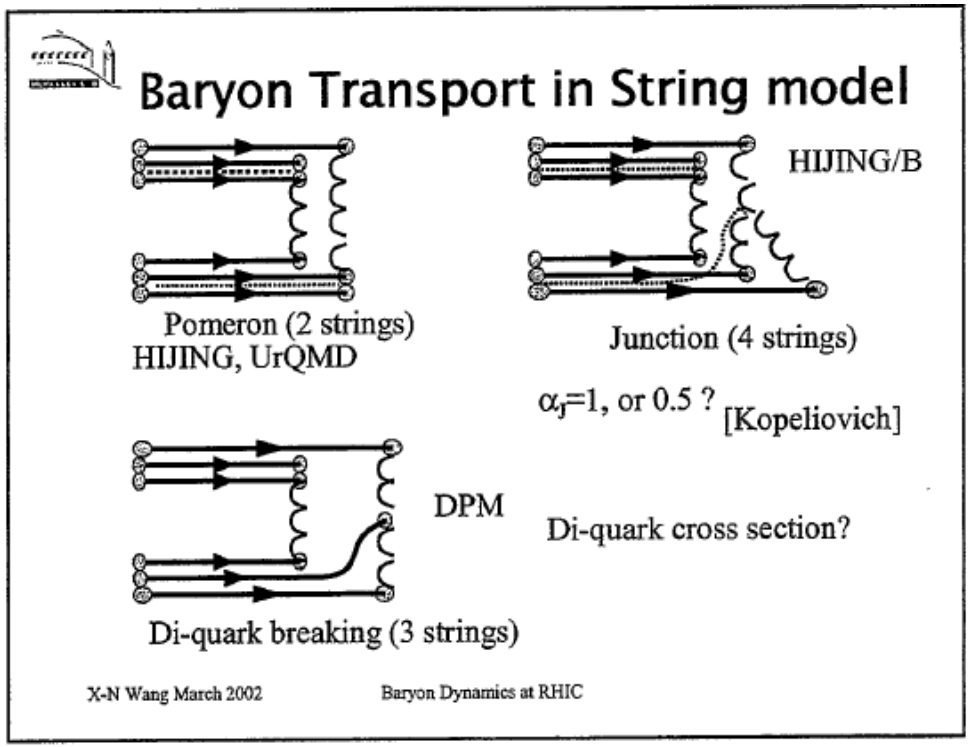
  D. Kharzeev, Physics Letters B **378**, 238-246 (1996)

- Specific rapidity dependence is predicted:

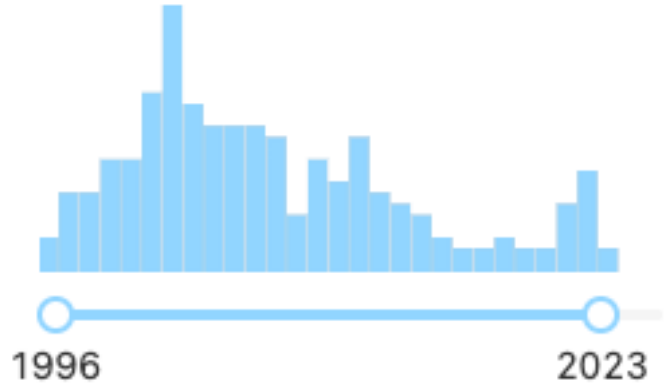
$$p = \sim e^{-\alpha_B y}$$

α_B ~=0.5

2003 RBRC Workshop on "Baryon Dynamics at RHIC"



D. Kharzeev, Physics Letters B **378**, 238-246 (1996)
"Can gluons trace baryon number?"



"Science, however, is never conducted as a popularity contest..." --- Michio Kaku

BUT citations ARE

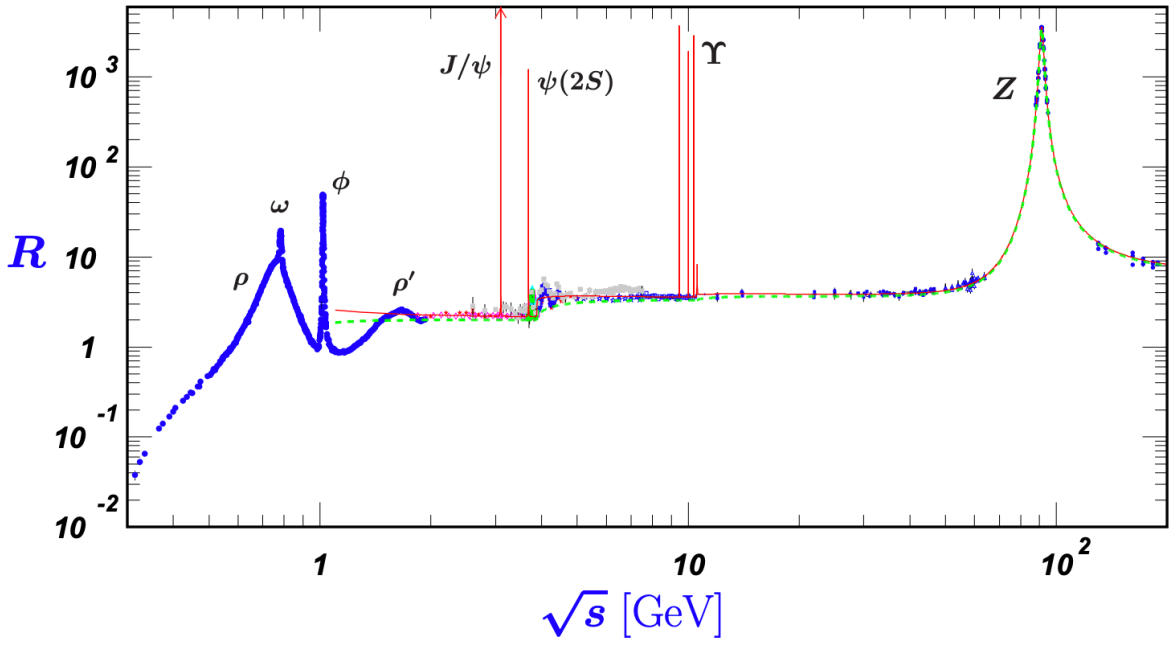
Measurements of quark electric charges

Scattering cross section $\sigma \propto e_q^2$

$(2/3)^2+(1/3)^2+(1/3)^2=2/3$

$(2/3)^2+(2/3)^2+(1/3)^2=1$

$(1/3)^2+(1/3)^2+(1/3)^2=1/3$



PDG

Figure 53.2: World data on the total cross section of $e^+e^- \to hadrons$ and the ratio $R(s) = \sigma(e^+e^- \to hadrons, s)/\sigma(e^+e^- \to \mu^+\mu^-, s)$. $\sigma(e^+e^- \to hadrons, s)$ is the experimental cross section corrected for initial state radiation and electron-positron vertex loops, $\sigma(e^+e^- \to \mu^+\mu^-, s) = 4\pi\alpha^2(s)/3s$. Data errors are total below 2 GeV and statistical above 2 GeV. The curves are an educative guide: the broken one (green) is a naive quark-parton model

Riordan, Science 1992

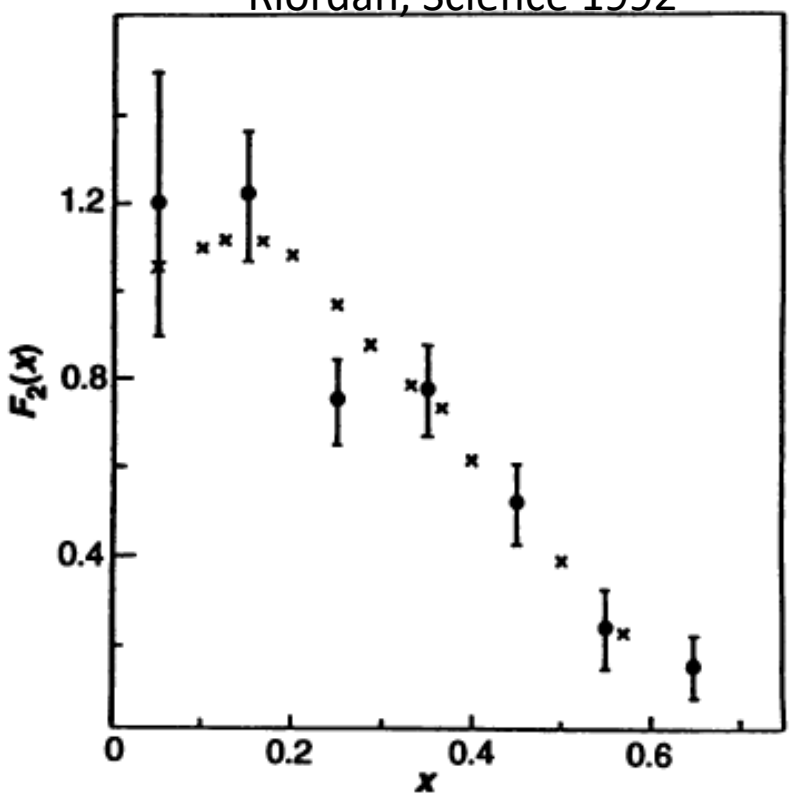
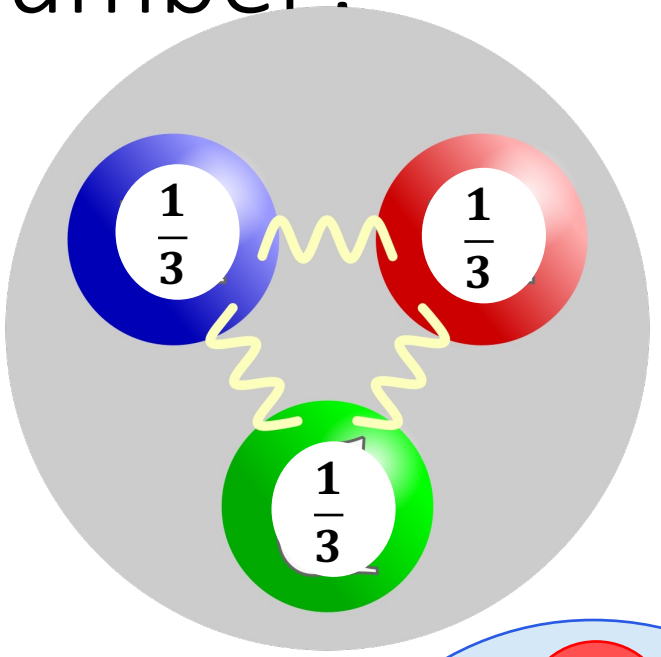
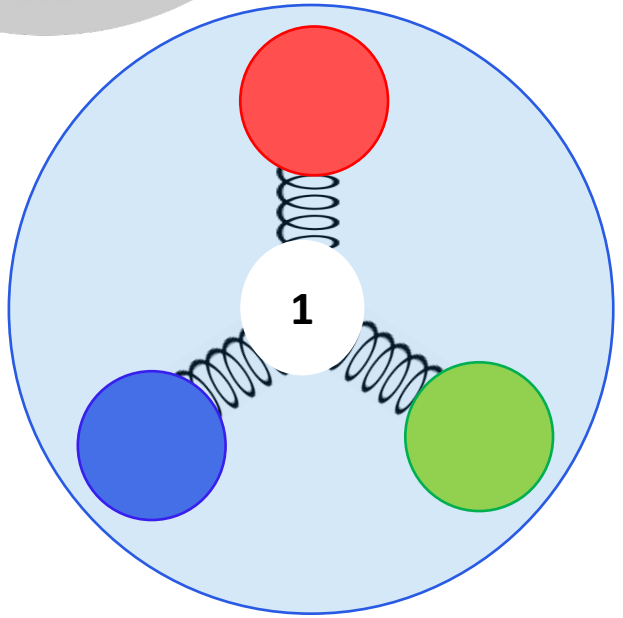


Fig. 8. Comparison of structure functions measured in deep inelastic neutrino-nucleon scattering experiments on the Gargamelle heavy-liquid bubble chamber with the MIT-SLAC data [(●), Gargamelle, $F_2^{\nu N}$; (×), MIT-SLAC, (18/5)F_2^{eN}]. When multiplied by 18/5, a number specified by the quark-parton model, the electron scattering data coincide with the neutrino data.

Measurements of quark baryon number?

- Textbook picture of a proton
  - Lightest baryon with strictly conserved baryon number
  - Each valence quark carries 1/3 of baryon number
  - Proton lifetime >10^{34} years
  - Quarks are connected by gluons
- Alternative picture of a proton
  - Proposed at the Dawn of QCD in 1970s
  - A Y-shaped gluon junction topology carries baryon number (B=1)
  - The topology number is the strictly conserved number
  - Quarks do not carry baryon number
  - Valence quarks are connected to the end of the junction always
- **Neither of these postulations has been verified experimentally**





[1]: Artru, X.; String Model with Baryons: Topology, Classical Motion. Nucl. Phys. B 85, 442–460 (**1975**).
[2]: Rossi, G. C. & Veneziano, G. A; Possible Description of Baryon Dynamics in Dual and Gauge Theories. Nucl. Phys. B 123, 507–545 (**1977**)